

Part I

FOUNDATIONS OF NUMBER THEORY

Chapter 1

THE GREATEST COMMON DIVISOR OF TWO NUMBERS

Until further notice, lower case italic letters will always represent integers, i.e.

$1, 2, 3, \dots$ (positive integers or natural numbers),

0 (zero),

$-1, -2, -3, \dots$ (negative integers).

The following facts will be used constantly: If a is an integer, then so are $-a$ and $|a|$; if a and b are integers, then so are $a + b$, $a - b$, and ab ; if $a > b$, then $a \geq b + 1$; and if $a < b$, then $a \leq b - 1$.

DEFINITION 1.1. Let $a \neq 0$; let b be arbitrary. Then b is said to be *divisible* by a if there exists a number q such that

$$b = qa.$$

This q , namely $q = \frac{b}{a}$, is then uniquely determined.

We also say that: b is a *multiple* of a , a is a *divisor* of b , a *divides* b , or a *goes into* b . In symbols,

$$a \mid b.$$

If $a \neq 0$ and b is not divisible by a , then we write

$$a \nmid b.$$

Examples: $2 \mid 6$, $4 \nmid 6$, $3 \nmid 4$, $2 \nmid -4$.

THEOREM 1.2. 1) *The number 1 divides every number.*

2) *Every number $a \neq 0$ divides itself, and divides 0.*

Proof. 1) $a = 1 \cdot a$.

2) $a = a \cdot 1$; $0 = a \cdot 0$. □

THEOREM 1.3. *If $a \mid b$ and $b \mid c$, then $a \mid c$.*

Proof.

$$b = q_1 a, \quad c = q_2 b = q_2 q_1 a.$$

□

THEOREM 1.4. *If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.*

Proof.

$$b = qa, \quad q \neq 0, \quad |q| \geq 1, \quad |b| = |q| \cdot |a| \geq |a|.$$

□

THEOREM 1.5. *If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$ and $a \mid (b - c)$.*

Proof.

$$b = q_1 a, \quad c = q_2 a, \quad b \pm c = (q_1 \pm q_2) a.$$

□

THEOREM 1.6. *If $a \mid b_1$, $a \mid b_2$, ..., $a \mid b_n$, then $a \mid (q_1 b_1 + \dots + q_n b_n)$ for arbitrary $q_1, \dots, q_n$.*

Proof. By Theorem 4, $a \mid q_m b_m$ ($m = 1, \dots, n$), and hence, by repeated application of Theorem 4, $a \mid \sum_{m=1}^n q_m b_m$. □

THEOREM 1.7. *Given any a and any $b \neq 0$, there exist uniquely determined numbers q and r such that*

$$a = qb + r, \quad 0 \leq r < |b|. \tag{1.1}$$

Proof. Among all the numbers of the form $a - xb$ (where x may be any integer) there are non-negative ones, for example $a + |a| \cdot b$; among the non-negative ones there is a smallest; call it r . Then for a suitable q we have

$$a - qb = r \geq 0.$$

Also $r < |b|$; for if $r \geq |b|$, then $r - |b| = a - (q \pm 1)b$ would be a smaller non-negative number of the form $a - xb$. Thus the numbers q and r exist.

If also $a = q_1b + r_1$, $0 \leq r_1 < |b|$, then $(q_1 - q)b = r - r_1$, $|r - r_1| < |b|$, so that by Theorem 3, $0 = q_1 - q$, $q_1 = q$, $r_1 = r$. $\square$

DEFINITION 1.8. The number r in (1.1) is called the *remainder* (or *residue*) upon division of a by b , and q is called the *incomplete quotient*.

Examples: $a = 7$, $b = 3$; $q = 2$, $r = 1$. $a = 7$, $b = -3$; $q = -2$, $r = 1$. $a = -7$, $b = 3$; $q = -3$, $r = 2$. $a = -7$, $b = -3$; $q = 3$, $r = 2$.

DEFINITION 1.9. If $r = 0$ in (1.1), then q is called the *quotient* upon exact division of a by b .

DEFINITION 1.10. A number d is called a *common divisor* of a and b if $d \mid a$ and $d \mid b$. Such a number obviously exists (for example, the number 1) provided a and b are not both $= 0$.

Among all the common divisors of a and b there is—if a and b are not both $= 0$ —a greatest, which we call the *greatest common divisor* of a and b , and denote by (a, b) .

Examples: $(6, 10) = 2$, $(0, b) = |b|$ for $b \neq 0$, $(0, 0) = 0$, $(a, b) = (b, a) = (|a|, |b|)$.

THEOREM 1.11. (a, b) is divisible by every common divisor of a and b .

Proof. Without loss of generality let $b \neq 0$. (Otherwise interchange a and b .) Set $(a, b) = d$. I assert that there exist numbers x and y such that

$$ax + by = d. \tag{1.2}$$

For if we divide a by b with a remainder r in accordance with Theorem 7, then $a = qb + r$, $0 \leq r < |b|$, and if we set $d_1 = (b, r)$, then

$d_1 \mid b$, $d_1 \mid r$, $d_1 \mid a$, so that d_1 is a common divisor of a and b , and therefore $d_1 \leq d$.

On the other hand, every common divisor of a and b divides $r = a - qb$ and is therefore $\leq d_1$; hence $d = (b, r) = d_1$.

This argument shows that the problem of finding the greatest common divisor of a and b has been reduced to that of finding the greatest common divisor of b and r , where r is smaller than $|b|$. Continuing in this way, after finitely many steps we reach two numbers of which one is a divisor of the other, and the greatest common divisor is then immediately obvious.

To complete the proof of (1.2), I remark that from the equation $a = qb + r$ it follows that $r = a - qb$, and if equations $b = q_1r + r_1$, $r = q_2r_1 + r_2$, ... are successively written down, then from each of them it follows that the remainder can be expressed in the form $ax + by$ as soon as this is possible for the two preceding remainders. Since this is possible for the last remainder (which is the greatest common divisor itself), it is possible for all the remainders.

Now let e be any common divisor of a and b . Then by Theorem 6 and by (1.2), $e \mid d$. □

THEOREM 1.12. *The numbers x and y of equation (1.2) may be chosen so that $|x| \leq |b|$ and $|y| \leq |a|$.*

Proof. If $a = b = 0$, then $d = 0$, and the numbers $x = y = 0$ do what is required. If a and b are not both $= 0$, then $d \neq 0$, and

$$\frac{a}{d} \cdot x + \frac{b}{d} \cdot y = 1.$$

By Theorem 7, let $x = \frac{b}{d}q + x_0$, $0 \leq x_0 < \left|\frac{b}{d}\right| \leq |b|$, $y = -\frac{a}{d}q + y_0$; then $\frac{a}{d}x_0 + \frac{b}{d}y_0 = 1$, so that by Theorem 6, since $\frac{a}{d}$ and $\frac{b}{d}$ are not both $= 0$, we have $y_0 \neq 0$ and hence $|y_0| \leq |a|$. □

THEOREM 1.13. *If $c \mid ab$ and $(c, a) = 1$, then $c \mid b$.*

Proof. By Theorem 10, $ax + cy = 1$, so that $abx + cb = b$; by hypothesis $c \mid ab$, and obviously $c \mid cb$, so that by Theorem 4, $c \mid b$. □

THEOREM 1.14. *Every common multiple of a and b is divisible by their smallest positive common multiple v (which obviously exists, since among the common multiples, for example ab and $-ab$, there are positive ones).*

Proof. Without loss of generality let $b \neq 0$. Among all positive common multiples let v be the smallest. Let n be any common multiple. By Theorem 7, $n = qv + r$, $0 \leq r < v$. Since $a \mid n$ and $a \mid v$, we have $a \mid r$; similarly $b \mid r$. If $r > 0$, then r would be a smaller positive common multiple than v , which is impossible; hence $r = 0$ and $v \mid n$. $\square$

THEOREM 1.15. *If v is the smallest positive common multiple of a and b , then every common multiple of a and b is of the form hv , where h is arbitrary; in particular, ab is such a multiple, so that $v \mid ab$.*

Proof. By Theorem 12, $v \mid n$, so that $n = hv$. Conversely, every hv is a common multiple. $\square$

THEOREM 1.16. *Let $ab \neq 0$, and let v be the smallest positive common multiple of a and b . Then*

$$v = \frac{|ab|}{(|a|, |b|)}.$$

Proof. Without loss of generality let $a > 0$ and $b > 0$. By Theorem 12, $\frac{ab}{(a,b)}$ is a common multiple of a and b , since a and b divide it; hence by Theorem 13, $v \leq \frac{ab}{(a,b)}$.

On the other hand, if we set $d = (a, b)$ and write $a = da_1$, $b = db_1$, then $(a_1, b_1) = 1$ and $v = ha = ha_1d = kb = kb_1d$, so that $ha_1 = kb_1$, and hence by Theorem 11, since $(a_1, b_1) = 1$, we have $a_1 \mid k$, so that $k = la_1$ and $v = la_1b_1d = \frac{ab}{d}l \geq \frac{ab}{d}$. $\square$

THEOREM 1.17. *If $d \mid ab$ and $(d, a) = 1$, then $d \mid b$.*

(Note: This is a restatement of Theorem 11, included for reference.)

Chapter 2

PRIME NUMBERS AND FACTORIZATION INTO PRIME FACTORS

The number 1 has only one positive divisor, namely 1; every number $a > 1$ has at least two positive divisors, namely 1 and a .

DEFINITION 2.1. A number $a > 1$ is called a *prime number* (or simply a *prime*) if it has only two positive divisors (namely 1 and a).

Examples: The first few primes are 2, 3, 5, 7, 11.

The letter p will be reserved for prime numbers only; likewise, symbols such as $p_1, p_2, \dots, p', p'', \dots$ will always represent primes.

Our next aim will be to prove that every number $a > 1$ can be represented as a product of primes (this will be easy) and that this representation is unique apart from the order of the factors (this is somewhat deeper).

THEOREM 2.2. *Every $a > 1$ can be represented as a product of prime numbers:*

$$a = \prod_{n=1}^r p_n, \quad r \geq 1. \quad (2.1)$$

Proof. 1) For $a = 2$ the assertion is true, since 2 is a prime.

2) Let $a > 2$ and assume the theorem true for 2, 3, $\dots$, $a - 1$.

21) If a is prime, the assertion is true.

22) Otherwise, by Definition 3, there exists a factorization $a = a_1 \cdot a_2$, $1 < a_1 < a$, $1 < a_2 < a$. Thus a_1 and a_2 , and therefore a also, are representable as products of primes. $\square$

Theorem 17 justifies the following definition:

DEFINITION 2.3. Every number > 1 which is not a prime is called a *composite number*.

The natural numbers thus fall into three classes:

- 1) The number 1;
- 2) The primes;
- 3) The composite numbers.

There are, of course, infinitely many composite numbers; for example, all numbers of the form 2^n , $n \geq 2$.

THEOREM 2.4. *There are infinitely many primes.*

Proof. We must show that to any finite set of primes there can be adjoined yet another prime.

Let $p_1, \dots, p_v$ be distinct prime numbers. Then

$$a = \prod_{n=1}^v p_n$$

is, to begin with, > 1 and, in addition, is not divisible by any of the prime numbers $p_1, \dots, p_v$, so that by Theorem 17 it is divisible by a prime number different from $p_1, \dots, p_v$. $\square$

Theorem 18 can be expressed as follows: For any $\xi > 0$, let $\pi(\xi)$ represent the number of primes $\leq \xi$. Then as ξ approaches infinity, so does $\pi(\xi)$; i.e., given $\omega > 0$ there exists $\eta = \eta(\omega)$ such that $\pi(\xi) > \omega$ if $\xi > \eta = \eta(\omega)$.

The question as to whether, and with what degree of accuracy, $\pi(\xi)$ can be approximated by the functions of analysis, can be answered only later on. In Part 7, Chapter 2, §3 of my *Vorlesungen über Zahlentheorie*, the reader will find a very accurate result, the methods used being those of complex function theory. This result contains as a special case the “Prime Number Theorem”

$$\lim_{\xi \rightarrow \infty} \frac{\pi(\xi)}{\frac{\xi}{\log \xi}} = 1;$$

this theorem can be found in Part 7, Chapter 1, §2 of the work cited.

Let us also note here that the question, for example, of whether there exist infinitely many primes whose decimal representations end in the digit 7 will be answered (in the affirmative) in Part Two, Chapter III; specifically, the answer will appear as a special case of Dirichlet's well-known Theorem on Arithmetic Progressions (Theorem 155).

None of this will be made use of, however, until it has first been proved.

THEOREM 2.5. *If $p \nmid a$, then $(p, a) = 1$.*

Proof. p has as positive divisors only 1 and p . Hence $(p, a) = 1$ or p and, since $p \nmid a$, the latter is impossible. $\square$

THEOREM 2.6. *If $p \mid \prod_{n=1}^v a_n$, then for at least one n we have $p \mid a_n$.*

Proof. If, for all n , $p \nmid a_n$, then by Theorem 19 we would always have $(p, a_n) = 1$, so that, by Theorem 16, $p \nmid \prod a_n$. $\square$

THEOREM 2.7. *If $p \mid \prod_{n=1}^v p_n$, then for at least one n we have $p = p_n$.*

Proof. By Theorem 20, $p \mid p_n$ for at least one n ; but since the prime p_n has 1 and p_n as its only positive divisors, and since $p \neq 1$, it follows that $p = p_n$. $\square$

THEOREM 2.8. *The representation (2.1) of any number $a > 1$ is unique up to the order of its factors.*

In words: Every prime number appearing in a decomposition into "prime factors" of a given number appears equally often in every such decomposition.

Every $a > 1$ is therefore of the form

$$a = \prod_{p|a} p^{l_p},$$

where l runs through the various primes that divide a ; and where every $l = l_{a,p} > 0$ and is uniquely determined by a and p . (This is the so-called *canonical decomposition* of a .)

Example: $12 = 2 \cdot 2 \cdot 3 = 2 \cdot 3 \cdot 2 = 3 \cdot 2 \cdot 2 = 2^2 \cdot 3 = 3 \cdot 2^2$.

Proof. It is obviously sufficient to prove the following: If

$$a = \prod_{n=1}^v p_n = \prod_{n=1}^{v'} p'_n, \quad p_1 \leq p_2 \leq \cdots \leq p_v, \quad p'_1 \leq p'_2 \leq \cdots \leq p'_{v'},$$

then $v = v'$, $p_n = p'_n$ for $1 \leq n \leq v$.

1) For $a = 2$ the assertion is true, since we merely have $v = v' = 1$, $p_1 = p'_1 = 2$.

2) Let $a > 2$ and suppose that the assertion has been proved for $2, 3, \dots, a-1$.

21) If a is a prime, then $v = v' = 1$, $p_1 = p'_1 = a$.

22) Otherwise, we have $v > 1$ and $v' > 1$. Since $\prod p_n = \prod p'_n$, it follows by Theorem 21 that $p_1 = p'_m$ for at least one m and $p'_1 = p_n$ for at least one n . Since $p_1 \leq p_n = p'_1 \leq p'_m = p_1$, we have $p_1 = p'_1$. Now (since $1 < p_1 < a$, $p_1 \mid a$) we have $1 < \frac{a}{p_1} < a$ and hence (by the induction hypothesis) $v-1 = v'-1$, $v = v'$ and $p_n = p'_n$ for $2 \leq n \leq v$. $\square$

THEOREM 2.9. *Let $a > 1$, let $T(a)$ be the number of positive divisors of a , and let*

$$a = \prod_{n=1}^r p_n^{l_n}$$

be the canonical decomposition of a (i.e., $p_1, \dots, p_r$ are distinct and every $l_n > 0$). Then a has for its positive divisors the numbers

$$\prod_{n=1}^r p_n^{m_n}, \quad 0 \leq m_n \leq l_n \quad \text{for} \quad 1 \leq n \leq r \quad (2.2)$$

and no others. Hence:

$$T(a) = \prod_{n=1}^r (l_n + 1).$$

Proof. 1) Each number of the form (2.2) obviously divides a .

2) If $d > 0$ and $d \mid a$, then $a = qd$, so that d cannot contain any prime factor that does not divide a , nor can d contain any prime factor of a a greater number of times than that factor appears in a itself. $\square$

DEFINITION 2.10. For any real number ξ , let $[\xi]$ denote the largest integer $\leq \xi$, that is, the integer g for which $g \leq \xi < g+1$.

Obviously

$$\xi - 1 < [\xi] \leq \xi,$$

and if $a \leq \xi$ then $a \leq [\xi]$, and if $a > \xi$ then $a \geq [\xi] + 1 > [\xi]$.

THEOREM 2.11. *The number q of Theorem 7 is equal to $\left[\frac{b}{a}\right]$.*

Proof. $qa \leq b = qa + r < (q+1)a$, so $\frac{b}{a} - 1 < q \leq \frac{b}{a}$. □

THEOREM 2.12. *If $k > 0$ and $n > 0$, then the number of positive multiples of k which are $\leq n$ is $\left[\frac{n}{k}\right]$.*

Proof. Since $h > 0$ and $hk \leq n$, it follows that $0 < h \leq \frac{n}{k}$, and conversely; but the number of natural numbers $\leq \xi$ is $[\xi]$ for every $\xi > 0$. □

THEOREM 2.13. *If $k > 0$ and $n = 0$, then $\left[\frac{n}{k}\right] = 0$. (For $n > 0$ this also follows from Theorem 25, for there are just as many positive multiples of k up to n as there are up to $[n]$.)*

Proof. From $q \leq \frac{n}{k} < q+1$ it follows that $kq \leq n < k(q+1)$, $kq \leq [n] < k(q+1)$, $[n] = kq + 1$. □

THEOREM 2.14. *Let $n > 0$, and let p be any prime. Then p divides $n!$ exactly*

$$\sum_{m=1}^{\infty} \left[\frac{n}{p^m} \right]$$

times.

(This infinite series converges, since the general term vanishes for sufficiently large m , and in particular for $m > \frac{\log n}{\log p}$. Since we then have $p^m > n$, $0 < \frac{n}{p^m} < 1$. The series can therefore also be written as $\sum_{m \geq 1} \left[\frac{n}{p^m} \right]$, where, in case $p > n$, the sum stands for zero—as shall every empty sum henceforth.)

In other words: We have

$$n! = \prod_{p \leq n} p^{\sum_m [n/p^m]} \quad (2.3)$$

(where, in case $n = 1$, the product represents the number 1—as shall every empty product henceforth); for the primes $p > n$ do not divide $n!$. We can equally well write

$$n! = \prod_p p^{\sum_m [n/p^m]}$$

where the product is taken over all primes arranged in increasing order of magnitude, for every factor is 1 for $p > n$.

Proof. In preparation for later on, I present two proofs.

1) The number of positive multiples of the number p up to n is, by Theorem 25, $\left[\frac{n}{p}\right]$; the number of positive multiples of p^2 up to n is $\left[\frac{n}{p^2}\right]$; etc. The multiplicity with which p divides $n!$ therefore

$$= \sum_{m=1}^{\infty} \text{number of positive multiples of } p^m \text{ up to } n = \sum_{m=1}^{\infty} \left[\frac{n}{p^m}\right],$$

for each of the numbers $1, \dots, n$ is counted l times (and so not at all if $l = 0$) as a multiple of p^m for $m = 1, 2, \dots, l$, if p divides it exactly l times ($l \geq 0$).

2) (This proof is longer than the former, and introduces the logarithmic function—which can, of course, be eliminated by the use of exponents—but the proof is otherwise useful.) Let us henceforth set

$$\Lambda(a) = \begin{cases} \log p & \text{for } a = p^c, \ c \geq 1, \\ 0 & \text{for all other } a > 0. \end{cases} \quad (2.4)$$

(thus $\Lambda(1) = 0$, $\Lambda(2) = \log 2$, $\Lambda(3) = \log 3$, $\Lambda(4) = \log 2$, $\Lambda(5) = \log 5$, $\Lambda(6) = 0$, $\dots$). Let the symbol

$$\sum_{d|a}$$

mean, on principle, for $a > 0$, that the sum is taken over all positive divisors d of a .

Then we have

$$\log a = \sum_{d|a} \Lambda(d). \quad (2.5)$$

For (2.5) is obvious if $a = 1$ ($0 = 0$), and if $a = \prod_{p|a} p^{l_p}$ ($l_p > 0$) is the canonical decomposition of $a > 1$, then

$$\log a = \sum_{p|a} l_p \log p = \sum_{p|a} \Lambda(p^{l_p}) = \sum_{d|a} \Lambda(d).$$

From (2.5) it now follows that

$$\log([\xi]!) = \sum_{a=1}^{[\xi]} \log a = \sum_{a=1}^{[\xi]} \sum_{d|a} \Lambda(d) = \sum_{d=1}^{[\xi]} \Lambda(d) \left[\frac{\xi}{d}\right]; \quad (2.6)$$

(I am generalizing the result somewhat, in that I replace n by any real $\xi > 0$); for $\Lambda(d)$ appears only for $1 \leq d \leq [\xi]$, and for each such d it appears as many times as there are positive multiples of d up to ξ , that is, $[\frac{\xi}{d}]$ times, by Theorem 25. By the definition (2.4) of Λ we have, by (2.6),

$$\log([\xi]!) = \sum_{p \leq \xi} \log p \left[\frac{\xi}{p} \right] + \sum_{p \leq \xi} \log p \left[\frac{\xi}{p^2} \right] + \cdots = \sum_{p, m} \log p \left[\frac{\xi}{p^m} \right] = \sum_{p \leq \xi} \log p \cdot \sum_m \left[\frac{\xi}{p^m} \right]$$

so that the assertion (2.3) is proved if we set $\xi = n$. $\square$

If we wish to apply Theorem 27, we should note that for every $n > 0$ and every p , the terms of $\sum_{m \geq 1} [\frac{n}{p^m}]$ can be most expeditiously computed one after the other by use of the result $[\frac{[n/p^m]}{p}] = [\frac{n}{p^{m+1}}]$ which follows from Theorem 26.

Example: $n = 1000$, $p = 3$; the calculations should not proceed as follows (every fraction is > 0 and < 1)

$$\frac{1000}{3} = 333 + \frac{1}{3}, \quad \frac{1000}{9} = 111 + \frac{1}{9}, \quad \frac{1000}{27} = 37 + \frac{1}{27}, \quad \frac{1000}{81} = 12 + \frac{28}{81}, \quad \dots$$

but rather

$$\frac{1000}{3} = 333 + \frac{1}{3}, \quad \frac{333}{3} = 111, \quad \frac{111}{3} = 37, \quad \frac{37}{3} = 12 + \frac{1}{3}, \quad \frac{12}{3} = 4, \quad \frac{4}{3} = 1 + \frac{1}{3}$$

in order to compute the terms 333, 111, 37, 12, 4, 1, and the final result of 498.

Chapter 3

THE GREATEST COMMON DIVISOR OF SEVERAL NUMBERS

THEOREM 3.1. *Let $a \geq 1$ and $b \geq 1$. Let their canonical decompositions be written*

$$a = \prod_{p|a} p^{l_p}, \quad b = \prod_{p|b} p^{m_p} \quad (l_p = l_p \geq 0, \quad m_p = m_p \geq 0)$$

(where, in case a or $b = 1$, the empty product shall mean 1). If l and m are allowed to assume the value 0, then a and b may be written in uniform notation as

$$a = \prod_{p|ab} p^{l_p}, \quad b = \prod_{p|ab} p^{m_p}.$$

Then

$$(a, b) = \prod_{p|ab} p^{\min(l_p, m_p)}. \quad (3.1)$$

If $y_1, \dots, y_r$ are real numbers, then $\min(y_1, \dots, y_r)$ represents here—as it shall from now on—the smallest and $\max(y_1, \dots, y_r)$ the largest, of the numbers $y_1, \dots, y_r$.

Examples: $\min(-3, 0, -3) = -3$; $\max(1, 0) = 1$.

Proof. The positive divisors of a are (by Theorem 23) the numbers $\prod_{p|ab} p^{t_p}$, $0 \leq t \leq l$; those of b are the numbers $\prod_{p|ab} p^{u_p}$, $0 \leq u \leq m$;

the common positive divisors are therefore the numbers $\prod_{p|ab} p^{v_p}$, $0 \leq v \leq \min(l, m)$ and the right-hand side of (3.1) is the largest of them. $\square$

Notation: If the numbers $a_1, \dots, a_r$ ($r \geq 2$) are not all 0, then their greatest common divisor (which of course exists) is denoted by $(a_1, \dots, a_r)$ (in agreement with our former notation for $r = 2$).

Examples: $(6, 10, 15) = 1$, $(2, 0, -4) = 2$.

DEFINITION 3.2. If $r \geq 2$ and $(a_1, \dots, a_r) = 1$, then $a_1, \dots, a_r$ are called *relatively prime*. (For $r = 2$ this is our old definition.)

THEOREM 3.3. If $r \geq 2$ and if $a_1, \dots, a_r$ are not all 0, then $(a_1, \dots, a_r)$ is divisible by every common divisor of $a_1, \dots, a_r$.

Proof. If only one of the numbers $a_1, \dots, a_r$ is different from 0, then the assertion is trivial.

Otherwise, without loss of generality, let $a_1, \dots, a_r$ all be > 0 ; for if they are not, then we merely discard those that equal zero and change the sign of those that are negative.

1) For $r = 2$ the assertion is true by Theorem 11.

2) For $r > 2$, I give two proofs.

21) Set $a_n = \prod_p p^{l_{n,p}}$ ($n = 1, \dots, r$). Then (compare the proof of Theorem 28), we obviously have

$$(a_1, \dots, a_r) = \prod_p p^{\min(l_1, \dots, l_r)},$$

and every common divisor is $\leq \prod_p p^{v_p}$, $0 \leq v \leq \min(l_1, \dots, l_r)$, and hence goes into $(a_1, \dots, a_r)$.

22) Let the assertion be already proved for $r-1$. Every common divisor of $a_1, \dots, a_r$ divides $a_1, \dots, a_{r-1}$ and hence $(a_1, \dots, a_{r-1})$; it also divides a_r , and hence $((a_1, \dots, a_{r-1}), a_r)$. This number divides $(a_1, \dots, a_{r-1})$ and a_r , and hence $a_1, a_2, \dots, a_{r-1}, a_r$; it is therefore equal to $(a_1, \dots, a_r)$.

We should make note of the relation

$$(a_1, \dots, a_r) = ((a_1, \dots, a_{r-1}), a_r) \quad \text{for } r > 2, a_1 > 0, \dots, a_r > 0, \quad (3.2)$$

which we found during the second proof. $\square$

THEOREM 3.4. Let $r \geq 2$, $a_1 > 0, \dots, a_r > 0$. Then every common multiple n of $a_1, \dots, a_r$ is divisible by their smallest positive common multiple v (which obviously exists).

Proof. 1) For $r = 2$, we know this by Theorem 9.

2) For $r > 2$, I give two proofs (as for Theorem 29).

21) In the notation of the previous proof, we clearly have

$$v = \prod_p p^{\max(l_1, \dots, l_r)}.$$

Either $n = 0$ (so that it is certainly divisible by v), or n contains every p at least l_1 times, $\dots$, at least l_r times, and therefore at least $\max(l_1, \dots, l_r)$ times.

22) Let the assertion already be proved for $r - 1$. Let w represent the smallest positive common multiple of $a_1, \dots, a_{r-1}$, so that n is divisible by $a_1, \dots, a_{r-1}$ and hence by w ; but it is also divisible by a_r , and hence by the smallest positive common multiple of w and a_r . Since this number is itself a positive common multiple of $a_1, \dots, a_{r-1}, a_r$, it must equal v . $\square$

Chapter 4

NUMBER-THEORETIC FUNCTIONS

DEFINITION 4.1. A function $F(a)$ which is defined for every $a > 0$ is called a *number-theoretic function*.

The value of the function is not required to be a positive integer, nor an integer, a rational number, or even a real number.

Examples: $F(a) = a!$, $F(a) = \sin a$, $F(a) = (a + 2)^{-2}$, $F(a) = T(a)$ (the number of positive divisors of Theorem 23), $F(a) = \Lambda(a)$ (Formula (2.4)), $F(a) = \sum_{d|a} d = S(a)$ (the sum of the positive divisors of a).

THEOREM 4.2. If $a > 1$ and $a = \prod_{p|a} p^{l_p}$ is its canonical decomposition, then

$$S(a) = \prod_{p|a} \frac{p^{l_p+1} - 1}{p - 1}.$$

Proof. If we add the positive divisors $p_1^{m_1} p_2^{m_2} \cdots p_r^{m_r}$ of a enumerated in (2.2), and use the fact that $\sum_{m=0}^l \frac{p^{m+1}-1}{p-1}$, then the result follows. $\square$

DEFINITION 4.3. Any divisor of a other than a itself is called a *proper divisor* of a .

DEFINITION 4.4. a is called *even* if $2 \mid a$; *odd*, if $2 \nmid a$.

Examples: 0 is even; of two successive numbers a and $a + 1$, exactly one is always even, the other odd; every $p > 2$ is odd.

DEFINITION 4.5. $a > 0$ is called a *perfect number* if a equals the sum of its proper divisors, that is, if $S(a) = 2a$.

Examples: $6 = 1 + 2 + 3$, $28 = 1 + 2 + 4 + 7 + 14$.

This old-fashioned concept of perfect number, and the questions associated with it, are not especially important; we consider them only because, in so doing, we will encounter two questions that remain unanswered to this day: Are there infinitely many perfect numbers? Is there an odd perfect number? Modern mathematics has solved many (apparently) difficult problems, even in number theory; but we stand powerless in the face of such (apparently) simple problems as these. Of course, the fact that they have never been solved is irrelevant to the rest of this work. We will leave no gaps; when we come to a bypath which leads to an insurmountable barrier, we will turn around, rather than—as is so often done—continue on beyond the barrier.

THEOREM 4.6. *If $p = 2^n - 1$ is a prime (so that $n > 1$; for example, $n = 2$, $p = 3$; $n = 3$, $p = 7$), then*

$$a = 2^{n-1}(2^n - 1)$$

is an (of necessity even) perfect number, and there are no other even perfect numbers.

Proof. 1) For $a = 2^{n-1}(2^n - 1)$, $2^n - 1 = p$ we have, by Theorem 31,

$$S(a) = \frac{2^n - 1}{2 - 1} \cdot \frac{p^2 - 1}{p - 1} = (2^n - 1)(p + 1) = (2^n - 1) \cdot 2^n = 2a.$$

2) If a is an even perfect number, then

$$a = 2^{n-1}u, \quad n > 1, \quad u > 0 \text{ and odd},$$

so that, by Theorem 31,

$$2a = S(a) = \frac{2^n - 1}{2 - 1} S(u) = (2^n - 1)S(u)$$

and

$$\frac{u}{2^n - 1} = \frac{S(u)}{2a} = \frac{S(u)}{u}.$$

In this formula, $\frac{u}{2^n - 1}$ ($= S(u) - u$) is an integer, and hence (since $n > 1$) it is a proper divisor of u . The sum $S(u)$ of all the divisors of u is therefore equal to the sum of u and a certain proper divisor. Hence u is a prime, and the proper divisor $\frac{u}{2^n - 1} = 1$, so that $u = 2^n - 1$. This proves the theorem. $\square$

Now, are there infinitely many perfect numbers? I do not know. It was already mentioned above that $2^n - 1$ is prime for $n = 2$ and $n = 3$. For $n = 4$, $2^n - 1 = 15$ is composite. More generally, $2^n - 1$ is always composite if n is composite; for if $n = bc$, with $b > 1$ and $c > 1$, then

$$2^n - 1 = 2^{bc} - 1 = (2^b - 1)(2^{b(c-1)} + 2^{b(c-2)} + \cdots + 2^b + 1),$$

where both factors are > 1 .

For $n = 5$, $2^n - 1 = 2^5 - 1 = 31$ is a prime that yields the perfect number $16 \cdot 31 = 496$; for $n = 7$, $2^n - 1 = 2^7 - 1 = 127$ is a prime that yields the perfect number $64 \cdot 127 = 8128$; for $n = 11$, $2^n - 1 = 2^{11} - 1 = 2047 = 23 \cdot 89$ is composite. The question is, therefore, whether there are infinitely many primes p for which $2^p - 1$ is a prime. Even this is not known.

Are there infinitely many odd perfect numbers? I do not even know whether there is a single one.

However, I should like to ask the reader not to meditate too long over these two questions; he will meet with many more promising and gratifying problems in his study of this work.

The analogous problem of finding all the numbers $a > 1$ which are equal to the product of their factors, i.e., for which

$$\prod_{d|a} d = a^2, \quad a > 1 \tag{4.1}$$

is trivial. For the following simple theorem holds:

THEOREM 4.7. (4.1) holds if and only if

$$a = p^3 \quad \text{or} \quad a = p_1 p_2, \quad p_1 \neq p_2.$$

Proof. 1) As d runs through all the positive divisors of a , so, obviously, does $\frac{a}{d}$. It therefore follows from (4.1) that

$$a^2 = \prod_{d|a} d \cdot \prod_{d|a} \frac{a}{d} = \prod_{d|a} \left(d \cdot \frac{a}{d} \right) = \prod_{d|a} a = a^{T(a)},$$

so that $T(a) = 4$; and hence, by Theorem 23, $(l_1 + 1) \cdots (l_r + 1) = 4$ in the canonical decomposition $p_1^{l_1} \cdots p_r^{l_r}$, so that either $r = 1$ and $l_1 = 3$, or $r = 2$ and $l_1 = l_2 = 1$.

2) Conversely, in these cases,

$$\prod_{d|a} d = 1 \cdot p \cdot p^2 \cdot p^3 = p^6 = (p^3)^2 \quad \text{and} \quad a \cdot p_1 \cdot p_2 \cdot p_1 p_2 = (p_1 p_2)^2,$$

respectively. □

DEFINITION 4.8. The number-theoretic function $\mu(a)$ (the *Möbius Function*) is defined by

$$\mu(a) = \begin{cases} 1 & \text{if } a = 1, \\ (-1)^r & \text{if } a \text{ is the product of } r \ (\geq 1) \text{ distinct primes,} \\ 0 & \text{otherwise, i.e., if the square of at least one prime divides } a. \end{cases}$$

The numbers $a = 1$ that are not divisible by the square of any prime (or, equivalently, by any perfect square > 1) are also called *square-free numbers*; this quite customary terminology is just as logical as saying that two numbers are prime to each other when they have exactly one positive common divisor (namely, 1). In this sense of square-free, we say: $\mu(a) = \pm 1$ if a is square-free, and $\mu(a) = 0$ otherwise.

Examples: $\mu(1) = 1$, $\mu(2) = -1$, $\mu(3) = -1$ ($\mu(p)$ is always $= -1$), $\mu(4) = 0$, $\mu(5) = -1$, $\mu(6) = 1$, $\mu(7) = -1$, $\mu(8) = 0$, $\mu(9) = 0$, $\mu(10) = 1$.

THEOREM 4.9. If $a > 0$, $b > 0$, and $(a, b) = 1$, then $\mu(ab) = \mu(a)\mu(b)$.

Proof. 1) If a or b is not square-free, then neither is ab , so that $\mu(ab) = 0 = \mu(a)\mu(b)$.

2) If a and b are square-free, then since $(a, b) = 1$, ab is also square-free. If $a = 1$ or $b = 1$, then the statement is obviously true; otherwise the number of prime factors of ab equals the sum of the number of prime factors of a and of b . □

THEOREM 4.10.

$$\sum_{d|a} \mu(d) = \begin{cases} 1 & \text{for } a = 1, \\ 0 & \text{for } a > 1. \end{cases}$$

Proof. 1) $\sum_{d|1} \mu(d) = \mu(1) = 1$.

2) If $a > 1$ and if $a = p_1 \cdots p_r$ is the canonical decomposition of a , then obviously

$$\sum_{d|a} \mu(d) = \sum_{s=0}^r \binom{r}{s} (-1)^s = (1-1)^r = 0,$$

for if $s = 1, 2, \dots, r$, then there are exactly $\binom{r}{s}$ divisors of $p_1 \cdots p_r$ which consist of exactly s prime factors, and for these we have $\mu(d) = (-1)^s$. $\square$

THEOREM 4.11. *If $\xi \geq 1$, then*

$$\sum_{n=1}^{[\xi]} \mu(n) \left[\frac{\xi}{n} \right] = 1.$$

Proof. Let the formula of Theorem 35 be summed over $a = 1, 2, \dots, [\xi]$. This gives

$$1 = \sum_{a=1}^{[\xi]} \sum_{d|a} \mu(d) = \sum_{d=1}^{[\xi]} \mu(d) \left[\frac{\xi}{d} \right],$$

for, by Theorem 25, the number of positive multiples of d up to ξ is $\left[\frac{\xi}{d} \right]$. $\square$

THEOREM 4.12. *If $x \geq 1$, then*

$$\left| \sum_{n=1}^x \frac{\mu(n)}{n} \right| \leq 1.$$

The infinite series $\sum_{n=1}^{\infty} \frac{\mu(n)}{n}$ therefore either converges, or else it oscillates between finite limits. The question as to which of these two alternatives holds does not interest us at the moment; the reader can learn the answer in Part Seven, Chapter 12, §1 of my *Vorlesungen über Zahlentheorie*.

Gordan used to say something to the effect that “Number Theory is useful since one can, after all, use it to get a doctorate with.” In 1899 I received my doctorate by answering this question.

Proof. We have

$$0 \leq \left\{ \frac{x}{n} \right\} < 1 \text{ for } 1 \leq n < x, \quad \left\{ \frac{x}{n} \right\} = 0 \text{ for } n = x.$$

Hence, by Theorem 36,

$$\sum_{n=1}^x \frac{\mu(n)}{n} = \sum_{n=1}^x \frac{\mu(n)}{n} \left\{ \frac{x}{n} \right\} + \frac{1}{x} \sum_{n=1}^x \mu(n) \left[\frac{x}{n} \right] = \sum_{n=1}^x \frac{\mu(n)}{n} \left\{ \frac{x}{n} \right\} + \frac{1}{x},$$

so that

$$\left| \sum_{n=1}^x \frac{\mu(n)}{n} \right| \leq \sum_{n=1}^x \frac{|\mu(n)|}{n} \cdot 1 + \frac{1}{x} \leq \sum_{n=1}^{[x]} \frac{1}{n} \cdot 1 + \frac{1}{x} < 1.$$

□

THEOREM 4.13. *Let $F(a)$ be any number-theoretic function whatever. Let $G(a)$ be the number-theoretic function*

$$G(a) = \sum_{d|a} F(d).$$

Then

$$F(a) = \sum_{d|a} \mu\left(\frac{a}{d}\right) G(d) = \sum_{d|a} \mu(d) G\left(\frac{a}{d}\right). \quad (4.2)$$

(This is the so-called *Möbius Inversion*.)

Remark: The fact that $F(a)$ is uniquely determined at all, in reverse, by $G(a)$ is clear to start with; for from $G(1) = F(1)$, $G(2) = F(2) + F(1)$, $G(3) = F(3) + F(1)$, ... we can successively compute $F(1)$, $F(2)$, $F(3)$, ...

Proof. For every positive $d \mid a$ we have $\frac{a}{d} \mid a$, so that

$$G\left(\frac{a}{d}\right) = \sum_{b|\frac{a}{d}} F(b).$$

Hence

$$\sum_{d|a} \mu(d) G\left(\frac{a}{d}\right) = \sum_{d|a} \mu(d) \sum_{b|\frac{a}{d}} F(b) = \sum_{b|a} F(b) \sum_{d|\frac{a}{b}} \mu(d) = F(a),$$

since, by Theorem 35, $\sum_{d|\frac{a}{b}} \mu(d) = 1$ for $b = a$ and $= 0$ for $b \mid a$, $b < a$. □

DEFINITION 4.14. The number-theoretic function $\varphi(a)$ (*Euler's Function*) represents the number of numbers n in the sequence $1, 2, \dots, a$ for which $(n, a) = 1$.

Examples: $\varphi(1) = 1$ ($n = 1$), $\varphi(2) = 1$ ($n = 1$), $\varphi(3) = 2$ ($n = 1, 2$), $\varphi(4) = 2$ ($n = 1, 3$), $\varphi(5) = 4$ ($n = 1, 2, 3, 4$), $\varphi(6) = 2$ ($n = 1, 5$), $\varphi(p) = p - 1$ ($n = 1, 2, \dots, p - 1$).

THEOREM 4.15.

$$\sum_{d|a} \varphi(d) = a.$$

Proof. Divide all the a numbers $n = 1, \dots, a$ into classes according to the value of $d = (n, a)$. Only those numbers $d > 0$ that divide a enter into consideration. To each $d | a$ let there belong the $n = kd$ for which $(kd, a) = d$, i.e. (by Theorems 13 and 14), $(k, \frac{a}{d}) = 1$ and moreover for which $0 < kd \leq a$, i.e., $0 < k \leq \frac{a}{d}$. But by Definition 12 there are exactly $\varphi(\frac{a}{d})$ such numbers.

Hence

$$a = \sum_{d|a} \varphi\left(\frac{a}{d}\right) = \sum_{d|a} \varphi(d),$$

since d runs through all the positive divisors of a when $\frac{a}{d}$ does. $\square$

THEOREM 4.16.

$$\varphi(a) = a \sum_{d|a} \frac{\mu(d)}{d}.$$

Proof. By Theorems 39 and 38 (with $F(a) = \varphi(a)$ and $G(a) = a$), we have

$$\varphi(a) = \sum_{d|a} \mu(d) \frac{a}{d} = a \sum_{d|a} \frac{\mu(d)}{d}.$$

$\square$

THEOREM 4.17.

$$\varphi(a) = a \prod_{p|a} \left(1 - \frac{1}{p}\right).$$

Proof. 1) For $a = 1$ we have $\varphi(1) = 1$ (the product in the statement of the theorem is empty).

2) For $a > 1$ let $a = \prod_{n=1}^r p_n^{l_n}$ be its canonical decomposition. Then by Theorem 40 we have

$$\varphi(a) = a \sum_{d|a} \frac{\mu(d)}{d} = a \prod_{p|a} \left(1 - \frac{1}{p}\right),$$

as is seen by calculating the 2^r terms of the product. $\square$

THEOREM 4.18. *For $a > 1$ we have, in the canonical notation,*

$$\varphi(a) = \prod_{n=1}^r p_n^{l_n-1} (p_n - 1).$$

Proof. By Theorem 41,

$$\varphi(a) = a \prod_{n=1}^r \left(1 - \frac{1}{p_n}\right) = \prod_{n=1}^r p_n^{l_n} \left(1 - \frac{1}{p_n}\right) = \prod_{n=1}^r p_n^{l_n-1} (p_n - 1).$$

□

THEOREM 4.19. *For $l > 0$ we have $\varphi(p^l) = p^l - p^{l-1}$.*

Two proofs: 1) Special case of Theorem 42.

2) (Direct proof.) Of the numbers $1, 2, \dots, p^l$, those not relatively prime to p^l are precisely all the multiples of p ; their number is p^{l-1} ; hence $\varphi(p^l) = p^l - p^{l-1}$.

All of Theorem 42 itself can be proven directly by counting the numbers n which are not relatively prime to a and for which $1 \leq n \leq a$; but this is somewhat more laborious and is a good exercise for the reader. (The solution of this exercise is, however, not essential for the remainder of this book.)

THEOREM 4.20. *If $a > 0$, $b > 0$, and $(a, b) = 1$, then $\varphi(ab) = \varphi(a)\varphi(b)$.*

Proof. Without loss of generality let (canonically) $a = \prod_{n=1}^r p_n^{l_n} > 1$ and $b = \prod_{n=1}^s q_n^{m_n} > 1$. From Theorem 42 it follows that

$$\varphi(a) = \prod_{n=1}^r p_n^{l_n-1} (p_n - 1), \quad \varphi(b) = \prod_{n=1}^s q_n^{m_n-1} (q_n - 1).$$

Since $(a, b) = 1$,

$$ab = \prod_{n=1}^r p_n^{l_n} \cdot \prod_{n=1}^s q_n^{m_n}$$

is the canonical decomposition of ab ; hence, by Theorem 42,

$$\varphi(ab) = \prod_{n=1}^r p_n^{l_n-1} (p_n - 1) \cdot \prod_{n=1}^s q_n^{m_n-1} (q_n - 1) = \varphi(a)\varphi(b).$$

□

The reader will find another proof of Theorem 44, one based directly on the definition of φ , in Theorem 74.

Chapter 5

CONGRUENCES

In this chapter m will always be > 0 .

DEFINITION 5.1. a is said to be *congruent* to b modulo m , written

$$a \equiv b \pmod{m},$$

if $m \mid (a - b)$.

a is called *incongruent* to b modulo m , written

$$a \not\equiv b \pmod{m},$$

if $m \nmid (a - b)$.

Examples: $31 \equiv -9 \pmod{10}$, $627 \equiv 587 \pmod{10}$, $5 \equiv 4 \pmod{2}$, $a \equiv b \pmod{1}$ for arbitrary a and b .

Any concept such as “congruent,” “equivalent,” “equal,” or “similar,” in mathematics must satisfy three properties (the so-called reflexivity, symmetry, and transitivity), which are expressed here by means of the following three theorems.

THEOREM 5.2 (Reflexivity). *We always have $a \equiv a \pmod{m}$.*

Proof. $m \mid 0$, $m \mid (a - a)$. □

THEOREM 5.3 (Symmetry). *If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$.*

Proof. $m \mid (a - b)$, and hence, by Theorem 1, $m \mid (b - a)$. □

THEOREM 5.4 (Transitivity). *If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$.*

Proof. $m \mid a - b$, $m \mid b - c$, $m \mid (a - b) + (b - c)$, $m \mid a - c$. $\square$

Thus, just like equations, congruences (with the same modulus) can be written in sequence as a congruence with more than two terms; for example, $a \equiv b \equiv c \pmod{m}$.

The following theorem (which, incidentally, makes Theorems 45–47 self-evident) provides a useful necessary and sufficient condition for the validity of a congruence.

THEOREM 5.5. *According to Theorem 7, given the numbers c and m , there is a uniquely determined number r such that $c = qm + r$, $0 \leq r < m$; let this number r be called the residue of c modulo m . Then $a \equiv b \pmod{m}$ holds if and only if a and b have the same residue modulo m .*

Proof. 1) If $a = q_1m + r$, $b = q_2m + r$, then $a - b = (q_1 - q_2)m$, $m \mid a - b$.

2) Let $a = q_1m + r$, $0 \leq r < m$, $a \equiv b \pmod{m}$; then $b = a + qm = (q_1 + q)m + r = q_1m + r$, $q_2 = q_1 + q$. $\square$

Theorem 48 shows that, given a number m , all the numbers fall into m classes (“residue classes”) in such a way that any two numbers in the same class are congruent, and any two numbers in different classes are incongruent. One of the classes consists of the multiples of m .

Theorems 49–56 which follow are analogous to the corresponding theorems on equalities; they make clear the usefulness of the congruence sign; looking at it intrinsically, one might have objected that no new symbol is needed for $m \mid (a - b)$. Since the modulus m in Theorems 49–56 remains the same throughout, we shall not bother to write it for the time being.

THEOREM 5.6. *If $a \equiv b$, $c \equiv d$, then $a + c \equiv b + d$, $a - c \equiv b - d$.*

Proof. $m \mid a - b$, $m \mid c - d$, $m \mid (a - b) + (c - d)$, $m \mid (a + c) - (b + d)$. $\square$

THEOREM 5.7. *If $a_n \equiv b_n$ for $n = 1, \dots, \nu$, then $\sum_{n=1}^{\nu} a_n \equiv \sum_{n=1}^{\nu} b_n$.*

Proof. Follows by induction from Theorem 49. $\square$

THEOREM 5.8. *If $a \equiv b$, then, for every c , $ac \equiv bc$.*

Proof. $m \mid (a - b)$, $m \mid (a - b)c$, $m \mid (ac - bc)$. $\square$

THEOREM 5.9. *If $a \equiv b$ and $c \equiv d$, then $ac \equiv bd$.*

Proof. By Theorem 51 it follows from the first part of the hypothesis that $ac \equiv bc$ and from the second that $bc \equiv bd$; hence, by Theorem 47, the conclusion follows. $\square$

THEOREM 5.10. *If $a_n \equiv b_n$ for $n = 1, \dots, \nu$, then $\prod_{n=1}^{\nu} a_n \equiv \prod_{n=1}^{\nu} b_n$.*

Proof. By induction, using Theorem 52. $\square$

THEOREM 5.11. *If $a \equiv b$, then $a^n \equiv b^n$ for every $n \geq 0$.*

Proof. Follows from Theorem 53. $\square$

THEOREM 5.12. *Let $f(x) = c_0 + c_1x + \dots + c_nx^n$ ($n \geq 0$) be any rational integral function with integer coefficients. If $a \equiv b$, then $f(a) \equiv f(b)$.*

The solutions (if any exist) of the congruence $f(x) \equiv 0$ thus fall into complete residue classes mod m .

Proof. By Theorem 54 it follows from the hypothesis that $a^\nu \equiv b^\nu$ for $0 \leq \nu \leq n$, so that, by Theorem 51, $c_\nu a^\nu \equiv c_\nu b^\nu$ for $0 \leq \nu \leq n$; since $c_\nu \equiv c_\nu$, our result $\sum_{\nu=0}^n c_\nu a^\nu \equiv \sum_{\nu=0}^n c_\nu b^\nu$ follows by Theorem 50. $\square$

Theorem 55 justifies:

DEFINITION 5.13. By the *number of solutions*, or *roots*, of a congruence $f(x) \equiv 0 \pmod{m}$ we shall mean the number of those numbers of the set $x = 0, \dots, m-1$ that satisfy the congruence, that is, the number of residue classes all of whose members satisfy the congruence.

Thus the number of solutions is always either 0 or some other finite number.

Example: $x^2 \equiv 1 \pmod{8}$ has four solutions, since $x = 1, 3, 5, 7$ (but not $x = 0, 2, 4, 6$) satisfy the congruence. This fact, that $8 \mid (x^2 - 1)$ for every odd number x , should be kept in mind.

THEOREM 5.14. *If $ac \equiv bc$ and $(c, m) = 1$, then $a \equiv b$.*

Proof. $m \mid (ac - bc)$, $m \mid (a - b)c$; since $(m, c) = 1$, it follows from Theorem 15 that $m \mid (a - b)$. $\square$

THEOREM 5.15. *If $ac \equiv bc \pmod{m}$, then $a \equiv b \pmod{\frac{m}{(c,m)}}$.*

(If $(c, m) = 1$, this reduces to Theorem 56.)

Proof. $m \mid (a - b)c$, hence, by Theorem 3, $\frac{m}{(c,m)} \mid (a - b)\frac{c}{(c,m)}$. By Theorem 13 it follows that $\left(\frac{m}{(c,m)}, \frac{c}{(c,m)}\right) = 1$, so that by Theorem 15, $\frac{m}{(c,m)} \mid (a - b)$. $\square$

THEOREM 5.16. *Let $c > 0$. If $a \equiv b \pmod{m}$, then $ac \equiv bc \pmod{cm}$, and conversely.*

Proof. Since $c > 0$, it follows from Theorem 3 that the relations $m \mid (a - b)$ and $cm \mid c(a - b)$ are equivalent. $\square$

THEOREM 5.17. *If $a \equiv b \pmod{m}$, $n > 0$, $n \mid m$, then $a \equiv b \pmod{n}$.*

Proof. $m \mid (a - b)$ and $n \mid m$; hence $n \mid (a - b)$. $\square$

THEOREM 5.18. *If $a \equiv b \pmod{m_n}$ for $n = 1, 2, \dots, \nu$ ($\nu \geq 2$), then, if m is the smallest common positive multiple of $m_1, \dots, m_\nu$, we have $a \equiv b \pmod{m}$.*

Proof. $a - b$ is divisible by $m_1, \dots, m_\nu$, and hence, according to Theorem 30, by m . $\square$

THEOREM 5.19. *If $a \equiv b \pmod{m}$, then $(a, m) = (b, m)$.*

In particular: If $(a, m) = 1$, then $(b, m) = 1$. Consequently the numbers in a residue class are either all relatively prime to m , or none of them is.

Proof. From $b = a + mq$ it follows that $(a, m) \mid b$, so that $(a, m) \mid (b, m)$; similarly, $(b, m) \mid (a, m)$. $\square$

DEFINITION 5.20. By a *complete set of residues mod m* is meant a set of m numbers each of which is congruent to exactly one of the numbers $0, 1, \dots, m - 1 \pmod{m}$, that is, which represents the m classes into which all the integers mod m fall.

It suffices, of course, to require that at least one of the m numbers belong to each class. "If m objects are put into m pigeon-holes and each pigeon-hole contains at least one object, then each pigeon-hole contains exactly one object."

Alternatively: It suffices to require that each pair of m numbers be incongruent. “If m objects are put into m pigeon-holes and each pigeon-hole contains at most one object, then each pigeon-hole contains exactly one object.”

Examples: Any m consecutive numbers, for example $1, \dots, m$, or the integers of the interval $-\frac{m}{2}$ (exclusive) to $\frac{m}{2}$ (inclusive) constitute a complete set of residues, since they are incongruent to each other.

Our old Definition 14 can now be expressed as follows: The number of solutions of $f(x) \equiv 0 \pmod{m}$ is the number of its solutions taken from any complete set of residues.

DEFINITION 5.21. By a *reduced set of residues mod m* is meant a set of $\varphi(m)$ numbers exactly one of which belongs to each of the classes all of whose numbers are relatively prime to m .

Once again it suffices, given $\varphi(m)$ numbers, to require either that at least one belong to each of the above-mentioned $\varphi(m)$ classes or that each of the $\varphi(m)$ numbers be relatively prime to m and that each pair of them be incongruent.

THEOREM 5.22. *If $(k, m) = 1$, then the numbers $0 \cdot k, 1 \cdot k, 2 \cdot k, \dots, (m-1) \cdot k$ constitute a complete set of residues mod m .*

More generally: If $(k, m) = 1$ and $a_1, \dots, a_m$ is any complete set of residues, then so is $a_1k, \dots, a_mk$.

Proof. From $a_rk \equiv a_sk \pmod{m}$, $1 \leq r \leq m$, $1 \leq s \leq m$ it follows by Theorem 56, since by assumption $(k, m) = 1$, that $a_r \equiv a_s \pmod{m}$ and $r = s$; the terms a_rk are therefore mutually incongruent. $\square$

THEOREM 5.23. *If $(k, m) = 1$ and if $a_1, \dots, a_{\varphi(m)}$ constitute a reduced set of residues mod m , then so do $a_1k, \dots, a_{\varphi(m)}k$.*

Proof. Each of these $\varphi(m)$ numbers is relatively prime to m (for any common factor of a_rk and m would have to go into a_r and m); hence any two are incongruent, by Theorem 62. $\square$

THEOREM 5.24. *If $(a, m) = 1$, then the congruence $ax + a_0 \equiv 0 \pmod{m}$ has exactly one solution.*

Proof. By Theorem 62, $a \cdot 0, a \cdot 1, \dots, a(m-1)$ constitute a complete set of residues; hence exactly one of these numbers is $\equiv -a_0 \pmod{m}$. $\square$

THEOREM 5.25. 1) *The congruence*

$$ax + a_0 \equiv 0 \pmod{m} \quad (5.1)$$

is solvable if and only if $(a, m) \mid a_0$.

2) *In that case the number of solutions $= (a, m)$, and the congruence is satisfied by precisely all of the numbers x in a certain residue class mod $\frac{m}{(a, m)}$.*

Remark: Theorem 64 is obviously a special case of this theorem, but is made use of to prove it.

Proof. 1) If (5.1) is solvable, then $ax + a_0 \equiv 0 \pmod{(a, m)}$, so $(a, m) \mid a_0$.

12) If $(a, m) \mid a_0$, then, by Theorem 64, the congruence

$$\frac{a}{(a, m)}x + \frac{a_0}{(a, m)} \equiv 0 \pmod{\frac{m}{(a, m)}} \quad (5.2)$$

is solvable. Hence, by Theorem 58, (5.1) is satisfied.

2) If $(a, m) \mid a_0$, then (5.2) has exactly one solution mod $\frac{m}{(a, m)}$, according to Theorem 64; since (5.1) and (5.2) have the same solutions, by Theorem 58, it follows that (5.1) has (a, m) solutions (solutions mod m , as usual), since if $d > 0$ and $d \mid m$, then a residue class mod $\frac{m}{d}$ breaks up into d residue classes mod m . $\square$

THEOREM 5.26. *Let $n > 1$ and let at least one of the numbers $a_1, \dots, a_n$ be different from 0; set $(a_1, \dots, a_n) = d$. We claim that the diophantine equation (i.e., equation with integral coefficients and unknowns)*

$$a_1x_1 + \dots + a_nx_n = c$$

is solvable if and only if $d \mid c$.

Hence, in particular: If $(a, b) = 1$, then

$$ax + by = 1 \quad (5.3)$$

is solvable.

Proof. 1) If exactly one coefficient does not vanish, say a_1 , then $a_1x_1 + 0 \cdot x_2 + \dots + 0 \cdot x_n = c$ is obviously solvable if $a_1 \mid c$, that is, if $(a_1, 0, \dots, 0) \mid c$.

2) If at least two coefficients do not vanish, then we may assume without loss of generality that no coefficient vanishes; for otherwise

we simply omit those terms a_mx_m for which $a_m = 0$, and this does not alter the value of the greatest common divisor of the coefficients; the number of terms that remain is then still ≥ 2 .

Without loss of generality we may even take all the coefficients to be > 0 ; for we merely have to replace each negative a_m by $-a_m$ (which does not alter the greatest common divisor) and the corresponding x_m by $-x_m$.

We may therefore assume that $n > 1$, $a_1 > 0$, $\dots$, $a_n > 0$.

21) If our diophantine equation is solvable, then obviously $d \mid a_1x_1 + \dots + a_nx_n$, so $d \mid c$.

22) Let $d \mid c$.

221) If $n = 2$, then we merely have to show that $a_1x_1 \equiv c \pmod{a_2}$ is solvable for x_1 . This follows from Theorem 65, since $(a_1, a_2) \mid c$.

222) Let $n > 2$, and assume the assertion proved for $2, \dots, n-1$; if we set $(a_1, \dots, a_{n-1}) = \delta$, then, by (3.2), $(\delta, a_n) = d$.

From what we showed in 221), it follows that $\delta\xi + a_nx_n = c$ for suitably chosen ξ, x_n . By our induction hypothesis for $n-1$, it follows in addition, since $\delta = (a_1, \dots, a_{n-1}) \mid \delta\xi$, that $a_1x_1 + \dots + a_{n-1}x_{n-1} = \delta\xi$ for suitably chosen $x_1, \dots, x_{n-1}$, so that, finally, $a_1x_1 + \dots + a_{n-1}x_{n-1} + a_nx_n = c$. $\square$

THEOREM 5.27. *If $(a, b) = d$ and $d \mid c$, then $ax + by = c$ is solvable, by Theorem 66; then, given any solution x_0, y_0 , all the solutions are of the form*

$$x = x_0 + h\frac{b}{d}, \quad y = y_0 - h\frac{a}{d},$$

where h is arbitrary.

Proof. 1) The fact that such a pair x, y satisfies the equation follows from the relation $a(x_0 + h\frac{b}{d}) + b(y_0 - h\frac{a}{d}) = ax_0 + by_0 = c$.

2) The fact that no other solutions exist is seen as follows. Without loss of generality, let $b \neq 0$. (Otherwise interchange a and b , and observe that as h runs through all the integers, so does $-h$.) Since $ax + by = c = ax_0 + by_0$, it follows that $ax - c \equiv 0 \pmod{|b|}$, $ax_0 - c \equiv 0 \pmod{|b|}$, and hence by Theorem 65 (with $a_0 = -c$, $m = |b|$), we have $x \equiv x_0 \pmod{\frac{|b|}{d}}$, $x = x_0 + h\frac{b}{d}$, so that $by = c - ax = c - a(x_0 + h\frac{b}{d}) = (c - ax_0) - h\frac{ab}{d} = by_0 - h\frac{ab}{d}$, $y = y_0 - h\frac{a}{d}$. $\square$

THEOREM 5.28. *If $(a, b) = 1$ and if x_0, y_0 is any solution of (5.3), then all the solutions are of the form $x = x_0 + hb, y = y_0 - ha$, where h is arbitrary.*

Proof. This follows from Theorem 67, with $d = c = 1$. □

THEOREM 5.29. 1) *The congruences*

$$x \equiv a_1 \pmod{m_1}, \quad (5.4)$$

$$x \equiv a_2 \pmod{m_2} \quad (5.5)$$

have a common solution if and only if

$$(m_1, m_2) \mid a_1 - a_2. \quad (5.6)$$

In particular, therefore, they always do if $(m_1, m_2) = 1$.

2) *If condition (5.6) is satisfied and if m represents the smallest common multiple of m_1 and m_2 , then the common solutions of (5.4) and (5.5) consist of all the numbers in a certain residue class mod m .*

Proof. 11) If we set $(m_1, m_2) = d$, then it follows from (5.4) and (5.5) that $x \equiv a_1 \pmod{d}$, $x \equiv a_2 \pmod{d}$, so $a_1 \equiv a_2 \pmod{d}$, i.e., $d \mid a_1 - a_2$.

12) If $d \mid a_1 - a_2$, then from among all solutions of (5.4) of the form $x = a_1 + ym_1$ (y arbitrary) we can certainly choose one for which (5.5) holds. For we need $a_1 + ym_1 \equiv a_2 \pmod{m_2}$; this is equivalent to

$$m_1 y + (a_1 - a_2) \equiv 0 \pmod{m_2} \quad (5.7)$$

which, by Theorem 65, 1), is solvable.

2) If (5.6) is satisfied, and therefore (5.4) and (5.5) along with it, then congruence (5.7) is satisfied for suitably chosen y_0 precisely by

$$y \equiv y_0 \pmod{\frac{m_2}{d}}$$

by virtue of Theorem 65, 2). Therefore, since $m_1 \cdot \frac{m_2}{d} = m$ (by Theorem 14), all the numbers x satisfying (5.4) and (5.5) are given by the formulas $x = a_1 + (y_0 + h \frac{m_2}{d})m_1 = a_1 + m_1 y_0 + h \cdot \frac{m_1 m_2}{d} = a_1 + m_1 y_0 + hm$, h arbitrary, but these constitute a certain residue class mod m . □

THEOREM 5.30. *Let $r > 1$, and let every pair from among the numbers $m_1, \dots, m_r$ be relatively prime. Then the congruences*

$$x \equiv a_n \pmod{m_n}, \quad n = 1, \dots, r \quad (5.8)$$

are consistent, and their common solutions consist of all the numbers in a certain residue class mod $m_1 m_2 \cdots m_r$.

Proof. 1) For $r = 2$ this follows from Theorem 69, since $m = m_1 m_2$ in that case.

2) Let $r > 2$, and assume the theorem proved for $r - 1$. Then the first $r - 1$ congruences (5.8) are covered by $x \equiv a \pmod{m_1 \cdots m_{r-1}}$ for a suitably chosen a . Hence, by Theorem 69, the conclusion follows, since $m_1 \cdots m_{r-1}$ is relatively prime to m_r . $\square$

THEOREM 5.31. *Let $r > 1$, and let each pair of numbers from among $m_1, \dots, m_r$ be relatively prime. Then the number of solutions of*

$$f(x) \equiv 0 \pmod{m_1 \cdots m_r} \quad (5.9)$$

equals the product of the numbers of solutions of

$$f(x) \equiv 0 \pmod{m_1}, \quad \dots, \quad f(x) \equiv 0 \pmod{m_r}. \quad (5.10)$$

In particular: If $m > 1$ and $m = \prod_{n=1}^r p_n^{l_n}$ is its canonical decomposition then, if $r > 1$, the number of solutions of $f(x) \equiv 0 \pmod{m}$ equals the product of the numbers of solutions of $f(x) \equiv 0 \pmod{p_n^{l_n}}$.

Proof. First of all, it is clear that (5.9) is satisfied if and only if the r congruences in (5.10) are simultaneously satisfied. Hence if one of these has no solution, then neither does (5.9). If the congruences in (5.10) are all solvable then, by Theorem 70, to each residue class mod m_1 that satisfies the first congruence in (5.10) there corresponds one-to-one a residue class mod $m_1 \cdots m_r$ that satisfies (5.9); similarly for $m_2, \dots, m_r$. $\square$

THEOREM 5.32. *If*

$$f(x) = c_0 + c_1 x + \cdots + c_n x^n, \quad p \nmid c_n,$$

then the congruence

$$f(x) \equiv 0 \pmod{p} \quad (5.11)$$

has at most n solutions.

Proof. 1) For $n = 0$ this is obvious, since for every x , $c_0 \not\equiv 0 \pmod{p}$, so that (5.11) has no root.

2) Let $n > 0$, and assume the theorem true for $n - 1$. If (5.11) had at least the $n + 1$ (incongruent) roots $x_0, x_1, \dots, x_n$, then if we note that

$$f(x) - f(x_0) = \sum_{\nu=0}^n c_\nu x^\nu - \sum_{\nu=0}^n c_\nu x_0^\nu = (x - x_0)g(x),$$

where

$$g(x) = b_1 + b_2x + \dots + b_{n-1}x^{n-2} + c_nx^{n-1},$$

it would follow that $(x_k - x_0)g(x_k) = f(x_k) - f(x_0) \equiv 0 - 0 = 0 \pmod{p}$ for $k = 1, \dots, n$, so that $g(x_k) \equiv 0 \pmod{p}$, contrary to the induction hypothesis for $n - 1$. $\square$

THEOREM 5.33. *Let $a > 0$, $b > 0$, and $(a, b) = 1$. Let x range over a complete set of residues mod b and y over a complete set of residues mod a . Then $ax + by$ ranges over a complete set of residues mod ab .*

Proof. Of the ab numbers $ax + by$, any two are incongruent mod ab . For if $ax_1 + by_1 \equiv ax_2 + by_2 \pmod{ab}$, then $ax_1 + by_1 \equiv ax_2 + by_2 \pmod{b}$, so $ax_1 \equiv ax_2 \pmod{b}$, and since $(a, b) = 1$, we have $x_1 \equiv x_2 \pmod{b}$, and similarly, by symmetry, $y_1 \equiv y_2 \pmod{a}$. $\square$

THEOREM 5.34. *Let $a > 0$, $b > 0$, and $(a, b) = 1$. Let x and y range over reduced sets of residues mod b and mod a , respectively. Then $ax + by$ ranges over a reduced set of residues mod ab .*

Remark: This is the direct proof of Theorem 44 which we announced earlier. Since Theorem 43 was also proved directly, there thus results a new, direct proof of Theorem 42, and consequently of Theorems 41 and 40; up to this point, everything had been obtained from the Möbius Inversion formula.

Proof. If $(x, b) > 1$, then certainly $(ax + by, ab) > 1$; for (x, b) divides $ax + by$ and ab , and hence divides $(ax + by, ab)$. If $(y, a) > 1$, then, by symmetry, $(ax + by, ab) > 1$ as well.

What remains to be shown, by Theorem 73, is that if $(x, b) = 1$ and $(y, a) = 1$, then $(ax + by, ab) = 1$. In fact, let $p \mid (ax + by, ab)$. Then we would have $p \mid ab$, so that, without loss of generality, $p \mid a$; moreover, $p \mid (ax + by)$, so that $p \mid by$, and consequently (since $(a, b) = 1$) $p \mid y$, contrary to the assumption that $(y, a) = 1$. $\square$

THEOREM 5.35 (The so-called Little Fermat Theorem). *If $(a, m) = 1$, then*

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

Remark: It is not known whether the so-called Last Theorem of Fermat, which is discussed in Parts 12 and 13 of my *Vorlesungen über Zahlentheorie*, is true or not. I would therefore rather refer to it as the Fermat Conjecture, and to Theorem 75 simply as Fermat's Theorem.

Proof. Let $a_1, \dots, a_{\varphi(m)}$ be a reduced set of residues mod m . Then, by Theorem 63, $aa_1, \dots, aa_{\varphi(m)}$ is also such a set. Hence the numbers a_n are congruent to the numbers aa_n ($n = 1, \dots, \varphi(m)$), apart from their order. Hence the product of the a_n is congruent to the product of the aa_n , or

$$\prod_{n=1}^{\varphi(m)} a_n \equiv \prod_{n=1}^{\varphi(m)} (aa_n) = a^{\varphi(m)} \prod_{n=1}^{\varphi(m)} a_n \pmod{m},$$

so that, by Theorem 56, $1 \equiv a^{\varphi(m)} \pmod{m}$. □

THEOREM 5.36. *If $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$; for any a at all we have $a^p \equiv a \pmod{p}$.*

Proof. The first statement follows from Theorem 75, since $\varphi(p) = p - 1$; the second follows from the first by Theorem 51 if $p \nmid a$; and if $p \mid a$, it is trivial, since $a^p \equiv 0 \equiv a \pmod{p}$. □

THEOREM 5.37 (The so-called Theorem of Wilson).

$$(p-1)! \equiv -1 \pmod{p}.$$

Two Proofs: 1) For $p = 2$ and $p = 3$, the statement is obvious. For $p > 3$, I consider the $p-3$ numbers

$$2, 3, \dots, p-3, p-2. \tag{5.12}$$

For each r in this sequence, $p \nmid r$, and hence, by Theorem 64, there is exactly one s in the sequence $0, 1, \dots, p-1$ for which

$$rs \equiv 1 \pmod{p}. \tag{5.13}$$

$s = 0$ does not obtain here; nor do $s = 1$ and $s = p-1$, since otherwise r would be $\equiv \pm 1$. The s therefore occurs in the sequence

(5.12) as well. Moreover, $s \neq r$; for $r^2 \equiv 1 \pmod{p}$ would give $p \mid (r-1)(r+1)$, $r \equiv \pm 1 \pmod{p}$.

Hence to each r in (5.12) there corresponds exactly one $s \neq r$ in (5.12) for which (5.13) holds. Since $rs = sr$, it follows, conversely, that r is uniquely determined by s . The $p-3$ numbers in (5.12) thus break up into $\frac{p-3}{2}$ pairs in such a way that the product of the numbers in each pair is $\equiv 1$. Hence

$$(p-2)! = 2 \cdot 3 \cdots (p-2) \equiv 1^{\frac{p-3}{2}} = 1 \pmod{p},$$

$$(p-1)! = (p-1)(p-2)! \equiv -(p-2)! \equiv -1 \pmod{p}.$$

Elementary Number Theory (Continued)

Edmund Landau

Chunk 2: Pages 51–100

Covering:

- Part One, Chapter V: Congruences (conclusion)
- Part One, Chapter VI: Quadratic Residues
- Part Two: Introduction
- Part Two, Chapter I: Some Elementary Inequalities of Prime-Number Theory
- Part Two, Chapter II: Brun's Theorem on Prime Pairs

Chapter V: Congruences (Concluded)

Proof of Theorem 76 (Wilson's Theorem) – continued:

2) If we set

$$f(x) = x^{p-1} - 1 - \prod_{m=1}^{p-1} (x - m),$$

then clearly

$$f(x) = c_0 + c_1x + c_2x^2 + \cdots + c_{p-2}x^{p-2}.$$

By Theorem 76, the congruence

$$f(x) \equiv 0 \pmod{p}$$

has at least the $p - 1$ roots $x = 1, 2, \dots, p - 1$. Hence, by Theorem 72,

$$c_0 \equiv c_1 \equiv c_2 \equiv \cdots \equiv c_{p-2} \equiv 0 \pmod{p}.$$

Our result then follows from the fact that

$$f(0) = -1 - (-1)^{p-1}(p-1)!.$$

Chapter 6

Quadratic Residues

Definition 6.1. If the congruence

$$x^2 \equiv n \pmod{m}$$

has a solution, then n is called a *quadratic residue* modulo m ; otherwise, n is called a *quadratic non-residue* modulo m .

Example 6.1. 0, 1, and all other perfect squares are quadratic residues modulo any number.

Definition 6.2 (Legendre's Symbol). If $p > 2$ and $p \nmid n$, let

$$\left(\frac{n}{p}\right) = \begin{cases} 1 & \text{if } n \text{ is a quadratic residue (mod } p), \\ -1 & \text{if } n \text{ is a quadratic non-residue (mod } p). \end{cases}$$

Example 6.2. $\left(\frac{n^2}{p}\right) = 1$ if $p > 2$ and $p \nmid n$; in particular, $\left(\frac{1}{p}\right) = 1$ if $p > 2$.

Theorem 6.1. *Let $p > 2$. If $n \equiv n' \pmod{p}$ and $p \nmid n$, then*

$$\left(\frac{n}{p}\right) = \left(\frac{n'}{p}\right).$$

Proof. By hypothesis, we certainly have $p \nmid n'$. From $x^2 \equiv n \pmod{p}$ it follows that $x^2 \equiv n' \pmod{p}$, and conversely. $\square$

Theorem 6.2. *Let $p > 2$. In every reduced set of residues mod p there are exactly $\frac{p-1}{2}$ numbers n for which $\left(\frac{n}{p}\right) = 1$ and hence exactly $\frac{p-1}{2}$ numbers n for which $\left(\frac{n}{p}\right) = -1$. The first set of $\frac{p-1}{2}$ numbers are represented by the residue classes to which the numbers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ belong.*

In particular, therefore: Given $p > 2$, there exists an n such that

$$\left(\frac{n}{p}\right) = -1.$$

Proof. The congruence

$$x^2 \equiv n \pmod{p},$$

if it is at all solvable, has at least one solution in the interval $0 \leq x \leq p-1$; but, by Theorem 72, it has at most two solutions in that interval, and in case $p \nmid n$ the number 0 is not one of them. Since

$$(p-x)^2 = (-x)^2 = x^2 \pmod{p},$$

there is consequently exactly one solution in the interval $1 \leq x \leq \frac{p-1}{2}$.

Hence any two of the numbers

$$1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$$

are incongruent.

The theorem has thus been proved. $\square$

Theorem 6.3 (Euler's Criterion). *If $p > 2$ and $p \nmid n$, then*

$$n^{\frac{p-1}{2}} \equiv \left(\frac{n}{p}\right) \pmod{p}.$$

Remark 6.1. The fact that

$$n^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$$

is a consequence of Fermat's Theorem to begin with; for from

$$n^{p-1} - 1 \equiv 0 \pmod{p}$$

it follows that

$$\left(n^{\frac{p-1}{2}} - 1\right) \left(n^{\frac{p-1}{2}} + 1\right) \equiv 0 \pmod{p}.$$

Proof. The modulus in the proof will be p throughout.

1) Let $\left(\frac{n}{p}\right) = 1$. Then there is an x such that

$$x^2 \equiv n \pmod{p}.$$

Hence, by Fermat's Theorem,

$$n^{\frac{p-1}{2}} \equiv x^{p-1} \equiv 1 = \left(\frac{n}{p}\right) \pmod{p}.$$

2) Let $\left(\frac{n}{p}\right) = -1$. The congruence

$$x^{\frac{p-1}{2}} - 1 \equiv 0 \pmod{p}$$

has at most $\frac{p-1}{2}$ solutions, by Theorem 72. By 1) and Theorem 79, it has at least $\frac{p-1}{2}$ solutions, namely the quadratic residues in any reduced set of residues; hence there are no further solutions. Thus, by (24), our number n , being a quadratic non-residue, satisfies the congruence

$$n^{\frac{p-1}{2}} + 1 \equiv 0 \pmod{p}.$$

□

Theorem 6.4. *If $p > 2$, $p \nmid n$, and $p \nmid n'$, then*

$$\left(\frac{nn'}{p}\right) = \left(\frac{n}{p}\right) \left(\frac{n'}{p}\right).$$

In words: The congruence $x^2 \equiv nn'$ is solvable if and only if the congruences $x^2 \equiv n$ and $x^2 \equiv n'$ are both solvable or both unsolvable. Expressed in yet another way: If n and n' are both quadratic residues or are both quadratic non-residues, then nn' is a quadratic residue; if one of them is a quadratic residue and the other is a quadratic non-residue, then the product is a quadratic non-residue. All of this under the assumption that $p > 2$, $p \nmid n$, and $p \nmid n'$.

Proof. By Theorem 80, we have

$$\left(\frac{nn'}{p}\right) \equiv (nn')^{\frac{p-1}{2}} = n^{\frac{p-1}{2}} (n')^{\frac{p-1}{2}} \equiv \left(\frac{n}{p}\right) \left(\frac{n'}{p}\right) \pmod{p}.$$

Since

$$\left(\frac{nn'}{p}\right) - \left(\frac{n}{p}\right)\left(\frac{n'}{p}\right)$$

is 0, ± 2 , and $p > 2$, we have

$$\left(\frac{nn'}{p}\right) - \left(\frac{n}{p}\right)\left(\frac{n'}{p}\right) = 0. \quad \square$$

Theorem 6.5. *If $p > 2$, $r \geq 2$, and $p \nmid m_1, \dots, p \nmid m_r$, then*

$$\left(\frac{m_1 m_2 \cdots m_r}{p}\right) = \prod_{r=1}^r \left(\frac{m_r}{p}\right).$$

Proof. Theorem 81. $\square$

If $p > 2$ and $p \nmid n$, then the symbol $\left(\frac{n}{p}\right)$ breaks up, by means of Theorem 82, into simpler symbols of the form $\left(\frac{-1}{p}\right)$, $\left(\frac{2}{p}\right)$ and $\left(\frac{n}{q}\right)$, where q is an odd prime different from p . The following three main theorems of the theory of quadratic residues (Theorems 83, 85, and 86) are concerned with these three types.

Theorem 6.6 (The so-called First Lemma of the Quadratic Reciprocity Law).

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} \quad \text{for } p > 2,$$

or, more explicitly,

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{for } p \equiv 1 \pmod{4}, \\ -1 & \text{for } p \equiv -1 \pmod{4}. \end{cases}$$

In words: Every odd prime divisor of $x^2 + 1$ is $\equiv 1 \pmod{4}$, and every $p \equiv 1 \pmod{4}$ divides $x^2 + 1$ for suitable numbers x .

Proof. By Euler's Criterion (Theorem 80), we have

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod{p};$$

since $p > 2$, we have equality. □

Theorem 6.7 (The so-called Gaussian Lemma). *Let $p > 2$ and $p \nmid n$. Let us consider the $\frac{p-1}{2}$ numbers*

$$n, 2n, \dots, \frac{p-1}{2}n$$

and determine their remainders mod p . We obtain (by Theorem 62) $\frac{p-1}{2}$ distinct numbers, which are > 0 and $< p$. Let m be the number of these remainders which are $> \frac{p}{2}$ (i.e., $> \left[\frac{p}{2}\right]$). (m can also be 0; for example, it always does if $n = 1$.)

We assert that

$$\left(\frac{n}{p}\right) = (-1)^m.$$

Example 6.3. $p = 7$, $n = 10$. The numbers 10, 20, and 30 leave remainders of 3, 6, and 2, respectively. In this case $m = 1$, and hence $\left(\frac{10}{7}\right) = -1$ by Theorem 84. And in fact, the congruence $x^2 \equiv 3 \pmod{7}$ is unsolvable.

Proof. Let the modulus be p throughout. m is the number of remainders which are $> \frac{p}{2}$ (i.e., $\leq p-1$). In case $l > 0$, denote these numbers by $a_1, \dots, a_l$; let the remainders $> \frac{p}{2}$ occurring in the theorem be $b_1, \dots, b_m$, in case $m > 0$. If we multiply all of the $\frac{p-1}{2}$ remainders (that is, all of the a_s, b_t), we obtain the congruence

$$\prod_{s=1}^l a_s \prod_{t=1}^m b_t \equiv \prod_{h=1}^{\frac{p-1}{2}} hn \equiv \left(\frac{p-1}{2}\right)! \cdot n^{\frac{p-1}{2}} \pmod{p}. \quad (25)$$

The “complements” of the numbers b_t (I mean the numbers $p - b_t$) belong to the interval from 1 to $\frac{p-1}{2}$. Any two of them are distinct, since this is true of the numbers b_t . Furthermore,

each a_s is distinct from each $p - b_t$; for

$$a_s = p - b_t$$

would give

$$xn = p - yn, \quad xn \equiv -yn, \quad x + y \equiv 0 \pmod{p},$$

in contradiction to $0 < x + y < p$.

Consequently (since there are $\frac{p-1}{2}$ numbers) the numbers a_s and the numbers $p - b_t$, taken together, are the same as the numbers $1, \dots, \frac{p-1}{2}$ in some order (the pigeon-hole principle), so that

$$\prod_{s=1}^l a_s \prod_{t=1}^m (p - b_t) = \left(\frac{p-1}{2}\right)!,$$

$$\prod_{s=1}^l a_s \prod_{t=1}^m b_t = (-1)^m \left(\frac{p-1}{2}\right)!,$$

and consequently, by (25),

$$\left(\frac{p-1}{2}\right)! \cdot n^{\frac{p-1}{2}} \equiv (-1)^m \left(\frac{p-1}{2}\right)! \pmod{p},$$

$$n^{\frac{p-1}{2}} \equiv (-1)^m \pmod{p},$$

and consequently, by Theorem 80,

$$\left(\frac{n}{p}\right) = (-1)^m.$$

□

Theorem 6.8 (The so-called Second Lemma of the Quadratic Reciprocity Law).

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} \quad \text{for } p > 2,$$

or (noting that $\frac{p^2-1}{8} = 8a^2 + 2a$ and $\frac{p^2-1}{8} \equiv \frac{p-1}{4} + \frac{p^2-1}{8} \pmod{2}$), more explicitly:

$$\left(\frac{2}{p}\right) = \begin{cases} 1 & \text{for } p \equiv \pm 1 \pmod{8}, \\ -1 & \text{for } p \equiv \pm 3 \pmod{8}. \end{cases}$$

Proof. For $p > 2$ and $n = 2$, Theorem 84 gives

$$\left(\frac{2}{p}\right) = (-1)^m.$$

For, the numbers $2, 4, \dots, 2\left[\frac{p}{4}\right]$ are themselves already > 0 and $< p$, and thus they are their own remainders; and

$$\frac{p}{2} < 2h < p$$

is true whenever

$$\frac{p}{4} < h < \frac{p}{2},$$

that is, $\left[\frac{p}{2}\right] - \left[\frac{p}{4}\right]$ times; if $p = 8a + r$, where $r = 1, 3, 5$, or 7 , then this is

$$4a - 2a = 0, \quad 4a + 1 - 2a = 1, \quad 4a + 2 - 2a - 1 = 1, \quad 4a + 3 - 2a - 1 = 0 \pmod{2},$$

respectively.

A more elegant proof of Theorem 85 is presented during the proof of the next theorem. $\square$

Theorem 6.9 (The Quadratic Reciprocity Law, first conjectured by Euler, first proved by Gauss). *Let $p > 2$ and $q > 2$ be primes, $p \neq q$. Then*

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

In words (since $\frac{p-1}{2} \cdot \frac{q-1}{2}$ is odd for $p \equiv q \equiv 3 \pmod{4}$, and is otherwise even): The congruences

$$x^2 \equiv p \pmod{q}, \quad y^2 \equiv q \pmod{p}$$

are both solvable or both unsolvable, unless $p \equiv q \equiv 3 \pmod{4}$; if $p \equiv q \equiv 3 \pmod{4}$, then one is solvable and the other unsolvable.

Proof. For the time being, let the value $q = 2$ be admitted; but let q still be a prime number $\neq p$. If $1 \leq k \leq \frac{p-1}{2}$, then

$$kq = q \left[\frac{kq}{p} \right] + r_k, \quad 1 \leq r_k \leq p-1, \tag{25}$$

where the numbers r_k are the numbers a_s and b_t of the proof of Theorem 84 (with $n = q$).

In this formula,

$$\sum_{k=1}^{\frac{p-1}{2}} kq = q \sum_{k=1}^{\frac{p-1}{2}} k = q \frac{p^2 - 1}{8}.$$

We already know that the numbers a_s and $p - b_t$, apart from their order, are $1, 2, \dots, \frac{p-1}{2}$. If, for the sake of brevity, we set

$$a = \sum_{s=1}^l a_s, \quad b = \sum_{t=1}^m b_t,$$

then

$$\begin{aligned} \sum_{k=1}^{\frac{p-1}{2}} r_k &= a + b, \\ \sum_{k=1}^{\frac{p-1}{2}} kq &= qa + qm - b. \end{aligned}$$

By addition of the equations (25), it follows that

$$\begin{aligned} \sum_{k=1}^{\frac{p-1}{2}} kq &= q \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p} \right] + a + b, \\ q \frac{p^2 - 1}{8} &= q \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p} \right] + a + b, \\ q \frac{p^2 - 1}{8} - q \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p} \right] - a &= b, \\ q \frac{p^2 - 1}{8} - q \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p} \right] - mp + 2b &= a + b. \end{aligned}$$

Hence

$$\begin{aligned} \sum_{k=1}^{\frac{p-1}{2}} r_k &\equiv \sum_{k=1}^{\frac{p-1}{2}} kq - q \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p} \right] - mp + 2b \pmod{2}, \\ \sum_{k=1}^{\frac{p-1}{2}} kq &\equiv q \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p} \right] + m \pmod{2}, \\ \sum_{k=1}^{\frac{p-1}{2}} k &\equiv \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p} \right] + m \pmod{2}, \end{aligned} \tag{26}$$

since q is odd.

1) (Alternate proof of Theorem 85.) Let $q = 2$. Then every $r_k = 0$, so that, by (26),

$$\sum_{k=1}^{\frac{p-1}{2}} k \equiv m \pmod{2},$$

and thus, by Theorem 84,

$$\left(\frac{2}{p}\right) = (-1)^m = (-1)^{\frac{p^2-1}{8}}.$$

2) Let $q > 2$. Then, by (26),

$$m \equiv \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p}\right] \pmod{2},$$

so that, by Theorem 84,

$$\left(\frac{q}{p}\right) = (-1)^m = (-1)^{\sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p}\right]}.$$

By symmetry, we have

$$\left(\frac{p}{q}\right) = (-1)^{\sum_{l=1}^{\frac{q-1}{2}} \left[\frac{lp}{q}\right]}.$$

Hence

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p}\right] + \sum_{l=1}^{\frac{q-1}{2}} \left[\frac{lp}{q}\right]}.$$

It therefore suffices to show that

$$\sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p}\right] + \sum_{l=1}^{\frac{q-1}{2}} \left[\frac{lp}{q}\right] = \frac{p-1}{2} \cdot \frac{q-1}{2} \pmod{2}.$$

It will even turn out that

$$\sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p}\right] + \sum_{l=1}^{\frac{q-1}{2}} \left[\frac{lp}{q}\right] = \frac{p-1}{2} \cdot \frac{q-1}{2}; \quad (27)$$

and we shall not make use of the fact that p and q are distinct odd primes but only of the fact that they are relatively prime odd numbers > 1 .

Indeed, let us consider the $\frac{p-1}{2} \cdot \frac{q-1}{2}$ numbers

$$lp - kq, \quad \text{where } k = 1, \dots, \frac{p-1}{2}, \quad l = 1, \dots, \frac{q-1}{2}.$$

(Whether they are distinct does not interest us: Exercise for the reader.)

None of these numbers is 0; for otherwise we would have

$$lp = kq, \quad q \mid lp, \quad q \mid l,$$

which is impossible since $l < q$.

The number of positive numbers among these $\frac{p-1}{2} \cdot \frac{q-1}{2}$ numbers is clearly $\sum_{l=1}^{\frac{q-1}{2}} \left[\frac{lp}{q}\right]$; for let

$$lp - kq > 0, \quad k < \frac{lp}{q};$$

for every $l = 1, \dots, \frac{q-1}{2}$. Since $\frac{lp}{q}$ is not an integer, it follows that $lp - kq > 0$ has exactly $\left[\frac{lp}{q}\right]$

solutions, so that $k < \frac{lp}{q}$ is automatically true.

The number of negative numbers among those is $\sum_{k=1}^{\frac{p-1}{2}} \left[\frac{kq}{p} \right]$, by symmetry.

Thus (27) is established. $\square$

Example 6.4 (of the application of the Reciprocity Law). By means of Theorems 82–86, we have

$$\left(\frac{438}{593} \right) = \left(\frac{2}{593} \right) \left(\frac{219}{593} \right) = \left(\frac{219}{593} \right) = \left(\frac{593}{219} \right) = \left(\frac{155}{219} \right) = \left(\frac{5}{219} \right) \left(\frac{31}{219} \right) = \left(\frac{219}{5} \right) \left(\frac{219}{31} \right) = \left(\frac{4}{5} \right) \left(\frac{12}{31} \right) =$$

Hence $x^2 \equiv 438 \pmod{593}$ is solvable.

More generally one sees that for a fixed odd prime q the symbol $\left(\frac{q}{p} \right)$ has the same value for all (provided there are any) that belong to the same reduced residue class mod $4q$. Indeed, $\left(\frac{q}{p} \right)$ has, by Theorem 78, the same value for all odd p belonging to a reduced residue class mod q , and $(-1)^{\frac{p-1}{2}}$ has the same value for all p belonging to a reduced residue class mod 4.

Theorem 6.10. *Let $l > 0$ and $p \nmid n$. Then the number of solutions of*

$$x^2 \equiv n \pmod{p^l} \tag{28}$$

has the following value:

$$\begin{array}{ll} 1 & \text{for } p = 2, l = 1, \\ 0 & \text{for } p = 2, l = 2, n \equiv 3 \pmod{4}, \\ 2 & \text{for } p = 2, l = 2, n \equiv 1 \pmod{4}, \\ 0 & \text{for } p = 2, l > 2, n \not\equiv 1 \pmod{8}, \\ 4 & \text{for } p = 2, l > 2, n \equiv 1 \pmod{8}, \\ 1 + \left(\frac{n}{p} \right) & \text{for } p > 2. \end{array}$$

Proof. 1) $x^2 \equiv n \pmod{2}$ has a root $x \equiv 1 \pmod{2}$ if $2 \nmid n$.

2) $x^2 \equiv 3 \pmod{4}$ has no root.

3) $x^2 \equiv 1 \pmod{4}$ has two roots, $x \equiv \pm 1 \pmod{4}$.

4) Let $p = 2, l > 2, 2 \nmid n$, and $n \not\equiv 1 \pmod{8}$. If

$$x^2 \equiv n \pmod{2^l} \tag{29}$$

were solvable, then x would be odd and we would have

$$x^2 \equiv n \pmod{8},$$

so that

$$x^2 \equiv 1 \pmod{8}.$$

However, the square of any odd number is $\equiv 1 \pmod{8}$.

5) Let $p = 2, l > 2$, and $n \equiv 1 \pmod{8}$. Without loss of generality, let $0 < n < 2^l$. The solutions of (29) need be sought merely among the 2^{l-2} odd numbers x satisfying $0 < x < 2^l$.

For every such x we certainly have

$$x^2 \equiv m \pmod{2^l}$$

for suitably chosen $m \equiv 1 \pmod{8}$ in the interval $0 < m < 2^l$. Each of these 2^{l-3} numbers m occurs at most four times. For from

$$x^2 \equiv a^2 \pmod{2^l}, \quad x \neq a,$$

$$2^l \mid (x-a)(x+a);$$

since x and a are odd, so that $x-a$ and $x+a$ are even, we have

$$\frac{x-a}{2} \cdot \frac{x+a}{2} \equiv 0 \pmod{2^{l-2}};$$

it follows that

$$2 \nmid \frac{x+a}{2},$$

so 2 does not divide both of the factors $\frac{x-a}{2}$ and $\frac{x+a}{2}$, since their sum x is odd; hence

$$\frac{x-a}{2} \equiv 0 \pmod{2^{l-2}} \quad \text{or} \quad \frac{x+a}{2} \equiv 0 \pmod{2^{l-2}},$$

that is,

$$x \equiv \pm a \pmod{2^{l-1}},$$

which yields at most four values for x .

Since the $2^{l-1} = 4 \cdot 2^{l-3}$ numbers x are distributed among 2^{l-3} pigeon-holes in such a way that there are at most four in each, it follows that there are exactly four in each, and therefore the given n -hole contains exactly four.

6₁) Let $\left(\frac{n}{p}\right) = -1$. Then we already have that

$$x^2 \equiv n \pmod{p}$$

is unsolvable, so that (28) $x^2 \equiv n \pmod{p^l}$ is certainly unsolvable, and the number of solutions of (28) is

$$0 = 1 + \left(\frac{n}{p}\right).$$

6₂) Let $\left(\frac{n}{p}\right) = 1$. Without loss of generality, let $0 < a < p^l$. The solutions of (28) need be sought merely among those $\varphi(p^l)$ numbers x in the interval $0 < x < p^l$ that are not divisible by p .

For every such x we certainly have

$$x^2 \equiv m \pmod{p^l}$$

for suitable m with $\left(\frac{m}{p}\right) = 1$, $0 < m < p^l$. Each of these $\frac{1}{2}\varphi(p^l)$ numbers occurs at most twice. For from

$$x^2 \equiv a^2 \pmod{p^l}, \quad p \nmid x,$$

it follows that

$$p^l \mid (x - a)(x + a).$$

p does not divide both of the factors $x - a$ and $x + a$, since their sum $2x$ is not divisible by p ; hence

$$x \equiv \pm a \pmod{p^l},$$

which yields at most two values for x .

Since the $\varphi(p^l)$ numbers x are distributed among $\frac{1}{2}\varphi(p^l)$ pigeon-holes in such a way that at most two are in each, it follows that there are exactly two in each, and therefore the given n -hole contains exactly two. Therefore the number of solutions of (28) is

$$2 = 1 + \left(\frac{n}{p}\right). \quad \square$$

Theorem 6.11. *Let $m > 0$ and $(n, m) = 1$. Then the number of solutions of*

$$x^2 \equiv n \pmod{m}$$

has the following value:

0, if $4 \mid m$, $8 \nmid m$, and $n \not\equiv 1 \pmod{4}$;

0, if $8 \mid m$ and $n \not\equiv 1 \pmod{8}$;

0, if a prime $p > 2$ for which $\left(\frac{n}{p}\right) = -1$ goes into m .

Otherwise, if s is the number of distinct odd $p \mid m$, then the number of solutions is

$$\begin{cases} 2^s & \text{for } 4 \nmid m, \\ 2^{s+1} & \text{for } 4 \mid m, 8 \nmid m, \\ 2^{s+2} & \text{for } 8 \mid m. \end{cases}$$

Proof. For $m = 1$ the statement is true (the number of solutions is 1); for $m > 1$ the number of solutions of (28) for the various primes $p \mid m$ and their accompanying multiplicities l appearing in the canonical decomposition of m is multiplicative, by Theorem 71. Our statements therefore follow. For if $p = 2$, then 0 is the number of solutions of (28) when $4 \mid m$, unless either $l = 2$ and $n \equiv 1 \pmod{4}$ or $l > 2$ and $n \equiv 1 \pmod{8}$; if $p > 2$ it is 0 when $\left(\frac{n}{p}\right) = -1$. Otherwise, the power of 2, if there is any, supplies a factor of 1 to the last formula if $l = 1$; 2 if $l = 2$; and 4 if $l \geq 3$; and every odd $p \mid m$ that occurs supplies a factor of 2. $\square$

The introduction, which is to follow, of the so-called Jacobi symbol, a generalization of that of Legendre, will, among other things, make the decomposition of $\left(\frac{n}{p}\right)$ into prime factors unnecessary for the complete analysis of $\left(\frac{n}{p}\right)$, where $p > 2$ and $p \nmid n$. In particular, the five main properties (Theorems 78, 81, 83, 85, and 86) of Legendre's symbol will turn out to be true for Jacobi's symbol as well.

Definition 6.3 (Jacobi's symbol). Let $m > 0$, let m be odd, and let $m = \prod_{r=1}^s p_r$ be the decomposition of m into prime factors (with repeated factors written an appropriate number of times); furthermore, let $(n, m) = 1$. Then

$$\left(\frac{n}{m}\right) = \prod_{r=1}^s \left(\frac{n}{p_r}\right).$$

(This represents 1 if $m = 1$.)

This definition is meaningful, since the factors on the right are defined as Legendre symbols, because $p_r \mid m$ and $p_r > 2$. And for $m = p > 2$ it agrees with Definition 18.

Example 6.5. $\left(\frac{n^2}{m}\right) = 1$ for odd $m > 0$; $\left(\frac{a^2}{m}\right) = 1$ for odd $m > 0$, if $(a, m) = 1$.

One should not be tempted to think that for odd $m > 0$ and $(n, m) = 1$, we have $\left(\frac{n}{m}\right) = -1$ precisely whenever n is a quadratic non-residue; but rather: whenever n is a quadratic non-residue of an odd number of p_r . Thus if n is a quadratic residue of m (and consequently a quadratic residue of all the p_r) then (but not: then only) $\left(\frac{n}{m}\right) = 1$.

Theorem 6.12 (Generalization of Theorem 78). *Let $m > 0$ be odd, $n \equiv n' \pmod{m}$, and $(n, m) = 1$. Then*

$$\left(\frac{n}{m}\right) = \left(\frac{n'}{m}\right).$$

Proof. By hypothesis, we certainly have $(n', m) = 1$. By Theorem 78, since $n \equiv n' \pmod{p_r}$ and $p_r \nmid n$, we have

$$\left(\frac{n}{p_r}\right) = \left(\frac{n'}{p_r}\right)$$

for every $p_r \mid m$, from which our result follows by multiplication with respect to all the r . $\square$

Theorem 6.13. *For $m > 0$ odd, $m' > 0$ odd, $(n, m) = 1$, and $(n, m') = 1$ (in short, if the left-hand side of the following equation is meaningful), we have*

$$\left(\frac{n}{m}\right) \left(\frac{n}{m'}\right) = \left(\frac{n}{mm'}\right).$$

Consequently, if $m = \prod p_r$ and $m' = \prod p'_r$, then

$$\left(\frac{n}{m}\right) \left(\frac{n}{m'}\right) = \prod_r \left(\frac{n}{p_r}\right) \cdot \prod_r \left(\frac{n}{p'_r}\right) = \left(\frac{n}{mm'}\right),$$

where the product is taken over the prime factors of mm' (with the proper multiplicities), and therefore

$$\left(\frac{n}{m}\right) \left(\frac{n}{m'}\right) = \left(\frac{n}{mm'}\right).$$

Theorem 6.14 (Generalization of Theorem 81). *Let $m > 0$ be odd, $(n, m) = 1$, and $(n', m) = 1$. Then*

$$\left(\frac{nn'}{m}\right) = \left(\frac{n}{m}\right) \left(\frac{n'}{m}\right).$$

Proof. We have $(nn', m) = 1$. For $p_r \mid m$ we have, by Theorem 81,

$$\left(\frac{nn'}{p_r}\right) = \left(\frac{n}{p_r}\right) \left(\frac{n'}{p_r}\right);$$

from which our result follows by multiplication with respect to all the r . $\square$

Theorem 6.15 (Generalization of Theorem 83; the so-called First Lemma of Jacobi's Reciprocity Law). *Let $m > 0$ be odd. Then*

$$\left(\frac{-1}{m}\right) = (-1)^{\frac{m-1}{2}}.$$

Proof. For $m = 1$ this is obvious; consequently, let $m > 1$.

For odd u and u' , we have

$$(u-1)(u'-1) \equiv 0 \pmod{4},$$

so that

$$uu' - 1 \equiv (u-1) + (u'-1) \pmod{4}.$$

For odd $u_1, \dots, u_s$, we therefore have

$$\begin{aligned} \prod_{r=1}^s u_r - 1 &\equiv \sum_{r=1}^s (u_r - 1) \pmod{4}, \\ \frac{\prod_{r=1}^s u_r - 1}{2} &\equiv \sum_{r=1}^s \frac{u_r - 1}{2} \pmod{2}, \\ (-1)^{\frac{\prod u_r - 1}{2}} &= \prod_{r=1}^s (-1)^{\frac{u_r - 1}{2}}. \end{aligned} \tag{30}$$

From $m = \prod p_r$ and since (by Theorem 83)

$$\left(\frac{-1}{p_r}\right) = (-1)^{\frac{p_r-1}{2}},$$

it therefore follows that

$$\left(\frac{-1}{m}\right) = \prod_{r=1}^s \left(\frac{-1}{p_r}\right) = \prod_{r=1}^s (-1)^{\frac{p_r-1}{2}} = (-1)^{\sum \frac{p_r-1}{2}} = (-1)^{\frac{m-1}{2}}. \quad \square$$

Theorem 6.16 (Generalization of Theorem 85; the so-called Second Lemma of Jacobi's Reciprocity Law). *Let $m > 0$ be odd. Then*

$$\left(\frac{2}{m}\right) = (-1)^{\frac{m^2-1}{8}}.$$

Proof. For $m = 1$ this is obvious; let $m > 1$.

For odd u and u' we have

$$(u^2 - 1)(u'^2 - 1) \equiv 0 \pmod{16} \quad (\text{and in fact even } \pmod{64});$$

hence

$$u^2 u'^2 - 1 \equiv (u^2 - 1) + (u'^2 - 1) \pmod{16}.$$

For odd $u_1, \dots, u_s$, we therefore have

$$\prod_{r=1}^s u_r^2 - 1 \equiv \sum_{r=1}^s (u_r^2 - 1) \pmod{16},$$

$$\frac{\prod u_r^2 - 1}{8} \equiv \sum_{r=1}^s \frac{u_r^2 - 1}{8} \pmod{2},$$

$$(-1)^{\frac{\prod u_r^2 - 1}{8}} = \prod_{r=1}^s (-1)^{\frac{u_r^2 - 1}{8}}.$$

From $m = \prod p_r$ and the fact that, by Theorem 85,

$$\left(\frac{2}{p_r}\right) = (-1)^{\frac{p_r^2 - 1}{8}},$$

it therefore follows that

$$\left(\frac{2}{m}\right) = \prod_{r=1}^s \left(\frac{2}{p_r}\right) = \prod_{r=1}^s (-1)^{\frac{p_r^2 - 1}{8}} = (-1)^{\sum \frac{p_r^2 - 1}{8}} = (-1)^{\frac{m^2 - 1}{8}}. \quad \square$$

Theorem 6.17 (Generalization of Theorem 86; Jacobi's Reciprocity Law). *Let n and m be > 0 , odd, and relatively prime. Then*

$$\left(\frac{n}{m}\right) \left(\frac{m}{n}\right) = (-1)^{\frac{n-1}{2} \cdot \frac{m-1}{2}}.$$

Proof. Without loss of generality, let $n > 1$ and $m > 1$; their decompositions into prime factors can be denoted by $n = \prod p$ and $m = \prod q$. Then, by Theorems 90 and 91, we have

$$\left(\frac{n}{m}\right) = \prod_q \left(\frac{n}{q}\right) = \prod_q \prod_p \left(\frac{p}{q}\right),$$

$$\left(\frac{m}{n}\right) = \prod_p \left(\frac{m}{p}\right) = \prod_p \prod_q \left(\frac{q}{p}\right),$$

so that, by Theorem 86 and (30), we have

$$\left(\frac{n}{m}\right) \left(\frac{m}{n}\right) = \prod_p \prod_q \left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = \prod_p \prod_q (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}} = (-1)^{\sum_p \sum_q \frac{p-1}{2} \cdot \frac{q-1}{2}} = (-1)^{\frac{n-1}{2} \cdot \frac{m-1}{2}}. \quad \square$$

Example 6.6 (Examples of the application of Jacobi's symbol).

$$\left(\frac{383}{443}\right) = -\left(\frac{443}{383}\right) = -\left(\frac{60}{383}\right) = -\left(\frac{2^2}{383}\right) \left(\frac{15}{383}\right) = -\left(\frac{15}{383}\right) = -\left(\frac{383}{15}\right) = -\left(\frac{8}{15}\right) = -\left(\frac{2}{15}\right) = -\left(\frac{2}{3}\right)$$

1. Legendre's symbol $\left(\frac{383}{443}\right)$ (443 is prime) can be computed rapidly by the application of the theorems on the Jacobi symbol, as follows:
2. The Jacobi symbol $\left(\frac{35}{87}\right)$ (87 is not prime, but is relatively prime to 35) can be similarly computed thus:

$$\left(\frac{35}{87}\right) = \left(\frac{87}{35}\right) = \left(\frac{17}{35}\right) = \left(\frac{35}{17}\right) = \left(\frac{1}{17}\right) = 1.$$

The computation is often simplified by the use of:

Theorem 6.18. *Let n and m be odd and relatively prime. Then*

$$\left(\frac{n}{m}\right) \left(\frac{m}{n}\right) = \begin{cases} -(-1)^{\frac{n-1}{2} \cdot \frac{m-1}{2}} & \text{if } n < 0 \text{ and } m < 0, \\ (-1)^{\frac{n-1}{2} \cdot \frac{m-1}{2}} & \text{otherwise.} \end{cases}$$

Proof. 1) If n and m are > 0 , then this is Theorem 94.

2) If n and m are < 0 , then by Theorems 92 and 94,

$$\left(\frac{n}{m}\right) \left(\frac{m}{n}\right) = \left(\frac{n}{|m|}\right) \left(\frac{|m|}{n}\right) \left(\frac{|n|}{m}\right) \left(\frac{m}{|n|}\right) = \left(\frac{n}{|m|}\right) \left(\frac{|n|}{m}\right) = \left(\frac{n}{|m|}\right) \left(\frac{|n|}{m}\right) \left(\frac{|n|}{|m|}\right) \left(\frac{|m|}{|n|}\right) = (-1)^{\frac{|n|-1}{2} \cdot \frac{|m|-1}{2}}.$$

After further simplification:

$$\left(\frac{n}{m}\right) \left(\frac{m}{n}\right) = -(-1)^{\frac{n-1}{2} \cdot \frac{m-1}{2}}.$$

3) If one of the numbers n and m is positive and the other negative, then without loss of generality let $n > 0$ and $m < 0$. Then, by Theorems 92 and 94, we have

$$\left(\frac{n}{m}\right) \left(\frac{m}{n}\right) = \left(\frac{n}{|m|}\right) \left(\frac{-1}{n}\right) \left(\frac{|m|}{n}\right) = (-1)^{\frac{n-1}{2}} (-1)^{\frac{n-1}{2} \cdot \frac{|m|-1}{2}} = (-1)^{\frac{n-1}{2} \cdot \frac{|m|+1}{2}} = (-1)^{\frac{n-1}{2} \cdot \frac{-m+1}{2}} = (-1)^{\frac{n-1}{2} \cdot \frac{m-1}{2}}.$$

□

Example 6.7.

$$\left(\frac{-3}{17}\right) = \left(\frac{-1}{17}\right) \left(\frac{3}{17}\right) = \left(\frac{3}{17}\right) = \left(\frac{17}{3}\right) = \left(\frac{2}{3}\right) = -1.$$

The reader is urged to work through these examples carefully.

Definition 6.4 (Kronecker's Symbol). (Throughout the rest of this chapter) let $d \equiv 0$ or $1 \pmod{4}$, and let d not be a perfect square (thus $d = 5, 8, 12, 13, 17, 20, 21, \dots$ or $-3, -4, -7, -8, \dots$). Let $m > 0$. Then $\left(\frac{d}{m}\right)$ is always given a meaning by means of the following:

$$\begin{aligned}\left(\frac{d}{2}\right) &= 0 && \text{if } 2 \mid d, \\ \left(\frac{d}{2}\right) &= 1 && \text{if } d \equiv 1 \pmod{8}, \\ \left(\frac{d}{2}\right) &= -1 && \text{if } d \equiv 5 \pmod{8}\end{aligned}$$

(hence $\left(\frac{d}{2}\right)$ = Jacobi's symbol $\left(\frac{2}{|d|}\right)$ for $2 \nmid d$),

$$\left(\frac{d}{p}\right) = \text{Legendre's symbol,} \quad \text{if } p > 2 \text{ and } p \nmid d,$$

$$\left(\frac{d}{m}\right) = \prod_r \left(\frac{d}{p_r}\right) \quad \text{for } m = \prod_r p_r \text{ (i.e., 1 for } m = 1\text{)}.$$

For those d and m for which the Kronecker and Jacobi symbols are defined (namely for the above d and for odd $m > 0$ relatively prime to d) both definitions obviously agree (as they should).

We notice at once that for $(d, m) > 1$ we always have

$$\left(\frac{d}{m}\right) = 0,$$

and for $(d, m) = 1$ we always have

$$\left(\frac{d}{m}\right) = \pm 1.$$

Theorem 6.19. *If $m_1 > 0$ and $m_2 > 0$, then*

$$\left(\frac{d}{m_1 m_2}\right) = \left(\frac{d}{m_1}\right) \left(\frac{d}{m_2}\right).$$

Proof. This follows immediately from Definition 20. □

Theorem 6.20. *Let $k > 0$ and let $(d, k) = 1$. The number of solutions of*

$$x^2 \equiv d \pmod{4k} \tag{31}$$

is

$$2 \sum_{f|k} \left(\frac{d}{f}\right),$$

where f runs through the square-free positive divisors of k .

Remark 6.2. Since whenever x_0 satisfies the congruence so does $x_0 + 2k$ (because $(x_0 + 2k)^2 = x_0^2 + 4kx_0 + 4k^2 \equiv x_0^2 \pmod{4k}$), it follows that $2 \sum_{f|k} \left(\frac{d}{f}\right)$ is the number of x in the interval $0 \leq x < 2k$ which satisfy congruence (31). It is in this form that we shall later apply Theorem 97.

Proof. 1) Let d be odd, and therefore $\equiv 1 \pmod{4}$; then $(d, 4) = 1$. For every p^l in the canonical decomposition of $4k$, the number of solutions of $x^2 \equiv d \pmod{p^l}$ is, by Theorem 87,

$$\begin{aligned} & 2 && \text{for } p = 2, l = 2, \\ & 2 \left(1 + \left(\frac{d}{p} \right) \right) && \text{for } p = 2, l > 2, \\ & 1 + \left(\frac{d}{p} \right) && \text{for } p > 2. \end{aligned}$$

($l = 1$ does not occur when $p = 2$, since $4 \mid 4k$), and

From Theorem 71 it follows that the number of solutions of (31) is

$$2 \prod_{p \mid k} \left(1 + \left(\frac{d}{p} \right) \right) = 2 \sum_{f \mid k} \left(\frac{d}{f} \right)$$

(since for $p = 2$ we have $l = 2$ if $2 \nmid k$ and $l > 2$ if $2 \mid k$).

2) Let d be even, and therefore $\equiv 0 \pmod{4}$. Then k is odd. The congruence

$$x^2 \equiv d \equiv 0 \pmod{4}$$

has two solutions; $x^2 \equiv d \pmod{p^l}$ has $1 + \left(\frac{d}{p} \right)$ solutions for $p^l \parallel k$, $l > 0$, so that in this case as well the number of solutions of (31) is

$$2 \prod_{p \mid k} \left(1 + \left(\frac{d}{p} \right) \right) = 2 \sum_{f \mid k} \left(\frac{d}{f} \right).$$

Let us set $|d| = a$ throughout the rest of this chapter. □

Theorem 6.21. *If $m > 0$ and $(d, m) = 1$, then*

1. *for d odd, we have*

$$\left(\frac{d}{m} \right) = \left(\frac{m}{a} \right)$$

(the symbol on the right is a Jacobi symbol);

2. *for d even, if 2^b goes into d exactly b times, so that $d = 2^b u$ and u is odd, and if we set $|u| = v$, then*

$$\left(\frac{d}{m} \right) = \left(\frac{m}{v} \right) \quad \text{or} \quad (-1)^{\frac{m-1}{2}} \left(\frac{m}{v} \right)$$

(both symbols on the right are Jacobi symbols).

Proof. 1) Let d be odd, so that $d \equiv 1 \pmod{4}$. We have $m = 2^r w$, where w is odd and > 0 , and $r \geq 0$. By Theorem 96 and Definition 20, we have

$$\left(\frac{d}{m} \right) = \left(\frac{d}{2^r} \right) \left(\frac{d}{w} \right) = \left(\frac{d}{2^r} \right) \left(\frac{d}{w} \right),$$

so that by Theorems 95 and 91 we have

$$\left(\frac{d}{m} \right) = \left(\frac{w}{a} \right) \left(\frac{d}{2^r} \right) = \left(\frac{m}{a} \right).$$

2) Let d be even. Then

$$\left(\frac{d}{m}\right) = \left(\frac{2^b}{m}\right) \left(\frac{u}{m}\right) = \left(\frac{2^b}{m}\right) \left(\frac{u}{m}\right),$$

so that by Theorem 95, since m is odd and > 0 , we have

$$\left(\frac{d}{m}\right) = \left(\frac{m}{v}\right) \quad \text{or} \quad (-1)^{\frac{m-1}{2}} \left(\frac{m}{v}\right).$$

□

Theorem 6.22. *The number-theoretic function $\left(\frac{d}{m}\right)$ of m has the following properties:*

1. $\left(\frac{d}{m}\right) = 0$ for $(d, m) > 1$.
2. $\left(\frac{d}{m_1}\right) = \left(\frac{d}{m_2}\right)$ for $m_1 \equiv m_2 \pmod{a}$.
3. $\left(\frac{d}{m_1 m_2}\right) = \left(\frac{d}{m_1}\right) \left(\frac{d}{m_2}\right)$.
4. $\left(\frac{d}{m}\right) = -1$ for suitable m .

Proof. 1) follows from the definition.

2) follows from Theorem 96.

3) follows from Theorem 96.

4) Let $(a, m_1) > 1$. Then we have $(a, m_2) > 1$, so that $\left(\frac{d}{m_1}\right) = 0 = \left(\frac{d}{m_2}\right)$.

4₂) Let $(a, m_1) = 1$, so that $(a, m_2) = 1$.

4₂₁) Let d be odd. By Theorems 98 and 89, we have

$$\left(\frac{d}{m_1}\right) = \left(\frac{m_1}{a}\right) = \left(\frac{m_2}{a}\right) = \left(\frac{d}{m_2}\right).$$

4₂₂) Let d be even. By Theorem 98, we have

$$\left(\frac{d}{m_1}\right) = \left(\frac{m_1}{v}\right) \quad \text{or} \quad (-1)^{\frac{m_1-1}{2}} \left(\frac{m_1}{v}\right),$$

$$\left(\frac{d}{m_2}\right) = \left(\frac{m_2}{v}\right) \quad \text{or} \quad (-1)^{\frac{m_2-1}{2}} \left(\frac{m_2}{v}\right).$$

Now $\left(\frac{m_1}{v}\right) = \left(\frac{m_2}{v}\right)$ by Theorem 89 (since $v \mid a$ and $m_1 \equiv m_2 \pmod{v}$); furthermore (since $4 \mid a$ and $m_1 \equiv m_2 \pmod{4}$), we have

$$(-1)^{\frac{m_1-1}{2}} = (-1)^{\frac{m_2-1}{2}}.$$

On the right, we have

$$(-1)^{\frac{u-1}{2}} \left(\frac{m}{v}\right) = (-1)^{\frac{u-1}{2} \cdot \frac{m-1}{2}} \left(\frac{m}{u}\right),$$

for this is obvious when $b = 2$, and when $b > 2$ it is a consequence of Theorem 93 and of the fact that $8 \mid a$ and $m_1 \equiv m_2 \pmod{8}$.

5₁) Let d be odd. Then a is not a perfect square; for if $d > 0$ we have $a = d$, and if $d < 0$ we have $a \equiv 3 \pmod{4}$. For suitable p we therefore have $a = p^l g$, where $p > 2$, l is odd, $p \nmid g$, and g

is odd. Let s be a quadratic non-residue mod p (s exists, by Theorem 79); choose $m > 0$, by Theorem 69, so that

$$m \equiv s \pmod{p}, \quad m \equiv 1 \pmod{g}.$$

□

Theorem 6.23.

$$\left(\frac{d}{-1}\right) = \begin{cases} 1 & \text{for } d > 0, \\ -1 & \text{for } d < 0. \end{cases}$$

Proof. 1) Let d be odd. Then, by Theorem 98, we have

$$\left(\frac{d}{-1}\right) = \left(\frac{-1}{a}\right) = (-1)^{\frac{a-1}{2}} = \begin{cases} 1 & \text{for } d > 0, \\ -1 & \text{for } d < 0. \end{cases}$$

2) Let d be even, so that $d = 2^b u$, $b = 2$, with u odd. Then, by Theorem 98,

$$\left(\frac{d}{-1}\right) = \left(\frac{-1}{v}\right) \quad \text{or} \quad (-1)^{\frac{-1-1}{2}} \left(\frac{-1}{v}\right) = \left(\frac{-1}{|u|}\right).$$

Here, on the one hand, we have

$$\left(\frac{-1}{|u|}\right) = (-1)^{\frac{|u|-1}{2}},$$

for if $b = 2$ this is obvious, and if $b = 3$ we have $a - 1 \equiv 7 \pmod{8}$; on the other hand, we have

$$(-1)^{\frac{|u|-1}{2}} = \begin{cases} 1 & \text{for } d > 0, \\ -1 & \text{for } d < 0. \end{cases}$$

□

Theorem 6.24. For $n > 0$, $m > 0$, and $n \equiv -m \pmod{a}$, we have

$$\left(\frac{d}{n}\right) = \begin{cases} \left(\frac{d}{m}\right) & \text{if } d > 0, \\ -\left(\frac{d}{m}\right) & \text{if } d < 0. \end{cases}$$

Proof.

$$\left(\frac{d}{n}\right) = \left(\frac{d}{-1}\right) \left(\frac{d}{-n}\right) = \left(\frac{d}{-1}\right) \left(\frac{d}{m}\right)$$

and Theorem 100.

□

Part I

Brun's Theorem and Dirichlet's Theorem

Introduction

In Part Two, my aim is to prepare the reader for the forthcoming applications of analysis to several important examples. Here, methods of real analysis will still suffice; even when, in Chapter III, complex numbers are employed (roots of unity), the variables still remain real; and the theorems used—on infinite series with complex terms (convergence, absolute convergence, and uniform convergence)—follow immediately from the corresponding theorems on real series. However, in Parts Five, Six, and Seven of *Vorlesungen uber Zahlentheorie* (which is a continuation of the present work) I repeatedly make use of the elements of the theory of functions of a complex variable.

Although the methods of this second part are elementary, I nevertheless discuss two theorems whose proofs, in spite of many simplifications obtained since their original discovery, are by no means short. Thus neither of these theorems, Brun's (1919) in Chapter II or Dirichlet's (1837) in Chapter III, is analytically deep, even though they both require extensive number-theoretic machinery. This is the reason why I introduce them at this stage. In order to orient the reader regarding the significance of Brun's Theorem, I first present, in Chapter I, the proofs of several simple asymptotic properties of prime number distribution, which also appear later (in Part Seven of my *Vorlesungen uber Zahlentheorie*) in considerably refined form, following the introduction of some complex function theory. Not everything presented in Chapter I is later applied in Chapter II (Chapter III is completely independent of Chapters I and II); but it is no extra trouble to prove the theorems stated there all at the same time.

Chapter 1

Some Elementary Inequalities of Prime-Number Theory

Throughout this chapter and the next, the symbols $a_1, \dots, a_{18}$ will be used to represent strictly positive constants the determination of whose values, or an estimate of whose values, will not matter (they will only appear in inequalities).

Theorem 1.1. *If $\xi \geq 2$, then*

$$a_1 \frac{\xi}{\log \xi} \leq \pi(\xi) \leq a_2 \frac{\xi}{\log \xi}.$$

This means: The number $\pi(\xi)$ of primes up to ξ has the order of magnitude $\frac{\xi}{\log \xi}$ as $\xi \rightarrow \infty$. This modest step toward proving the Hadamard–de la Vallée Poussin Prime-Number Theorem (1896),

$$\lim_{\xi \rightarrow \infty} \frac{\pi(\xi)}{\frac{\xi}{\log \xi}} = 1,$$

was first made by Chebyshev (in 1852).

Proof. For every $\xi \geq 0$, we have

$$\xi - 2 \left\lfloor \frac{\xi}{2} \right\rfloor < 1; \tag{39}$$

for

$$0 \leq \xi - 2 \left\lfloor \frac{\xi}{2} \right\rfloor < 2$$

and the left-hand side is an integer.

Let $n \geq 2$. For every $p \leq 2n$, let r denote the largest rational number for which $p^r \leq 2n$ (in other words, $r = \left\lfloor \frac{\log(2n)}{\log p} \right\rfloor$). First I will show that

$$n^{\pi(2n) - \pi(n)} < \frac{(2n)!}{(n!)^2} \leq \prod_{p \leq 2n} p^r. \tag{40}$$

(The expression in the middle is an integer, being the binomial coefficient $\binom{2n}{n}$. The right-hand side obviously represents the smallest common multiple of all the natural numbers $\leq 2n$.)

The first half of (40) follows from the fact that every p for which $n < p \leq 2n$ divides $(2n)!$, but not $n!n!$. The second half of (40) is proved as follows. $\frac{(2n)!}{n!n!}$ contains only prime factors

which are $\leq 2n$, and contains each $p \leq 2n$ exactly

$$\sum_{m=1}^r \left(\left\lfloor \frac{2n}{p^m} \right\rfloor - 2 \left\lfloor \frac{n}{p^m} \right\rfloor \right)$$

times, by Theorem 27; therefore, by (39), it contains each such p at most

$$\sum_{m=1}^r 1 = r$$

times.

From (40) (where the left-hand side has $\pi(2n) - \pi(n)$ factors which are all $> n$, and the right-hand side $\pi(2n)$ factors which are all $< 2n$), it follows further that

$$n^{\pi(2n) - \pi(n)} < \prod_{n < p \leq 2n} p \leq \frac{(2n)!}{(n!)^2} \leq \prod_{p \leq 2n} p^r \leq (2n)^{\pi(2n)},$$

$$(\pi(2n) - \pi(n)) \log n < \log \frac{(2n)!}{(n!)^2} < n \pi(2n) \log(2n).$$

Now, on the one hand, we have

$$\frac{(2n)!}{(n!)^2} = \binom{2n}{n} < \sum_{k=0}^{2n} \binom{2n}{k} = 2^{2n};$$

and, on the other hand,

$$\frac{(2n)!}{(n!)^2} = \binom{2n}{n} \geq \frac{(n+1)(n+2) \cdots (2n)}{n!} \geq \prod_{a=1}^n \frac{n+a}{a} \geq \prod_{a=1}^n 2 = 2^n;$$

consequently, on the one hand, we have

$$(\pi(2n) - \pi(n)) \log n \leq \log(2^{2n}) = 2n \log 2,$$

$$\pi(2n) - \pi(n) \leq a_3 \frac{n}{\log n}, \quad (41)$$

and, on the other hand,

$$n \pi(2n) \log(2n) \geq \log(2^n) = n \log 2,$$

$$\pi(2n) \geq a_4 \frac{n}{\log n}. \quad (42)$$

If $\xi \geq 4$, then by (42) we have

$$\pi(\xi) \geq \pi \left(2 \left\lfloor \frac{\xi}{2} \right\rfloor \right) \geq a_4 \frac{\left\lfloor \frac{\xi}{2} \right\rfloor}{\log \left\lfloor \frac{\xi}{2} \right\rfloor} \geq a_5 \frac{\xi}{\log \xi},$$

and for $\xi \geq 2$ (since $\pi(\xi) \geq 1$ whenever $2 \leq \xi < 4$) we therefore have

$$\pi(\xi) > a_1 \frac{\xi}{\log \xi},$$

which proves the first part of our theorem.

On the other hand, since $n = 2^{n_1} + 2^{n_2} + \cdots + 2^{n_t}$ with $n_1 > n_2 > \cdots > n_t \geq 0$, it follows from (41) that

$$\pi(n) - \pi\left(\frac{n}{2}\right) \leq a_3 \frac{\left[\frac{n}{2}\right]}{\log \left[\frac{n}{2}\right]} \leq a_6 \frac{n}{\log n},$$

$$\pi\left(\frac{n}{2}\right) - \pi\left(\frac{n}{4}\right) \leq a_6 \frac{\frac{n}{2}}{\log \frac{n}{2}},$$

.....

for $n \geq 2$; consequently, for $n \geq 2$, we have

$$\begin{aligned} \log \pi(n) - \log \frac{n}{2} &= \log n \left(\pi(n) - \pi\left(\frac{n}{2}\right) \right) + \log \frac{n}{2} \left(\pi\left(\frac{n}{2}\right) - \pi\left(\frac{n}{4}\right) \right) + \cdots + \log 2 \cdot \pi(2) \\ &< \log n + a_6 \log n + a_6 \log n + \cdots < a_7 \log n. \end{aligned}$$

Consequently, if $m \geq 0$ and $a_8 m \leq \xi$, then

$$\log \xi \leq a_9 \left(\pi(\xi) - \pi\left(\frac{\xi}{2}\right) \right) + a_{10} \pi\left(\frac{\xi}{2}\right) < a_{11} \pi(\xi) \log \xi$$

for $\xi \geq 2$. This is then summed over all the m under consideration; since, for the largest of these m (call it ν), we have

$$2^{\nu+1} > \frac{\xi}{a_8}, \quad \nu + 1 < \log_2 \xi + a_{12}, \quad \pi\left(\frac{\xi}{2^{\nu+1}}\right) = 0,$$

it follows that

$$\sum_{m=0}^{\nu} \left(\pi\left(\frac{\xi}{2^m}\right) - \pi\left(\frac{\xi}{2^{m+1}}\right) \right) = \pi(\xi) < a_{13} \frac{\xi}{\log \xi},$$

which proves the second part of our theorem. □

Theorem 1.2. *If p_r represents the r -th prime, then for $r > 1$ we have*

$$a_{14} r \log r < p_r < a_{15} r \log r.$$

Proof. 1) By the second half of Theorem 112, upon replacing ξ by p_r , we have

$$\pi(p_r) < a_2 \frac{p_r}{\log p_r},$$

$$r < a_2 \frac{p_r}{\log p_r},$$

$$p_r > a_{16} r \log p_r.$$

Now certainly $p_r > r$ (for successive primes differ by at least 1 and $p_1 = 2 > 1$); consequently, for $r > 0$, we have

$$p_r > a_{14} r \log r.$$

2) By the first half of Theorem 112 we have

$$a_1 \frac{p_r}{\log p_r} \leq \pi(p_r) = r. \tag{43}$$

Hence, for $r > a_{17}$, we have

$$\frac{p_r}{\log p_r} \leq \frac{r}{a_1},$$

$$p_r < r,$$

$$\log p_r < 2 \log r;$$

and hence, for $r > a_{17}$, it follows from (43) that

$$a_1 \frac{p_r}{2 \log r} < r,$$

$$p_r < a_{18} r \log r. \quad (44)$$

It follows from (44), for $r > 1$, that

$$p_r < a_{15} r \log r. \quad \square$$

Theorem 1.3. *The series*

$$\sum_r \frac{1}{p_r},$$

summed over all the primes in ascending order, diverges.

Two Proofs:

1) (Based on Theorem 113.) From the second half of Theorem 113 it follows, for $r > 1$, that

$$\frac{1}{p_r} > \frac{1}{a_{15} r \log r}.$$

Since the series

$$\sum_{r=2}^{\infty} \frac{1}{r \log r}$$

diverges, it follows that $\sum_r \frac{1}{p_r}$ also diverges.

2) (Direct.) From Theorem 22 it follows, for $\xi \geq 2$, that

$$\prod_{p \leq \xi} \frac{1}{1 - \frac{1}{p}} = \prod_{p \leq \xi} \left(1 + \frac{1}{p} + \frac{1}{p^2} + \cdots \right) \geq \sum_{a=1}^{\xi} \frac{1}{a},$$

where the symbol $\sum^*$ signifies that the sum is taken over all those numbers a that are not divisible by any $p > \xi$. These a include all of the natural numbers $\leq \xi$. Consequently we have

$$\prod_{p \leq \xi} \frac{1}{1 - \frac{1}{p}} \geq \sum_{a=1}^{\xi} \frac{1}{a}. \quad (45)$$

Because of the divergence of the harmonic series $\sum_{a=1}^{\infty} \frac{1}{a}$, it follows that as $\xi \rightarrow \infty$,

$$\prod_{p \leq \xi} \frac{1}{1 - \frac{1}{p}} \rightarrow \infty,$$

$$-\sum_{p \leq \xi} \log \left(1 - \frac{1}{p} \right) \rightarrow +\infty.$$

For $0 < \eta < 1$ we now have

$$-\log(1 - \eta) = \eta + \frac{\eta^2}{2} + \frac{\eta^3}{3} + \cdots < \eta + \eta^2 + \eta^3 + \cdots = \frac{\eta}{1 - \eta} < 2\eta;$$

and hence

$$\sum_{p \leq \xi} \frac{1}{p} > \frac{1}{2} \sum_{p \leq \xi} \left(-\log \left(1 - \frac{1}{p} \right) \right) \rightarrow +\infty.$$

Theorem 1.4. For $\xi = 3$ we have

$$\sum_{p \leq \xi} \frac{1}{p} > a_{19} \log \log \xi, \quad (46)$$

$$\sum_{p \leq \xi} \frac{1}{p} < a_{20} \log \log \xi. \quad (47)$$

To clarify: (46) implies that there are few, and (47) that there are many, primes. Accordingly, (46) can be shown to result from the first half of Theorem 113. On the other hand, (47) can be proved at once directly; the second half of Theorem 113 would not give a close enough estimate.

One should note that $\log \log \xi \geq \log \log 3 > 0$.

Proof. 1) For $r > 1$ it follows, by Theorem 113, that

$$\frac{1}{p_r} > \frac{1}{a_{15} r \log r},$$

so that, for $\xi \geq 3$, $r \leq \pi(\xi)$,

$$\begin{aligned} \sum_{p \leq \xi} \frac{1}{p} &> \frac{1}{a_{15}} \sum_{r=2}^{\pi(\xi)} \frac{1}{r \log r} > \frac{1}{a_{15}} \int_2^{\pi(\xi)+1} \frac{dn}{n \log n} \\ &> \frac{1}{a_{15}} (\log \log(\pi(\xi) + 1) - \log \log 2) > a_{19} \log \log \xi. \end{aligned}$$

2) For $\xi \geq 3$, it follows from (45) that

$$\begin{aligned} \log \sum_{a=1}^{\xi} \frac{1}{a} &\leq - \sum_{p \leq \xi} \log \left(1 - \frac{1}{p} \right) < \sum_{p \leq \xi} \frac{2}{p}, \\ \sum_{p \leq \xi} \frac{1}{p} &> \frac{1}{2} \log \sum_{a=1}^{\xi} \frac{1}{a} > \frac{1}{2} \log \log(\xi + 1) - a_{21} > a_{22} \log \log \xi. \end{aligned}$$

Wait—we want the upper bound. For $\xi \geq 3$, from (45):

$$\log \sum_{a=1}^{\xi} \frac{1}{a} \geq - \sum_{p \leq \xi} \log \left(1 - \frac{1}{p} \right) > \sum_{p \leq \xi} \frac{1}{p},$$

so

$$\sum_{p \leq \xi} \frac{1}{p} < \log \sum_{a=1}^{\xi} \frac{1}{a} + a_{23} < a_{20} \log \log \xi. \quad \square$$

Chapter 2

Brun's Theorem on Prime Pairs

This chapter is somewhat difficult; the reader may skip it entirely. Brun's Theorem will be applied neither in the remainder of this book nor in my *Vorlesungen über Zahlentheorie*.

There is only one pair of prime numbers that differ by 1—namely, the primes 2 and 3—since all of the succeeding primes are odd.

No matter how far out we go in a table of primes, we find pairs of primes that differ by 2, for example,

$$3, 5; \quad 7, 11, 13; \quad 17, 19; \quad 29, 31; \quad \dots; \quad 101, 103;$$

(Incidentally, each of these pairs, aside from the second, begins with a number greater than the second number of the preceding pair; this is because one of the numbers n , $n + 2$, and $n + 4$ is always divisible by 3, so that if these are all to be primes, we must have $n = 3$.)

The question now arises whether there are infinitely many “prime pairs,” that is, whether there are infinitely many numbers for which n and $n + 2$ are prime. The methods of number theory and of analysis have, to this day, not proven powerful enough to answer this question. (One would certainly place one's bet on a yes answer.)

All the more remarkable is the following theorem of Brun (Theorem 120) which is proved by elementary methods (the so-called method of the Sieve of Eratosthenes): If there are infinitely many prime pairs, then the series

$$\frac{1}{3} + \frac{1}{5} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \frac{1}{13} + \frac{1}{17} + \frac{1}{19} + \frac{1}{29} + \frac{1}{31} + \dots,$$

which is summed over all of these primes, converges. (In order to understand the significance of this theorem, compare it with Theorem 114.)

In any case, it follows that there are not “too many” prime pairs.

I begin with three simple lemmas (Theorems 116–118). The entire difficulty, then, lies in Theorem 119; Theorem 120 follows easily from it.

Theorem 2.1. *If $b \geq 0$ and if $m > 0$ is odd, then*

$$\sum_{j=0}^{m-1} (-1)^j \binom{b}{j} \geq 0.$$

Two Proofs:

1) It is easily seen by induction, for $b \geq 0$ and $n \geq 0$, that

$$\sum_{j=0}^n (-1)^j \binom{b}{j} = (-1)^n \binom{b-1}{n}. \quad (48)$$

For if $n = 0$, it is true ($1 = 1$); and in the induction from n to $n + 1$ the right-hand side increases by

$$(-1)^{n+1} \binom{b-1}{n+1} - (-1)^n \binom{b-1}{n} = (-1)^{n+1} \left(\binom{b-1}{n+1} + \binom{b-1}{n} \right) = (-1)^{n+1} \binom{b}{n+1}.$$

(At the bottom of this, of course, lie the series expansions

$$(1-x)^{b-1} = \sum_{j=0}^{\infty} (-1)^j \binom{b-1}{j} x^j, \quad (1-x)^b = \sum_{j=0}^{\infty} (-1)^j \binom{b}{j} x^j,$$

$$(1-x)^b = (1-x)^{b-1}(1-x) = \sum_{j=0}^{\infty} (-1)^j \binom{b-1}{j} x^j - \sum_{j=0}^{\infty} (-1)^j \binom{b-1}{j} x^{j+1},$$

which hold for $|x| < 1$, from a comparison of whose coefficients (48) follows.)

If $n = m - 1$ is even, then our assertion follows from (48).

2) The binomial coefficients $\binom{b}{j}$ for $0 \leq j \leq b$ increase until midway (where, if b is odd, namely for $j = \frac{b-1}{2}$ and $j = \frac{b+1}{2}$ two equal ones occur) and then they decrease; also, they are alternating in sign.

For $m - 1 \geq b$ it follows that the sum in the statement of the theorem is > 0 , since the last term is > 0 .

For $m - 1 = b$ it is $(1 - 1)^b = 0$.

For $b < m - 1$ it is

$$\sum_{j=0}^b (-1)^j \binom{b}{j} - \sum_{j=m}^b (-1)^j \binom{b}{j} = 2 \sum_{j=0}^{m-1} (-1)^j \binom{b}{j},$$

where the first term on the right is > 0 and the terms are alternate in sign and decrease in absolute value; thus here, too, the sum is > 0 .

Theorem 2.2. If S_n is the n -th elementary symmetric function of s positive numbers $\xi_1, \xi_2, \dots, \xi_s$ ($1 \leq n \leq s$), then

$$S_n \leq \frac{(S_1)^n}{n!}.$$

Proof. In the expansion of

$$(\xi_1 + \xi_2 + \dots + \xi_s)^n$$

each of the $\binom{s}{n}$ products $\xi_{i_1} \xi_{i_2} \dots \xi_{i_n}$ where $1 \leq i_1 < i_2 < \dots < i_n \leq s$, has the coefficient $n!$. □

Theorem 2.3. Let $d > 0$ and $\xi > 0$. Then the number of positive numbers $n \leq \xi$ which belong to any given residue class mod d differs from $\frac{\xi}{d}$ by less than 1.

Proof. Of each set of d consecutive numbers n , exactly one is to be counted; up to ξ there are $\left[\frac{\xi}{d}\right]$ complete sets of residues and, if ξ is not integral, usually (but not always) a partial set. The desired number is thus $\left[\frac{\xi}{d}\right]$ or $\left[\frac{\xi}{d}\right] + 1$; the latter does not occur if ξ is integral; hence the number is $\frac{\xi}{d} + \vartheta$ where $|\vartheta| < 1$. □

Theorem 2.4. *For $\xi \geq 3$ we have*

$$P(\xi) < a_{24} \frac{\xi(\log \log \xi)^2}{\log^2 \xi},$$

where $P(\xi)$ is the number of primes $p \leq \xi$ for which $p + 2$ is also prime.

Proof. Let $\xi > 5$ and let η be chosen anywhere in the interval $5 < \eta < \xi$. Later on, I shall choose η as a particular function of ξ ; the proof will be clearer, however, if this is not done immediately.

The number of positive numbers $n \leq \xi$ having a particular property is at most η plus the number of n in the interval $\eta < n \leq \xi$ having that property (since I have counted all of the numbers $1, 2, \dots, [\eta]$ and thus have not lost any there). It follows that for our $P(\xi)$ the relation

$$P(\xi) \leq \eta + Q(\xi) \tag{50}$$

holds, where $Q(\xi)$ is the number of n in the interval $\eta < n \leq \xi$ for which n and $n + 2$ are prime.

Let p_h be the h -th prime and let $\pi(\eta) = r$; then $p_2, p_3, \dots, p_r$ are the odd primes $\leq \eta$, and we have $r(\xi) \geq 3$, since $\eta > 5$. (I write $r(\xi)$ without further ado, since I think of η as a function of ξ .) By $A(\xi)$, I shall mean the number of numbers for which

$$0 < n \leq \xi, \quad n \not\equiv 0, \text{ and } n \not\equiv -2 \pmod{p_h} \text{ for } h = 2, \dots, r$$

holds. We obviously have

$$Q(\xi) \leq A(\xi); \tag{51}$$

for every n counted in $Q(\xi)$ is $> \eta$, and hence $> p_h$ for $h = 2, \dots, r$, so that neither the prime n nor the prime $n + 2$ is divisible by p_h .

From (50) and (51) it follows that

$$P(\xi) \leq \eta + A(\xi). \tag{52}$$

Let $\Omega(d)$ be the number of distinct prime factors of $d > 0$. For every odd, square-free number $d > 0$, I shall denote by $B(d, \xi)$ the number of positive numbers $n \leq \xi$ which for every $p \mid d$ (if any) satisfies one of the conditions

$$n \equiv 0 \text{ or } n \equiv -2 \pmod{p}. \tag{53}$$

(Then, in particular, we have

$$B(1, \xi) = [\xi],$$

since no n in the interval $0 < n \leq \xi$ is excluded.) By Theorem 118, we have

$$B(d, \xi) = 2^{\Omega(d)} \frac{\xi}{d} + \vartheta 2^{\Omega(d)}, \quad |\vartheta| < 1; \tag{54}$$

for n ranges over the interval $0 < n \leq \xi$ and, by Theorem 70, belongs to $2^{\Omega(d)}$ residue classes mod d (namely, two residue classes modulo each of the $\Omega(d)$ prime factors of $d = \prod_{p \mid d} p$; $p = 2$ does not occur, since d is odd, so that (53) does actually represent two distinct residue classes).

As regards the requisite determination of an upper bound for $A(\xi)$, I shall first show how it can *not* be done; afterwards, however, by going somewhat out of our way, we shall achieve our goal. This excursion will be of use to the reader who wants to learn not only what is already known, but also how to do research on his own.

Using the abbreviation

$$p_2 p_3 \cdots p_r = k,$$

I shall first prove the identity

$$A(\xi) = \sum_{d|k} \mu(d) B(d, \xi) = B(1, \xi) - B(p_2, \xi) - \cdots - B(p_r, \xi) + B(p_2 p_3, \xi) + \cdots + (-1)^r B(p_2 p_3 \cdots p_r, \xi); \quad (55)$$

then I shall show that it does not suffice, and I shall finally replace it by a useful inequality.

(55) is proved as follows: Every n in the interval $0 < n \leq \xi$ which is counted in $A(\xi)$ is also counted in $B(1, \xi)$ on the right, but not in any $B(d, \xi)$ for which $d > 1$, since for no p_h ($h = 2, \dots, r$) do we have $n \equiv 0$ or $-2 \pmod{p_h}$. Every n in the interval $0 < n \leq \xi$ which is not counted in $A(\xi)$ is $\equiv 0$ or -2 for b distinct primes p_h ($h = 2, \dots, r$), where $1 \leq b \leq r-1$; say for $p_{h_1}, \dots, p_{h_b}$ ($2 \leq h_1 < h_2 < \cdots < h_b \leq r$). Then this n is counted on the right-hand side of (55) in all of the terms $B(d, \xi)$ for which $d \mid p_{h_1} p_{h_2} \cdots p_{h_b}$; altogether, therefore,

$$\sum_{d \mid p_{h_1} \cdots p_{h_b}} \mu(d) = 0$$

times, by Theorem 35.

Thus (55) is proved; it does not, however, lead us to our goal. We may, of course, conclude from (55) and (54) that

$$\begin{aligned} A(\xi) &< \xi \sum_{d|k} \frac{\mu(d) 2^{\Omega(d)}}{d} + 2^r = \xi \prod_{2 < p \leq p_r} \left(1 - \frac{2}{p}\right) + 2^r, \\ A(\xi) &< \xi \prod_{2 < p \leq \eta} \left(1 - \frac{2}{p}\right) + 2^{\pi(\eta)}; \end{aligned} \quad (56)$$

(since for every h for which $0 \leq h \leq r-1$ there are exactly $\binom{r-1}{h}$ divisors of $k = p_2 \cdots p_r$ for which $\Omega(d) = h$). However, because of the fact that

$$\sum_{h=0}^{r-1} \binom{r-1}{h} = 2^{r-1},$$

we obtain not even a single estimate of the form

$$P(\xi) < a_{25} \frac{\xi}{\log \xi}$$

for the right-hand side of (56), no matter what our choice of $\eta(\xi)$.

(and even such an estimate would be useless; see the Remark above). For η must be so large that

$$\prod_{2 < p \leq p_r} \left(1 - \frac{2}{p}\right) = \prod_{2 < p \leq \eta} \left(1 - \frac{2}{p}\right) < \frac{a_{26}}{\log \xi} \quad (57)$$

holds and, at the same time, so small that

$$2^{\pi(\eta)} < \frac{a_{27} \xi}{\log \xi}$$

holds. This can never happen, however, for the following reasons. By Theorem 113 we would

have

$$\log p_r < a_{28} r \log r < a_{29} \eta \log \eta < a_{30} \left(\frac{\xi}{\log \xi} \right) \log \left(\frac{\xi}{\log \xi} \right) < a_{31} \frac{\xi \log \xi}{\log \xi} = a_{31} \xi; \quad (58)$$

by (57) we would have

$$\begin{aligned} \prod_{2 < p \leq p_r} \left(1 - \frac{2}{p} \right) &< \frac{a_{26}}{\log \xi}, \\ \sum_{2 < p \leq p_r} \log \left(1 - \frac{2}{p} \right) &< \log a_{26} - \log \log \xi, \\ - \sum_{2 < p \leq p_r} \left(\frac{2}{p} + \frac{4}{p^2} + \cdots \right) &< a_{32} - \log \log \xi, \\ a_{33} \log \log p_r &> a_{34} \log \log \xi - a_{35}, \end{aligned}$$

and therefore, by (46) and (58),

$$a_{34} \log \log \xi < a_{36} \log \log p_r < a_{37} \log \log(a_{31} \xi) < a_{38} \log \log \xi + a_{39};$$

for $\xi > a_{40}$ we would therefore have

$$\log \log \xi < a_{41} \log \log \log \xi,$$

which is certainly false for large ξ .

Now for our excursion!

Brun's essential idea lies in replacing equation (55) by the inequality

$$A(\xi) \leq \sum_{\substack{d|k \\ \Omega(d) < m}} \mu(d) B(d, \xi), \quad (59)$$

where m is an arbitrary odd positive integer (which will ultimately be taken as a function of ξ). (59) is seen as follows: Every n in the interval $0 < n \leq \xi$ which is counted in $A(\xi)$ is counted on the right-hand side exactly once (namely for $d = 1$), as was already remarked above; every n in the interval $0 < n \leq \xi$ which is not counted in $A(\xi)$ is counted on the right-hand side precisely in those terms $B(d, \xi)$ (where $d | k$) for which (in our old notation) $d | p_{h_1} \cdots p_{h_b}$, and for which, moreover, $\Omega(d) < m$. The complete count on the right for these n is therefore, by Theorem 116,

$$\sum_{\substack{d|p_{h_1} \cdots p_{h_b} \\ \Omega(d) < m}} \mu(d) = \sum_{j=0}^{m-1} (-1)^j \binom{b}{j} \geq 0.$$

Thus (59) is proved.

Now it follows further, from (59) and (54), that

$$A(\xi) \leq \xi \sum_{\substack{d|k \\ \Omega(d) < m}} \frac{\mu(d) 2^{\Omega(d)}}{d} + \sum_{\substack{d|k \\ \Omega(d) < m}} 2^{\Omega(d)}. \quad (60)$$

In this formula, on the one hand, we have

$$\begin{aligned} \sum_{\substack{d|k \\ \Omega(d) < m}} 2^{\Omega(d)} &\leq \sum_{h=0}^{m-1} \binom{r-1}{h} 2^h \leq \sum_{h=0}^{m-1} (2(r-1))^h \\ &\leq \sum_{h=0}^{m-1} (2r)^h = \frac{(2r)^m - 1}{2r - 1} < (2r)^m \leq (2\pi(\eta))^m. \end{aligned} \quad (61)$$

On the other hand, we have

$$\sum_{\substack{d|k \\ \Omega(d) < m}} \frac{\mu(d)2^{\Omega(d)}}{d} = \sum_{\substack{d|k \\ \Omega(d) < m}} \prod_{p|d} \left(1 - \frac{2}{p}\right) = \prod_{2 < p \leq p_r} \left(1 - \frac{2}{p}\right) + \sum_{\substack{d|k \\ \Omega(d) \geq m}} \prod_{p|d} \left(1 - \frac{2}{p}\right).$$

Now

$$\sum_{\substack{d|k \\ \Omega(d) \geq m}} \prod_{p|d} \left(1 - \frac{2}{p}\right) = \sum_{n=m}^{r-1} (-1)^n 2^n S_n,$$

where S_n is the n -th elementary symmetric function of $\frac{1}{p_2}, \dots, \frac{1}{p_r}$. By Theorem 117 and (46), we have (since $\sum_{h=0}^{\infty} \frac{x^h}{h!} = e^x$)

$$S_n \leq \frac{1}{n!} \left(\sum_{h=2}^r \frac{1}{p_h} \right)^n < \frac{1}{n!} (a_{42} \log \log \eta)^n.$$

Hence

$$\left| \sum_{n=m}^{r-1} (-1)^n 2^n S_n \right| \leq \sum_{n=m}^{r-1} \frac{(2a_{42} \log \log \eta)^n}{n!} \leq \sum_{n=m}^{\infty} \frac{(a_{43} \log \log \eta)^n}{n!}.$$

Now, for $n \geq m$,

$$\frac{(a_{43} \log \log \eta)^n}{n!} \leq \frac{(a_{43} \log \log \eta)^m}{m!} \left(\frac{a_{43} \log \log \eta}{m} \right)^{n-m},$$

and if $m > 2a_{43} \log \log \eta$, then

$$\left| \sum_{n=m}^{r-1} (-1)^n 2^n S_n \right| < \frac{(a_{43} \log \log \eta)^m}{m!} \sum_{t=0}^{\infty} \frac{1}{2^t} = 2 \frac{(a_{43} \log \log \eta)^m}{m!}. \quad (63)$$

From (62), (47), and (63), it follows that

$$\sum_{\substack{d|k \\ \Omega(d) < m}} \frac{\mu(d)2^{\Omega(d)}}{d} < \frac{a_{44}}{\log^2 \eta} + 2 \frac{(a_{43} \log \log \eta)^m}{m!}, \quad (64)$$

and from (52), (60), (64), and (61), it follows that

$$P(\xi) < \eta + \xi \left(\frac{a_{44}}{\log^2 \eta} + 2 \frac{(a_{43} \log \log \eta)^m}{m!} \right) + (2\pi(\eta))^m. \quad (65)$$

In these formulas, we let $\eta(\xi)$ and $m(\xi)$ be subject to the conditions

$$5 < \eta < \xi, \quad m > 2a_{43} \log \log \eta, \quad m \text{ odd.} \quad (66)$$

For $\xi > a_{45}$, these conditions are satisfied, provided I choose

$$\eta = \frac{\xi^{1/2}}{\log \xi}, \quad m = [2a_{43} \log \log \xi] + 1.$$

This is non-trivial only for the condition $m > 2a_{43} \log \log \eta$, but this condition is a consequence of the fact that, for $\xi > a_{46}$, we have

$$2a_{43} \log \log \eta = 2a_{43} \log \log \frac{\xi^{1/2}}{\log \xi} < 2a_{43} \log \log \xi - 3 < m.$$

For this choice of η and m , I obtain, using (65),

$$P(\xi) < a_{47} \left(\frac{\xi^{1/2}}{\log \xi} + \frac{\xi}{\log^2 \xi} + \xi \frac{(a_{48} \log \log \xi)^m}{m!} + \left(\frac{\xi^{1/2}}{\log \xi} \right)^m \right)$$

for $\xi > a_{49}$. Each of the four terms in the parentheses is

$$< a_{50} \frac{\xi}{(\log \log \xi)^2},$$

for we have

$$\frac{\xi^{1/2}}{\log \xi} = \frac{\xi}{(\log \log \xi)^2} \cdot \frac{(\log \log \xi)^2}{\xi^{1/2} \log \xi} < \frac{\xi}{(\log \log \xi)^2}$$

(since $\xi > 2$, $2 \log 2 > 1$),

$$\frac{\xi}{\log^2 \xi} = \frac{\xi}{(\log \log \xi)^2} \cdot \frac{(\log \log \xi)^2}{\log^2 \xi} < \frac{\xi}{(\log \log \xi)^2},$$

$$\frac{(a_{48} \log \log \xi)^m}{m!} < \left(\frac{ea_{48} \log \log \xi}{m} \right)^m < \left(\frac{e}{2} \right)^m < a_{51},$$

$$\left(\frac{\xi^{1/2}}{\log \xi} \right)^m < \xi^{m/2} < \xi^{a_{52} \log \log \xi} < \xi^{a_{53}}.$$

Hence, for $\xi > a_{54}$, we have

$$P(\xi) < a_{55} \frac{\xi}{(\log \log \xi)^2};$$

therefore, for $\xi \geq 3$, we have

$$P(\xi) < a_{24} \frac{\xi (\log \log \xi)^2}{\log^2 \xi}. \quad \square$$

Theorem 2.5. *If there are infinitely many primes p for which $p + 2$ is prime, then the series*

$$\sum_p \frac{1}{p},$$

summed over all of those p , converges.

Remark: This agrees, of course, with the wording used in the beginning of this chapter, even

though only the first prime of each pair is used in the sum; for we have $p > a_{14}$ and $\frac{1}{p} + \frac{1}{p+2} < \frac{2}{p}$.

Proof. I shall use Theorem 119 only in the weaker form

$$P(n) < a_{56} \frac{n}{\log^2 n} \quad \text{for } n \geq 3.$$

Then, if $p_1, p_2, \dots$ are the primes for which $p+2$ is also prime, we have for $r \geq 2$:

$$r = P(p_r) < a_{56} \frac{p_r}{\log^2 p_r},$$

$$\frac{1}{p_r} < \frac{a_{56}}{r \log^2 p_r} < \frac{a_{57}}{r \log^2 r},$$

and the series

$$\sum_{r=2}^{\infty} \frac{1}{r \log^2 r}$$

converges. □

Elementary Number Theory

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THEOREM 120 From this it follows that if p_r is the r -th prime for which $p_r + 2$ is also prime (if there are infinitely many such primes at all), then for all $r \geq 1$ we have

$$\frac{1}{p_r} < \frac{A_9}{\log^2 p_r} < \frac{A_9}{\log^2(r+1)},$$

and from the convergence of

$$\sum_{r=1}^{\infty} \frac{1}{r \log^2(r+1)}$$

it follows that

$$\sum_{r=1}^{\infty} \frac{1}{p_r}$$

converges.

CHAPTER III

DIRICHLET'S THEOREM ON THE PRIME NUMBERS OF AN ARITHMETIC PROGRESSION

§ 1 Further Theorems on Congruences

Definition 2.120. Let $m > 0$ and $(a, m) = 1$. We say that a “belongs to the exponent $f \bmod m$ ” if a^f is the first of all the powers $a^1, a^2, \dots$ of a with positive integral exponents for which

$$a^f \equiv 1 \pmod{m}.$$

(f exists, since by Fermat's Theorem we have $a^{\varphi(m)} \equiv 1$.)

THEOREM 2.121. *If a belongs to the exponent f , then, if $b_1 \geq 0$ and $b_2 \geq 0$,*

$$a^{b_1} \equiv a^{b_2} \pmod{m}$$

if and only if

$$b_1 \equiv b_2 \pmod{f}.$$

Proof: 1) From

$$a^{b_1} \equiv a^{b_2} \pmod{m}, \quad b_1 \geq b_2$$

it follows that

$$a^{b_2}(a^{b_1-b_2} - 1) \equiv 0, \quad a^{b_1-b_2} \equiv 1 \pmod{m},$$

so that, by the definition of f , we have

$$f \mid b_1 - b_2.$$

2) From

$$b_1 - b_2 = qf$$

it follows (since $a^f \equiv 1$) that

$$a^{b_1} = a^{b_2+qf} = a^{b_2}(a^f)^q \equiv a^{b_2} \pmod{m}.$$

THEOREM 2.122. *Let q be a prime, let $l > 0$, and let $q^l \mid p-1$. Then there exists a number a which belongs to $q^l \pmod{p}$.*

Proof: By Theorem 72, the congruence

$$x^{\frac{p-1}{q}} \equiv 1 \pmod{p}$$

has at most $\frac{p-1}{q} < p-1$ solutions (since $q \geq 2$). Hence there is at least one number c for which $1 \leq c \leq p-1$ such that

$$c^{\frac{p-1}{q}} \not\equiv 1 \pmod{p}.$$

Let us set

$$a = c^{\frac{p-1}{q^l}}.$$

Then

$$a^{q^l} = c^{p-1} \equiv 1 \pmod{p},$$

and if a belonged to a smaller exponent than q^l , say q^λ where $\lambda < l$, then it would follow that

$$c^{\frac{p-1}{q^{l-\lambda}}} = a^{q^\lambda} \equiv 1 \pmod{p},$$

and for $l - \lambda \geq 1$ this would contradict the choice of c .

THEOREM 2.123. *There exists a number belonging to $p - 1 \pmod{p}$ (a so-called primitive root mod p).*

Proof: Let $p - 1 = p_1^{l_1} \cdots p_r^{l_r}$. By Theorem 122 we may choose a number a_n belonging to $p_n^{l_n}$ for each $n = 1, \dots, r$. If we then set

$$g = \prod_{n=1}^r a_n,$$

then g belongs to f . By Theorem 121, we have $f \mid p - 1$. If we did not have $f = p - 1$ then, without loss of generality, we would have $f \mid \frac{p-1}{p_1}$ so that

$$1 \equiv g^{\frac{p-1}{p_1}} = a_1^{\frac{p-1}{p_1}} \prod_{n=2}^r a_n^{\frac{p-1}{p_1}} \equiv a_1^{\frac{p-1}{p_1}}$$

(since $p_n^{l_n} \mid \frac{p-1}{p_1}$ for $n = 2, \dots, r$), and consequently, by Theorem 121,

$$\frac{p-1}{p_1} \equiv 0 \pmod{p_1^{l_1}},$$

which is not the case.

Definition 2.124. Every g belonging to $p - 1 \pmod{p}$ is called a *primitive root mod p* .

THEOREM 124–125 I shall now prove that the g determined by (67) (which is, incidentally, a primitive root mod p) satisfies the condition of the theorem.

To this end, I shall first prove, by mathematical induction, that for all $l \geq 1$ we have

$$g^{p^{l-1}(p-1)} \equiv 1 + hp^l \pmod{p^{l+1}}, \quad p \nmid h. \quad (2.1)$$

For $l = 1$ this is assured by the Fermat Theorem, together with (67). If (2.1) holds for l , then it follows for $l + 1$ thus:

$$g^{p^l(p-1)} = (1 + hp^l)^p = 1 + \binom{p}{1}hp^l + \binom{p}{2}h^2p^{2l} + \cdots \equiv 1 + hp^{l+1} \pmod{p^{l+2}}.$$

THEOREM 2.125. *For $l > 2$, every odd number a satisfies the relation*

$$a \equiv (-1)^{\frac{a-1}{2}} 5^b \pmod{2^l}, \quad b \geq 0$$

for precisely those numbers $b \geq 0$ belonging to a particular residue class mod 2^{l-2} .

Proof: 1) Let $a \equiv 1 \pmod{4}$. For $0 \leq b < 2^{l-2}$, 5^b represents exactly 2^{l-2} numbers incongruent mod 2^l , by Theorem 125. All of these are $\equiv 1 \pmod{4}$; however, every reduced set of residues mod 2^l contains exactly 2^{l-2} numbers that are $\equiv 1 \pmod{4}$; hence it follows

$$a \equiv 5^b \pmod{2^l}.$$

2) Let $a \equiv 3 \pmod{4}$. Then $-a \equiv 1 \pmod{4}$, so that by 1) we have $-a \equiv 5^{b_1}$, $a \equiv (-1)5^{b_1}$; and $(-1) \equiv (-1)5^0$, so that by 1), (-1) corresponds to a definite residue class mod 2^{l-2} .

§ 2 Characters

i will now represent the familiar complex number, but all other lower case italic letters except e will still, as before, represent integers.

Let $k > 0$ be fixed. Let us set $\varphi(k) = h$.

Definition 2.126. A number-theoretic function $\chi(a)$ is called a *character* mod k , provided that

- I) $\chi(a) = 0$ for $(a, k) > 1$,
- II) $\chi(1) \neq 0$,
- III) $\chi(a_1 a_2) = \chi(a_1) \chi(a_2)$ for $(a_1, k) = 1$ and $(a_2, k) = 1$ (and therefore, by I), always),
- IV) $\chi(a_1) = \chi(a_2)$ for $a_1 \equiv a_2 \pmod{k}$ and $(a_1, k) = 1$ (and therefore, by I), whenever $a_1 \equiv a_2 \pmod{k}$).

THEOREM 2.127. *We always have*

$$\chi(1) = 1.$$

Proof: By II), $\chi(1) \neq 0$. By III) with $a_1 = a_2 = 1$ we have $\chi(1) = \chi(1)^2$, whence $\chi(1) = 1$.

THEOREM 2.128. *If $a \geq 1$, then*

$$\chi(a)^h = \chi(a^h) = \chi(1) = 1 \quad \text{for } (a, k) = 1.$$

Proof: For $(a, k) = 1$, by III) we have $\chi(a)^h = \chi(a^h)$. By Fermat's theorem (Theorem 77), $a^h \equiv 1 \pmod{k}$, so that by IV) and Theorem 127, $\chi(a^h) = \chi(1) = 1$.

THEOREM 2.129. *For every k there is a finite number of characters and, moreover, at least one.*

Proof: 1) For any a in the interval $1 \leq a \leq k$, it follows from I) and Theorem 128 that $\chi(a)$ must be chosen from a finite collection of possible values (0 or an h -th root of unity); from IV) it follows further that the value of $\chi(a)$ for $1 \leq a \leq k$ determines the character uniquely.

2) The function

$$\chi_0(a) = \begin{cases} 0 & \text{for } (a, k) > 1, \\ 1 & \text{for } (a, k) = 1 \end{cases}$$

is obviously a character (the so-called *principal character*).

Definition 2.130. The number of distinct characters mod k is denoted by c or, more explicitly, by $c(k)$.

THEOREM 2.131. $c \geq 1$.

Proof: This is contained in Theorem 129.

THEOREM 2.132. *If $\chi(a)$ is a character, then $\overline{\chi(a)}$ is also a character.*

THEOREM 2.133. *If $\chi_1(a)$ and $\chi_2(a)$ are characters, then $\chi_1(a)\chi_2(a)$ is also a character.*

In particular: If $\chi(a)$ is a character, then $\chi^2(a)$ is also a character.

THEOREM 2.134. *If $\chi_1(a)$ is a character, then as $\chi(a)$ runs over all c characters, so does $\chi(a)\chi_1(a)$.*

Proof: If

$$\chi_2(a)\chi_1(a) = \chi_3(a)\chi_1(a),$$

then it follows for $(a, k) = 1$, since $\chi_1(a) \neq 0$, that

$$\chi_2(a) = \chi_3(a),$$

and, by I), this is satisfied for $(a, k) > 1$ as well. The c functions $\chi(a)\chi_1(a)$ are therefore distinct characters; since there are only c characters altogether, these are all of them.

THEOREM 2.135. *We always have*

$$\sum_{a=1}^k \chi(a) = \begin{cases} h & \text{for the principal character,} \\ 0 & \text{for every other character.} \end{cases}$$

Proof: For the principal character, we have

$$\sum_{a=1}^k \chi_0(a) = h \cdot 1 + (k - h) \cdot 0 = h.$$

If $\chi(a)$ is not the principal character, then choose b so that $(b, k) = 1$, $\chi(b) \neq 1$; then as a runs through a complete residue system mod k , so does ab , and therefore

$$\chi(b) \sum_{a=1}^k \chi(a) = \sum_{a=1}^k \chi(ab) = \sum_{a=1}^k \chi(a),$$

so that, since $\chi(b) \neq 1$, we have $\sum_{a=1}^k \chi(a) = 0$.

THEOREM 2.136. *For every prime power p^l ($l > 0$) occurring in the canonical decomposition of k , and for every $h(p^l)$ -th root of unity $e \neq 1$, there exists a character for which*

$$\chi(g) = e$$

where g is a suitably chosen primitive root mod p^l (for $p = 2$, $l \geq 3$, one primitive root each mod 2^l of the two kinds occurring in Theorem 126).

Proof: The construction is divided into cases.

1) Let $d \equiv 1 \pmod{p^l}$, $p \nmid d$, and therefore $p^l \mid k$ (so that $l \geq 1$), $p > 2$. Then $d \equiv g^r \pmod{p^l}$ where r is uniquely determined mod $\varphi(p^l)$. The function

$$\chi(d) = e^r, \quad \chi(a) = 0 \text{ for } (a, k) > 1$$

is a character.

Verification:

II) $\chi(1) = e^0 = 1$;

III) for $(a_1, k) = 1$, $(a_2, k) = 1$, $a_1 \equiv g^{r_1}$, $a_2 \equiv g^{r_2} \pmod{p^l}$ we have $a_1 a_2 \equiv g^{r_1+r_2} \pmod{p^l}$,

$$\chi(a_1 a_2) = e^{r_1+r_2} = e^{r_1} e^{r_2} = \chi(a_1) \chi(a_2);$$

IV) is obvious; for $a_1 \equiv a_2 \pmod{k}$ yields $a_1 \equiv a_2 \pmod{p^l}$.

Finally, it follows from $d \equiv 1 \pmod{p^l}$ and $p \nmid d$, that $d \equiv g^r \pmod{p^l}$, $\varphi(p^l) \mid r$, so $\chi(d) = e^r = 1$.

2) Let $d \equiv 1 \pmod{2^l}$, $l > 0$, and $2^l \mid k$, and therefore $l \geq 1$ (since k is even, so that $d \equiv 1 \pmod{2}$).

2a) Let $d \equiv 1 \pmod{4}$, so that $l \geq 2$. Then

$$\chi(d) = (-1)^{\frac{d-1}{4}}$$

is a character, since conditions I)–IV) are obviously satisfied.

THEOREM 2.137. *For fixed $a > 0$,*

$$\sum_{\chi} \chi(a) = \begin{cases} c & \text{for } a \equiv 1 \pmod{k}, \\ 0 & \text{for } a \not\equiv 1 \pmod{k}, \end{cases}$$

where the sum is taken over all of the c characters.

Proof: 1) For $a \equiv 1 \pmod{k}$, the sum (as follows from Theorem 127) has c terms each of which = 1.

2) For $(a, k) > 1$ all of the terms vanish.

3) For $(a, k) = 1$ and $a \not\equiv 1 \pmod{k}$, choose a character χ_1 for which $\chi_1(a) \neq 1$ (possible by Theorem 134); then, by Theorem 133,

$$\chi_1(a) \sum_{\chi} \chi(a) = \sum_{\chi} \chi_1(a) \chi(a) = \sum_{\chi} \chi(a),$$

so that $\sum_{\chi} \chi(a) = 0$.

THEOREM 2.138.

$$\sum_x \chi(a) \overline{\chi(b)} = \begin{cases} c & \text{for } b \equiv a \pmod{k}, \\ 0 & \text{otherwise,} \end{cases}$$

where the sum is taken over all c characters.

Proof: Let us choose $l > 0$ in such a way that $lb \equiv 1 \pmod{k}$. Since $\overline{\chi(b)} = \chi(l)$, $\chi(l) \neq 0 = \chi(lb) = \chi(l)\chi(b)$, it follows that

$$\sum_x \chi(a) \overline{\chi(b)} = \sum_x \chi(a) \chi(l) = \sum_x \chi(al),$$

so that, by Theorems 135 and 136,

$$\sum_x \chi(a) \overline{\chi(b)} = \begin{cases} c & \text{for } al \equiv 1 \text{ i.e. } a \equiv l \pmod{k}, \\ 0 & \text{otherwise.} \end{cases}$$

Definition 2.139. $\chi(a)$ is called a character of the first kind if it is the principal character; of the second kind, if it is real but is not the principal character (so that its value is always 0, 1, or -1 , and -1 actually occurs); of the third kind, if it is not real (so that it also assumes non-real values).

§ 3 *L*-Series

From now on, the letter s need no longer stand for an integer.

THEOREM 2.140. *For each of the h characters mod k , the series*

$$F(s) = L(s, \chi) = \sum_{a=1}^{\infty} \frac{\chi(a)}{a^s} \quad (2.2)$$

is absolutely convergent for $s > 1$.

(The easily answered question as to whether these h functions of s are, moreover, distinct, is unimportant; as an Exercise, the reader may think it over.)

Proof: By Definition 23, I), and Theorem 128, we have

$$|\chi(a)| \leq 1,$$

from which the theorem follows because of the convergence of $\sum_{a=1}^{\infty} \frac{1}{a^s}$.

THEOREM 2.141. *For $s > 1$, $L(s, \chi)$ is a continuous function of s and*

$$L'(s, \chi) = - \sum_{a=1}^{\infty} \frac{\chi(a) \log a}{a^s}.$$

THEOREM 2.142. *If χ is not the principal character, then the series (71) converges for $s > 0$, and indeed uniformly for $s \geq \epsilon$ (for any $\epsilon > 0$).*

Proof: By Theorem 135, for $u \leq v$ we have

$$\left| \sum_{a=u}^v \chi(a) \right| = \left| \sum_{\chi} \chi(a) - \sum_{a=1}^{u-1} \chi(a) \right| \leq h.$$

The assertion therefore follows from Theorem 139.

THEOREM 2.143. *Let $u \leq v$, let v_a be an arbitrary complex number for each a satisfying $u \leq a \leq v$, and let*

$$\sum_{a=u}^w v_a = R(w) \quad \text{for } u \leq w \leq v, \quad \max_{u \leq w \leq v} |R(w)| = \rho.$$

Then if $u \leq w \leq v$ and $\epsilon_a \geq \epsilon_{a+1} \geq 0$, we have

$$\left| \sum_{a=u}^v \epsilon_a v_a \right| \leq \epsilon_u \rho.$$

Proof: Let $R(u-1)$ denote 0. Then we have

$$\sum_{a=u}^v \epsilon_a v_a = \sum_{a=u}^v \epsilon_a (R(a) - R(a-1)) = \sum_{a=u}^{v-1} R(a) (\epsilon_a - \epsilon_{a+1}) + R(v) \epsilon_v,$$

so that

$$\left| \sum_{a=u}^v \epsilon_a v_a \right| \leq \rho \left(\sum_{a=u}^{v-1} (\epsilon_a - \epsilon_{a+1}) + \epsilon_v \right) = \epsilon_u \rho.$$

THEOREM 2.144. *If χ is not the principal character, then the series (71) converges uniformly for $s \geq 1$, and for these values of s its sum is $< h$ in absolute value.*

Proof: 1) Let $s \geq 1 + \epsilon$. Since $\frac{\log a}{a^s} \leq \frac{\log a}{a^{1+\epsilon}}$ and $\sum_{a=1}^{\infty} \frac{\log a}{a^{1+\epsilon}}$ converges.

2) This follows from (71) and 1).

THEOREM 2.145. *If χ is not the principal character, then the series (73) converges uniformly for $s \geq 1$, and for these values of s its sum is $< h$ in absolute value.*

Proof: 1) Let $s \geq 1$. Since $\frac{d}{ds} \frac{\log a}{a^s} = \frac{1-s \log a}{a^s}$ is a decreasing function for $s > \frac{1}{\log a}$ and, since $1 > \frac{1}{\log 2}$, $\frac{\log a}{a^s}$ is decreasing for $a \geq 3$; hence, by Theorems 139 and 140, we have, for $v \geq u \geq 3$,

$$\left| \sum_{a=u}^v \frac{\chi(a) \log a}{a^s} \right| \leq \frac{\log u}{u^s} h \leq \frac{\log u}{u} h.$$

THEOREM 2.146. *For $s > 1$, we have*

$$\prod_p \frac{1}{1 - \frac{\chi(p)}{p^s}} = L(s, \chi).$$

Proof: From the absolute convergence of both series on the left-hand side of the next formula, together with Theorem 35, we have

$$\prod_p \frac{1}{1 - \frac{\chi(p)}{p^s}} = \prod_p \sum_{m=0}^{\infty} \frac{\chi(p^m)}{p^{ms}} = \sum_{a=1}^{\infty} \frac{\chi(a)}{a^s} = L(s, \chi).$$

THEOREM 2.147. *For $s > 1$, we have*

$$\log L(s, \chi) = \sum_p \frac{\chi(p)}{p^s} + \sum_p \sum_{m=2}^{\infty} \frac{\chi(p^m)}{mp^{ms}}.$$

THEOREM 2.148. *As $s \rightarrow 1$ (from the right),*

$$\frac{L'(s, \chi_0)}{L(s, \chi_0)} + \sum_p \frac{1}{p^s} \rightarrow \infty.$$

Proof: By Theorem 147, we have

$$\frac{L'(s, \chi_0)}{L(s, \chi_0)} = - \sum_p \frac{\chi_0(p) \log p}{p^s - 1} = - \sum_{p \nmid k} \frac{\log p}{p^s - 1} = - \sum_{p \nmid k} \frac{\log p}{p^s} - \sum_{p \nmid k} \frac{\log p}{p^s(p^s - 1)}$$

As $s \rightarrow 1$, the second term approaches a finite value. Thus we merely have to show that the first term approaches infinity.

I should like to give two proofs.

1) If Theorem 147 is applied to $k = 1$, it follows that the first term

$$-\sum_p \frac{\log p}{p^s - 1} = \frac{\zeta'(s)}{\zeta(s)} \rightarrow -\infty$$

as $s \rightarrow 1$.

2) We use Theorem 114.

THEOREM 2.149. *For $0 < \eta < 1$ and $\eta \neq 0$, we have*

$$(1 - \eta)^3 |1 - \eta e^{i\varphi}|^4 |1 - \eta e^{2i\varphi}|^2 \leq 1.$$

Proof: The geometric mean of three positive numbers is at most equal to their arithmetic mean; therefore, since

$$2 \cos \varphi + \cos 2\varphi = 2 \cos^2 \varphi + 2 \cos \varphi - 1 = 2 \left(\cos \varphi + \frac{1}{2} \right)^2 - \frac{3}{2} \geq -\frac{3}{2},$$

we have

$$\begin{aligned} |1 - \eta e^{i\varphi}|^4 |1 - \eta e^{2i\varphi}|^2 &= (1 - 2\eta \cos \varphi + \eta^2)^2 (1 - 2\eta \cos 2\varphi + \eta^2) \\ &\leq \left(\frac{2(1 - 2\eta \cos \varphi + \eta^2) + (1 - 2\eta \cos 2\varphi + \eta^2)}{3} \right)^3 \leq \left(\frac{3 + 3\eta^2}{3} \right)^3 = (1 + \eta^2)^3 \end{aligned}$$

THEOREM 2.150. *For $s > 1$ we have*

$$(L(s, \chi_0))^3 |L(s, \chi)|^4 |L(s, \chi^2)|^2 \geq 1.$$

(By Theorem 132, χ^2 is a character.)

Proof: In Theorem 149, let me set

$$\chi(p) = e^{i\varphi}, \quad \eta = \frac{1}{p^s}$$

for $p \nmid k$. This yields

$$\left(1 - \frac{\chi_0(p)}{p^s}\right)^3 \left|1 - \frac{\chi(p)}{p^s}\right|^4 \left|1 - \frac{\chi^2(p)}{p^s}\right|^2 \leq 1;$$

this holds also if $p \mid k$ ($l = 1$). Multiplication with respect to p , by Theorem 146, yields the result of the theorem.

THEOREM 2.151. *If χ is of the second or third kind, then*

$$L(1, \chi) \neq 0.$$

Proof: From Theorem 150, for $s > 1$,

$$\left| \frac{1}{L(s, \chi_0)} \right|^3 |L(s, \chi)|^4 |L(s, \chi^2)|^2 \geq 1.$$

On the other hand, for $1 < s < 2$, we have

$$|L(s, \chi_0)| = \left| \sum_{\substack{a=1 \\ (a,k)=1}}^{\infty} \frac{1}{a^s} \right| < \sum_{a=1}^{\infty} \frac{1}{a^s} < 1 + \int_1^{\infty} \frac{da}{a^s} = 1 + \frac{1}{s-1} = \frac{s}{s-1} < \frac{2}{s-1}$$

Hence

$$|L(s, \chi)|^4 \geq \frac{(s-1)^3}{8} |L(s, \chi^2)|^{-2}.$$

If we had $L(1, \chi) = 0$, then it would follow by Theorem 143 that, for $s > 1$,

$$|L(s, \chi)| = |L(s, \chi) - L(1, \chi)| = \left| \int_1^s L'(t, \chi) dt \right| < h(s-1).$$

Hence, for $1 < s < 2$, we would have

$$(s-1)^4 > \frac{(s-1)^3}{8h^2}.$$

This, however, is false for $s = 1 + \frac{1}{128h^2}$.

THEOREM 2.152. *For every character of the second kind we have*

$$L(1, \chi) \neq 0.$$

This is the deepest of all of the lemmas that are necessary for Dirichlet's proof. Dirichlet proved it only by the considerably roundabout method of using the so-called theory of the class number of quadratic forms.

Proof: Let us set, for $a > 0$,

$$H(a) = \sum_{d|a} \chi(d).$$

Then $H(a) \geq 0$ always, and $H(a) \geq 1$ if a is a perfect square. For

$$\chi(p) = 0, \ 1, \text{ or } -1,$$

and

$$1 + 1 + \cdots + 1 = t + 1 \text{ for } \chi(p) = 1,$$

$$1 - 1 + \cdots + 1 \text{ or } 0 \text{ for } \chi(p) = -1, \ 2 \nmid t,$$

$$1 - 1 + \cdots + (-1)^t = \begin{cases} 1 & \text{for } 2 \mid t, \\ 0 & \text{for } 2 \nmid t. \end{cases}$$

Hence we have

$$H(a) \begin{cases} = 0 & \text{always,} \\ \geq 1 & \text{if } a \text{ is a square.} \end{cases}$$

Let $a_1 > 0$, $a_2 > 0$, and $(a_1, a_2) = 1$. There is a one-to-one correspondence between all positive numbers $d|a_1a_2$, and all products of positive numbers $d_1|a_1$ by positive numbers $d_2|a_2$. We therefore have

$$H(a_1a_2) = \sum_{d|a_1a_2} \chi(d) = \sum_{d_1|a_1} \sum_{d_2|a_2} \chi(d_1d_2) = \sum_{d_1|a_1} \chi(d_1) \sum_{d_2|a_2} \chi(d_2) = H(a_1)H(a_2) \quad (2.3)$$

From (75) and (76) it follows that

$$H(a) \begin{cases} = 0 & \text{always,} \\ \geq 1 & \text{for quadratic } a. \end{cases}$$

Let us set

$$m = (4h)^8$$

and

$$T = \sum_{a=1}^m H(a) = \sum_{a=1}^m \sum_{d|a} \chi(d) = \sum_{b=1}^{\sqrt{m}} \chi(b) \sum_{\substack{a=1 \\ b^2|a}}^m 1 = \sum_{b=1}^{\sqrt{m}} \chi(b) \left[\frac{m}{b^2} \right]$$

for the sake of brevity. Then, by (77), we have

$$T \geq \sum_{b=1}^{\sqrt{m}} \chi(b) \left(\frac{m}{b^2} - 1 \right) = m \sum_{b=1}^{\sqrt{m}} \frac{\chi(b)}{b^2} - \sum_{b=1}^{\sqrt{m}} \chi(b). \quad (2.4)$$

On the other hand (and the reader should verify this), from Theorem 139 and Theorem 140 (with d in place of a , $y_d = \chi(d)$, and $v_a = 2(m - ab)$ for $m/b < b$, $v_a = 1$, $v_a = 2m$), it follows that

$$\max_{1 \leq a \leq \sqrt{m}} \left| \sum_{b=1}^{\sqrt{m}} 2(m - ab) \chi(b) \right| < L \cdot 2m \cdot 2m^{1/4} \cdot h. \quad (2.5)$$

THEOREM 2.153. *For any character of the second or third kind,*

$$\frac{L'(s, \chi)}{L(s, \chi)}$$

is bounded for $s > 1$.

(The bound may of course depend on k and χ . However, χ need not be mentioned here, since for every k there are only $\varphi(k)$ possible values of χ .)

Proof: By Theorem 141, $L(s, \chi)$ is continuous for $s \geq 1$; by Theorem 145, it is never 0 for $1 < s \leq 2$; and by Theorems 151 and 152, it is never 0 for $s = 1$. Hence $\frac{L'(s, \chi)}{L(s, \chi)}$ is bounded for $1 \leq s \leq 2$; and for $s > 2$ the same is true, by Theorems 144 and 145. Finally, by Theorem 143, $L'(s, \chi)$ is bounded for $s = 1$.

§ 4 Dirichlet's Proof

THEOREM 2.154. *Let $(l, k) = 1$ and $l > 0$. Then for $s > 1$ we have*

$$\frac{1}{h} \sum_{\chi} \frac{\overline{\chi(l)}}{\log L(s, \chi)} = \sum_{p \equiv l \pmod{k}} \frac{1}{p^s} + g(s) \quad (2.6)$$

where $g(s)$ is bounded as $s \rightarrow 1$.

(The term on the right is summed over all numbers $p \equiv l \pmod{k}$ in increasing—or, of course, arbitrary—order. The fact that not all of the terms vanish will later follow from Theorem 155 and is, for the moment, irrelevant.)

Proof: By Theorems 147 and 137, we have

$$\frac{1}{h} \sum_{\chi} \overline{\chi(l)} \log L(s, \chi) = \frac{1}{h} \sum_{\chi} \sum_p \frac{\overline{\chi(l)} \chi(p)}{p^s} + O(1) = \sum_{p \equiv l \pmod{k}} \frac{1}{p^s} + O(1).$$

THEOREM 2.155. *Let $(l, k) = 1$. Then there exist infinitely many primes $p \equiv l \pmod{k}$.*

Proof: Without loss of generality, let $l > 0$. As $s \rightarrow 1$, the left-hand side of (82) approaches ∞ ; for the term of the sum that involves χ_0 approaches $-\infty$, by Theorem 148, and the other $h-1$ terms remain bounded, by Theorem 153. Hence we have

$$\sum_{p \equiv l \pmod{k}} \frac{1}{p} = \infty.$$

PART THREE

DECOMPOSITION INTO TWO, THREE, AND FOUR SQUARES

Introduction

These investigations really belong to the most elementary parts of number theory. I go into them only now, because at one point I intend to apply Dirichlet's Theorem on Arithmetic Progressions. Part Three will answer the questions ($n > 0$):

1) When can n be written as the sum of two squares?

Answer (Theorem 164): If and only if n has no prime factor $p \equiv 3 \pmod{4}$ that occurs with odd multiplicity.

2) When this is the case, how many solutions does the diophantine equation

$$n = x^2 + y^2$$

have?

Answer (Theorem 163): The number of solutions (and this is true even in the case of unsolvability) equals four times the excess of the number of positive $d|n$ of the form $d \equiv 1 \pmod{4}$ over the number of positive $d|n$ of the form $d \equiv 3 \pmod{4}$.

3) When can n be written as the sum of three squares?

Answer (Theorem 180): If and only if n is not of the form $4^a(8b+7)$.

4) When can n be written as the sum of four squares?

Answer (Theorem 169): Always.

The question that immediately suggests itself in consequence of 4), as to whether a finite number of summands (depending only on k) always suffices also for third, fourth, . . . , and, in general, k -th powers of integers ≥ 0 (the qualification ' ≥ 0 ' is unnecessary if k is even), will constitute one of the topics of Part Six of *Vorlesungen über Zahlentheorie* and is one of the most important that I have to discuss with my readers; not only will this question be answered there affirmatively (by the Hilbert Theorem), but we shall also delve a good deal further into the subject (the Hardy-Littlewood Theorem).

CHAPTER I

FAREY FRACTIONS

This subject, which is over 100 years old, has more recently shown itself to be extraordinarily useful in the development of number theory. The reader will first come across its main applications in Parts Five and Six of *Vorlesungen über Zahlentheorie*.

Definition 2.156. For a fixed number $n > 0$, let all of the reduced fractions with positive denominators $\leq n$, that is, all of the rational numbers

$$\frac{a}{b}, \quad (a, b) = 1, \quad 0 < b \leq n,$$

be arranged in increasing order of magnitude; the sequence thus obtained is called the *Farey sequence* belonging to n .

THEOREM 2.157. *If $\frac{a}{b}$ and $\frac{a'}{b'}$ are two successive terms in the Farey sequence belonging to n , then*

$$ba' - ab' = 1.$$

Proof: By symmetry, we may let $\frac{a}{b} < \frac{a'}{b'}$.

By Theorem 68, we can determine numbers x and y corresponding to a and b for which

$$ba - ay = 1, \quad n - b < y \leq n; \quad (2.7)$$

for we have $(b, -a) = 1$ and $b > 0$, so that there is a number y in the complete set of residues mod b indicated above and then a suitable x corresponding to y . Then we have

$$\frac{x}{y} = \frac{a}{b} + \frac{1}{by} > \frac{a}{b}, \quad (x, y) = 1, \quad y \leq n.$$

If I succeed in showing that

$$\frac{x}{y} = \frac{a'}{b'},$$

then I shall be through. For then

$$\frac{a'}{b'} = \frac{x}{y}, \quad y \mid b', \quad b' \mid y, \quad y = b', \quad a' = x,$$

and therefore, by (83),

$$ba' - ab' = 1.$$

Suppose that

$$\frac{x}{y} \neq \frac{a'}{b'}.$$

Then, since $\frac{a'}{b'} \leq \frac{x}{y}$, we would have

$$\frac{a'}{b'} < \frac{x}{y}.$$

Likewise (since $\frac{a}{b} < \frac{a'}{b'}$), we would have

$$\frac{a'}{b'} - \frac{a}{b} = \frac{ba' - ab'}{bb'} \geq \frac{1}{bb'}.$$

By addition, and by the use of (83) and the fact that $b' \leq n$, we would obtain

$$\frac{1}{bb'} \leq \frac{a'}{b'} - \frac{a}{b} = \frac{ba' - ab'}{bb'} = \frac{1}{b'y} \leq \frac{1}{b'(n - b + 1)} \leq \frac{1}{bb'},$$

which is a contradiction.

Definition 2.158. If $\frac{a}{b}$ and $\frac{a'}{b'}$ are two successive terms in the Farey sequence belonging to n , then $\frac{a+a'}{b+b'}$ is called their *mediant*.

THEOREM 2.159. *The mediant $\frac{a+a'}{b+b'}$ lies between $\frac{a}{b}$ and $\frac{a'}{b'}$ (and thus is certainly not a term in the Farey sequence); its distance from $\frac{a}{b}$ and $\frac{a'}{b'}$ is $\frac{1}{b(b+b')}$ and $\frac{1}{b'(b+b')}$, respectively.*

Proof: Without loss of generality, let $\frac{a}{b} < \frac{a'}{b'}$; then, by Theorem 156,

$$\frac{a+a'}{b+b'} - \frac{a}{b} = \frac{ba' - ab'}{b(b+b')} = \frac{1}{b(b+b')} > 0,$$

$$\frac{a'}{b'} - \frac{a+a'}{b+b'} = \frac{ba' - ab'}{b'(b+b')} = \frac{1}{b'(b+b')} > 0.$$

THEOREM 2.160. *Given any number $n > 0$ and any real number ξ , there exists a fraction $\frac{a}{b}$ such that*

$$(a, b) = 1, \quad 0 < b \leq n, \quad \left| \frac{a}{b} - \xi \right| \leq \frac{1}{b(n+1)}.$$

THEOREM 2.161. *Given $n \geq 1$ and $\xi \geq 0$, there exists a fraction $\frac{a}{b}$ such that*

$$(a, b) = 1, \quad 0 < b \leq \sqrt{n}, \quad \left| \frac{a}{b} - \xi \right| < \frac{1}{b\sqrt{n}}.$$

Proof: Theorem 158, with $n = \lfloor \sqrt{n} \rfloor$.

CHAPTER II

DECOMPOSITION INTO TWO SQUARES

The letters n, x, y, m, d, d_1 , and d_2 will in this chapter always represent positive numbers.

THEOREM 2.162. *If*

$$n > 1, \quad l^2 \equiv -1 \pmod{n},$$

then

$$n = x^2 + y^2, \quad x > 0, \quad y > 0, \quad (x, y) = 1, \quad y \equiv lx \pmod{n} \quad (2.8)$$

is always solvable, and uniquely so.

Proof: 1) (Solvability.) By Theorem 159 with $n = \sqrt{n}$, corresponding to given m and l there exist two numbers a and b for which

$$(a, b) = 1, \quad 0 < b \leq \sqrt{n}, \quad \left| \frac{a}{b} - \frac{l}{n} \right| < \frac{1}{b\sqrt{n}}.$$

If we set

$$lb + na = c,$$

then it follows that

$$c \equiv lb \pmod{n}, \quad |c| < \sqrt{n},$$

so that

$$0 < b^2 + c^2 < 2n.$$

Since

$$b^2 + c^2 \equiv b^2 + l^2 b^2 \equiv (1 + l^2)b^2 \equiv 0 \pmod{n},$$

it follows that

$$b^2 + c^2 = n.$$

Furthermore, we have $(b, c) = 1$; for from

$$n = b^2 + (lb + na)^2 = (1 + l^2)b^2 + 2lnba + n^2a^2$$

it follows that

$$1 = \frac{1 + l^2}{n}b^2 + 2lba + na^2 = ub + ac.$$

Furthermore $c \neq 0$; for otherwise we would have $b^2 = n > 1$ and $(b, c) > 1$.

In the case $c > 0$, the choice

$$x = b, \quad y = c$$

accomplishes what we want.

In the case $c < 0$, the choice

$$x = -c, \quad y = b$$

does it. For,

$$n = (-c)^2 + b^2, \quad -c > 0, \quad b > 0, \quad (-c, b) = 1, \quad b \equiv -lb \equiv -lc \equiv l(-c)$$

2) (Uniqueness.) Let x_1, y_1 and x_2, y_2 satisfy the conditions in (84). Then we have

$$n^2 = (x_1^2 + y_1^2)(x_2^2 + y_2^2) = (x_1x_2 + y_1y_2)^2 + (x_1y_2 - y_1x_2)^2,$$

$$x_1x_2 + y_1y_2 \equiv (1 + l^2)x_1x_2 \equiv 0 \pmod{n},$$

so that, since $x_1x_2 + y_1y_2 > 0$, we have

$$x_1x_2 + y_1y_2 = n, \quad x_1y_2 - y_1x_2 = 0,$$

$$y_1 = (x_2^2 + y_2^2)y_1 = x_2(x_1x_2 + y_1y_2) - x_1(x_1y_2 - y_1x_2) = nx_2,$$

$$y_1 = ny_2, \quad x_1 = x_2, \quad y_1 = y_2.$$

THEOREM 2.163. *Let $V(n)$ be the number of solutions of*

$$l^2 \equiv -1 \pmod{n}.$$

Then the number of solutions of

$$n = x^2 + y^2, \quad (x, y) = 1 \tag{2.9}$$

equals $4V(n)$.

Proof: 1) For $n = 1$ the statement is trivial; we have

$$V(1) = 1,$$

and the four decompositions read

$$1 = (\pm 1)^2 + 0^2 = 0^2 + (\pm 1)^2.$$

2) For $n > 1$, we necessarily have $x \neq 0$ and $y \neq 0$ (since $(x, y) = 1$), and therefore the number of solutions of (86) must be four times the number of solutions with the supplementary conditions that $x > 0$ and $y > 0$. It follows from Theorem 160 that for each l satisfying (85) there exists a solution of (86) for which

$$x > 0, \quad y > 0, \quad \text{and} \quad y \equiv lx \pmod{n}.$$

Conversely, every solution of (86) for which $x > 0$ and $y > 0$ yields exactly one $l \pmod{n}$ satisfying (85) for which

$$y \equiv lx \pmod{n}. \quad (2.10)$$

For since $(x, n) = 1$, we have $(x, n) = 1$, and therefore (87) is uniquely solvable for $l \pmod{n}$, so that

$$0 \equiv n \equiv x^2 + y^2 \equiv x^2 + l^2 x^2 \equiv (1 + l^2)x^2 \pmod{n},$$

$$0 \equiv 1 + l^2 \pmod{n}.$$

THEOREM 2.164. *The number $U(n)$ of solutions of*

$$n = x^2 + y^2$$

is a multiplicative function:

$$U(n_1 n_2) = U(n_1)U(n_2) \quad \text{for } (n_1, n_2) = 1.$$

THEOREM 2.165.

$$U(n) = 4 \sum_{d|n} \chi(d), \quad (2.11)$$

where $\chi(d)$ is the non-principal character mod 4 (see Part Two, Chapter III, § 2), that is,

$$\chi(d) = \begin{cases} 0 & \text{for } d \equiv 0 \pmod{2}, \\ 1 & \text{for } d \equiv 1 \pmod{4}, \\ -1 & \text{for } d \equiv 3 \pmod{4}. \end{cases}$$

Written differently,

$$U(n) = 4 \sum_{\substack{d|n \\ d \text{ odd}}} (-1)^{\frac{d-1}{2}}.$$

Remark: Thus, this theorem also justifies the wording of the answer to Question 2) in the Introduction.

Proof: If $(n_1, n_2) = 1$, then by Theorem 71 we have

$$V(n_1 n_2) = V(n_1) V(n_2);$$

so that by Theorem 162 we have

$$U(n_1 n_2) = 4V(n_1 n_2) = 4V(n_1) V(n_2) = U(n_1) U(n_2).$$

If we let

$$W(n) = 4 \sum_{d|n} \chi(d) = U(n),$$

then $W(n)$ also has the property that

$$W(n_1 n_2) = W(n_1) W(n_2).$$

It therefore suffices, since (89) is obvious for $n = 1$ ($4 = 4 \cdot 1$), to prove (89) for $n = p^l$, $l > 0$, where this statement now reads:

$$\frac{U(p^l)}{4} = \sum_{m=0}^l \chi(p^m).$$

We do, in fact, have $V(1) = 1$, and by Theorem 87 (the value of $\left(\frac{-1}{p}\right)$ being -1 here), or else by the Remark of Theorem 161, we also have

$$V(p^m) = \begin{cases} 1 & \text{for } p = 2, \ m = l, \\ 1 & \text{for } p = 2, \ m > 1, \\ 0 & \text{for } p \equiv 3 \pmod{4}, \ m > 0, \\ 2 & \text{for } p \equiv 1 \pmod{4}, \ m > 0. \end{cases}$$

It follows from Theorem 162 that, for even l ,

$$\frac{U(p^l)}{4} = V(1) + V(p) + \cdots + V(p^l) = \begin{cases} 1 & \text{for } p = 2, \\ 2 \cdot \frac{l}{2} + 1 = l + 1 & \text{for } p \equiv 1 \pmod{4}, \\ 1 & \text{for } p \equiv 3 \pmod{4}, \end{cases} \quad (2.12)$$

and that, for odd l ,

$$\frac{U(p^l)}{4} = V(1) + V(p) + \cdots + V(p^l) = \begin{cases} 1 & \text{for } p = 2, \\ 1 + 1 + \cdots + 1 = l + 1 & \text{for } p \equiv 1 \pmod{4}, \\ 0 & \text{for } p \equiv 3 \pmod{4}, \end{cases} \quad (2.13)$$

on the other hand, it follows by definition that

$$\sum_{m=0}^l \chi(p^m) = \begin{cases} 0 + \cdots + 0 + 1 = 1 & \text{for } p = 2, \\ 1 + \cdots + 1 + 1 = l + 1 & \text{for } p \equiv 1 \pmod{4}, \\ 1 + (-1) + \cdots + (-1) = 0 & \text{for } p \equiv 3 \pmod{4}, \ 2 \mid l, \\ -1 + 1 - \cdots + 1 = 0 & \text{for } p \equiv 3 \pmod{4}, \ 2 \nmid l. \end{cases}$$

THEOREM 2.166.

$$U(n) = \begin{cases} 0, & \text{if there is a prime } p \equiv 3 \pmod{4} \text{ that divides } n \text{ with} \\ & \text{(precisely) odd multiplicity,} \\ T(n) & \text{otherwise,} \end{cases}$$

where $T(n)$ is the product of the powers of the primes $p|n$ of the form $p \equiv 1 \pmod{4}$ appearing in the canonical decomposition of n .

Proof: For $n = 1$ the statement is obvious ($1 = 1^2$). For $(n_1, n_2) = 1$ the equation

$$U(n_1 n_2) = U(n_1) U(n_2)$$

holds for $U(n)$ as well as (obviously) for the right-hand side of the statement to be proved. It therefore suffices to prove the statement for $n = p^l$, $l > 0$. In this case, it follows from (91) and (92) that we do, in fact, have

$$U(2^l) = 4 \cdot 1 = T(2^l) \quad \text{for } p = 2,$$

$$\frac{U(p^l)}{4} = l + 1 = T(p^l) \quad \text{for } p \equiv 1 \pmod{4},$$

$$\frac{U(p^l)}{4} = 0 = T(p^l) \quad \text{for } p \equiv 3 \pmod{4}, \quad 2 \nmid l,$$

$$\frac{U(p^l)}{4} = 0 \quad \text{for } p \equiv 3 \pmod{4}, \quad 2 \nmid l.$$

THEOREM 2.167. *Every prime $p \equiv 1 \pmod{4}$ can be written as a sum of two squares, and indeed uniquely so (up to the order of the components).*

CHAPTER III

DECOMPOSITION INTO FOUR SQUARES

Introduction

I shall consider the diophantine equation

$$n = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

before the equation

$$n = x_1^2 + x_2^2 + x_3^2,$$

for the proof of the solvability of the former equation, as well as the determination of the number of its roots, is easier than the answer, which follows in Chap. IV, to the question as to when the latter equation is solvable.

§ 1 Lagrange's Theorem

The following proof could be shortened somewhat if we were to bring in some of the above theorems on quadratic residues and on decomposition into a sum of two squares. However, the presentation of an as brief as possible direct proof, however, seems to me to be of interest.

THEOREM 2.168 (Euler's identity).

$$\begin{aligned}
 & (x_1^2 + x_2^2 + x_3^2 + x_4^2)(y_1^2 + y_2^2 + y_3^2 + y_4^2) \\
 &= (x_1y_1 + x_2y_2 + x_3y_3 + x_4y_4)^2 + (x_1y_2 - x_2y_1 + x_3y_4 - x_4y_3)^2 \\
 &+ (x_1y_3 - x_3y_1 + x_4y_2 - x_2y_4)^2 + (x_1y_4 - x_4y_1 + x_2y_3 - x_3y_2)^2.
 \end{aligned}
 \tag{2.14}$$

(We shall use (93) only when $x_1, \dots, y_4$ are integers; this identity holds, however, for arbitrary complex numbers.)

Proof: Let us verify the assertion. On the left, after multiplying out, we have sixteen expressions of the form $2x_\alpha y_\beta$ ($\alpha = 1, \dots, 4$; $\beta = 1, \dots, 4$). These also appear, among others, in the squares on the right; and the coefficients on both sides agree.

THEOREM 2.169. *Let $p > 2$. Then there exist numbers x_1, x_2, x_3, x_4 , no two of which are congruent mod p , such that*

$$mp = x_1^2 + x_2^2 + x_3^2 + x_4^2, \quad 0 < m < p.$$

Remark: Without the condition $m < p$ this would be trivial:

$$p^2 = p^2 + 0^2 + 0^2 + 0^2;$$

but it would also be quite useless.

Proof: The numbers $2x_1$, $0 \leq x_1 < \frac{p+1}{2}$ are incongruent to each other (mod p) in pairs (for from $4x_1^2 \equiv 4x_2^2$ it would follow that $p|(4x_1 - 4x_2)(4x_1 + 4x_2)$ and $x_1 \equiv \pm x_2 \pmod{p}$); the same is true of the $\frac{p+1}{2}$ numbers $-1 - y^2$, $0 \leq y < \frac{p+1}{2}$. Therefore, since this makes $p + 1$ numbers altogether, and since there are only p residue classes mod p , it follows (pigeon-hole principle!) that there exists a pair x, y for which

$$4x^2 \equiv -1 - y^2 \pmod{p}, \quad |x| < \frac{p}{2}, \quad |y| < \frac{p}{2}.$$

Then we have

$$4x^2 + y^2 + 1 = mp, \quad 0 < mp < \frac{p^2}{4} + \frac{p^2}{4} + 1 = \frac{p^2}{2} + 1 < p^2.$$

THEOREM 2.170. *For every prime p ,*

$$mp = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

is solvable.

Proof: For $p = 2$ this is obvious ($2 = 1^2 + 1^2 + 0^2 + 0^2$); therefore let $p > 2$. Let $m = m(p)$ be the smallest positive number for which

$$mp = x_1^2 + x_2^2 + x_3^2 + x_4^2 \tag{2.15}$$

is solvable. By Theorem 167, $m < p$. We assert that

$$m = 1.$$

In any case, m is odd; for otherwise it would follow from (94) that

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 \equiv 0 \pmod{2},$$

so that, without loss of generality,

$$x_1 + x_2 \equiv 0, \quad x_3 + x_4 \equiv 0 \pmod{2}$$

(that is, if the x_i were all odd or all even, this would certainly be true; and if two were odd and two even, the same holds).

We may therefore set

$$y_i \equiv x_i \pmod{m}, \quad |y_i| < \frac{m}{2} \quad (i = 1, 2, 3, 4),$$

where not all $y_i = 0$ (for otherwise $m^2 | x_i^2$ for every i , $m^2 | mp$, $m | p$, contradicting $1 < m < p$). Furthermore we have $n < m$, for by (95),

$$mn = y_1^2 + y_2^2 + y_3^2 + y_4^2 < 4 \left(\frac{m}{2} \right)^2 = m^2.$$

From (94) and (95), it follows, by (93), that

$$m^2 np = (\text{right-hand side of (93)}). \quad (2.16)$$

Each parenthesized term on the right-hand side of (93) is $\equiv 0 \pmod{m}$. For the first one is

$$\sum x_i y_i = \sum x_i^2 = mp \equiv 0 \pmod{m},$$

and for the other three terms we merely have to notice, twice for each term, that

$$x_i y_j - x_j y_i \equiv x_i x_j - x_j x_i = 0 \pmod{m}.$$

From (96) it therefore follows that

$$np = z_1^2 + z_2^2 + z_3^2 + z_4^2,$$

which, since $0 < n < m$, contradicts the minimality of m .

THEOREM 2.171 (Lagrange's Theorem). *The diophantine equation*

$$n = x_1^2 + x_2^2 + x_3^2 + x_4^2 \quad (2.17)$$

is solvable for every $n \geq 0$.

Proof: For $n = 0$ and $n = 1$ this is obvious; therefore let $n > 1$. By Theorem 166 it follows that the statement is true for every prime; and by (93), if it is true for two numbers, it is also true for their product. Therefore, by Theorem 12, it is true for every $n > 1$.

§ 2 Determination of the Number of Solutions

Since the number of solutions of (97) is obviously 1 for $n = 0$, we may assume that $n > 0$. Let $Q(n)$ denote the number of solutions of (97). The fact that we already know $Q(n)$ to be > 0 , by § 1, is irrelevant; we shall not use § 1, but we shall indeed use Chap. II (the number of decompositions into two squares).

Throughout the present section (§ 2) the symbols $u, u_1, u_2, u_3, u_4, v, l, m, a, c, b, \alpha, \beta, d_1, d, \delta, b$, and f_1 will denote positive odd numbers.

THEOREM 2.172. *Let $A(u)$ be the number of solutions of*

$$u = u_1^2 + u_2^2 + u_3^2 + u_4^2, \quad (2.18)$$

and (as in Part One, Chap. IV) let $S(u) = \sum_{d|u} d$. Then we have

$$A(u) = 8S(u).$$

Example: For $p > 2$, the theorem states that

$$4p = u_1^2 + u_2^2 + u_3^2 + u_4^2$$

has exactly $4(p + 1)$ solutions. For example, for $p = 3$ the four solutions are

$$12 = 3^2 + 1^2 + 1^2 + 1^2 = 1^2 + 3^2 + 1^2 + 1^2 = 1^2 + 1^2 + 3^2 + 1^2 = 1^2 + 1^2 + 1^2 + 3^2.$$

Proof: By (93), applied to the case in which the numbers x on the left are fixed as a decomposition

$$2u = a^2\alpha^2 + b^2\beta^2 + c^2 + d^2, \quad a\alpha + b\beta = 2u,$$

and in which the numbers y on the right are determined from

$$u = u_1^2 + u_2^2 + u_3^2 + u_4^2,$$

we obtain

$$A(u) = \sum_{\substack{a\alpha+b\beta=2u \\ a,b,\alpha,\beta \text{ odd} > 0}} \chi(a\beta),$$

where χ is the non-principal character mod 4. That is, the last term is summed over all quadruplets (of odd, positive numbers) a, α, b, β for which $a\alpha + b\beta = 2u$.

Let us first count the contribution of the quadruples in which $a = b$. In this case, $a|u$; the equation

$$a\alpha + a\beta = 2u, \quad \alpha + \beta = \frac{2u}{a}$$

clearly has $\frac{u}{a}$ solutions ($\alpha = 1, 3, \dots, \frac{2u}{a} - 1$ and the β determined therefrom); since $\chi(a\alpha) = 1$, the contribution of each is thus

$$\sum_{a|u} \frac{u}{a} = \sum_{d|u} d = S(u).$$

It remains to be shown that

$$\sum_{\substack{a\alpha+b\beta=2u \\ a>b}} \chi(a\beta) = 0.$$

By symmetry, it suffices to show that

$$\sum_{\substack{a\alpha+b\beta=2u \\ a>b}} \chi(a\beta) = 0.$$

And for this it suffices to pair off the solutions of

$$a\alpha + b\beta = 2u, \quad a > b$$

one to one in such a way that for every pair $a, b, \alpha, \beta; a_1, b_1, \alpha_1, \beta_1$ we have

$$\chi(a\beta) + \chi(a_1\beta_1) = 0. \quad (2.19)$$

The rule is: Setting

$$n = \left[\frac{b}{a} \right]$$

for brevity, let

$$a_1 = (n+2)a + (n+1)b, \quad b_1 = -(n+1)a + nb,$$

$$\alpha_1 = -(n+1)\alpha + n\beta, \quad \beta_1 = n\alpha + (n+1)\beta.$$

We have, indeed, that

(11) each of these four numbers is odd, since

$$a_1 \equiv n+2+n+1 \equiv 1, \quad b_1 \equiv -(n+1)+n \equiv 1, \quad \alpha_1 \equiv -(n+1)+n \equiv 1, \quad \beta_1$$

(12) each of these numbers is > 0 : a_1 and b_1 , obviously; α_1 , since

$$\beta > \frac{b}{a}\alpha \geq n\alpha, \quad \text{so } \beta \geq n\alpha+1, \quad \alpha_1 = -(n+1)\alpha+n\beta \geq -(n+1)\alpha+n(n\alpha+1)$$

β_1 , since $n\alpha > 0$;

$$(13) \quad a_1\alpha_1 + b_1\beta_1 = 2u;$$

$$(14) \quad a_1 > b_1 \text{ (obvious).}$$

For the inverse mapping:

$$a = ((n+2)a_1 - (n+1)b_1) \cdot 1 = a, \quad \text{etc.}$$

THEOREM 2.173. $Q(2u) = 3Q(u)$.

Proof: In the equation

$$2u = x_1^2 + x_2^2 + x_3^2 + x_4^2 \quad (2.20)$$

two of the x_i must be even and two odd, since every perfect square is $\equiv 0$ or $1 \pmod{4}$. The number of solutions in which x_1 and x_2 are even and x_3 and x_4 are odd is therefore $\frac{1}{4}Q(2u)$.

It then follows from (100), if we set

$$x_1 = y_1 + y_2, \quad x_2 = y_1 - y_2, \quad x_3 = y_3 + y_4, \quad x_4 = y_3 - y_4, \quad (2.21)$$

that y_1, y_2, y_3 , and y_4 are integers and that

$$u = y_1^2 + y_2^2 + y_3^2 + y_4^2, \quad y_1 + y_2 \equiv 0, \quad y_3 + y_4 \equiv 1 \pmod{2}. \quad (2.22)$$

Conversely, it follows from (102), if $x_1, \dots, x_4$ are determined from (101), that is, if we set

$$x_1 = y_1 + y_2, \quad x_2 = y_1 - y_2, \quad x_3 = y_3 + y_4, \quad x_4 = y_3 - y_4, \quad (2.23)$$

that (100) holds, and that x_1 and x_2 are even, and x_3 and x_4 odd.

$\frac{1}{4}Q(2u)$ is therefore the number of solutions of (102).
 $Q(u)$ is the number of solutions of

$$u = y_1^2 + y_2^2 + y_3^2 + y_4^2. \quad (2.24)$$

The number of solutions of (102) having $y_1 + y_2 \equiv 0 \pmod{2}$, $y_3 + y_4 \equiv 1 \pmod{2}$ equals $\frac{1}{2}Q(u)$, since for the solutions of (104) the four cases

$$y_1 + y_2 \equiv 0, 1; \quad y_3 + y_4 \equiv 0, 1 \pmod{2}$$

are equally distributed. Hence

$$\frac{1}{4}Q(2u) = \frac{1}{2}Q(u), \quad Q(2u) = 3Q(u).$$

THEOREM 2.174. *For $n > 0$, we have*

$$Q(2n) = Q(4n). \quad (2.25)$$

For

$$x_1 \equiv x_2 \equiv x_3 \equiv x_4 \pmod{2}, \quad (2.26)$$

so that

$$n = y_1^2 + y_2^2 + y_3^2 + y_4^2 \quad (2.27)$$

where the y_i are determined as integers by (101); conversely, (106) follows from (107), when the x_i are determined by (103).

Furthermore, we have

$$Q(4u) = 16A(u) + Q(u). \quad (2.28)$$

For in the equation

$$4u = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

either all of the x_i are even—this, $Q(u)$ times, since the equation is then equivalent to

$$u = \left(\frac{x_1}{2}\right)^2 + \left(\frac{x_2}{2}\right)^2 + \left(\frac{x_3}{2}\right)^2 + \left(\frac{x_4}{2}\right)^2$$

or else they are all odd—this (because of the coefficient), $16A(u) = 16S(u)$ times, by Theorem 170.

From Theorem 171, (105), and (108), it follows that

$$\begin{aligned} 3Q(u) &= Q(2u) = Q(4u) = 16S(u) + Q(u), \\ Q(u) &= 8S(u), \end{aligned} \quad (2.29)$$

and from (109) and Theorem 171 that

$$Q(2u) = 24S(u). \quad (2.30)$$

For $l > 0$, finally, it follows from (109) and (110) that

$$Q(2^l) = \begin{cases} 24S(2^{l-1}) = 24\sigma(2^{l-1}) & \text{for } l \geq 1, \\ 1 & \text{for } l = 0. \end{cases}$$

CHAPTER IV

DECOMPOSITION INTO THREE SQUARES

§ 1 Equivalence of Quadratic Forms

In this chapter I shall assume that the reader is acquainted to a certain extent with the theory of determinants and their applications. That portion of the theory of the so-called quadratic forms which I shall require, involves only two or three variables; it is more expeditious, however, to work it out once, for r variables, instead of twice, for two and for three variables. Consequently, let r be an arbitrary number ≥ 2 .

Definition 2.175. If $x_1, \dots, x_r$ are integral variables, and if the numbers a_{kl} for $1 \leq k \leq l \leq r$ are integral coefficients, then

$$F(x_1, \dots, x_r) = \sum_{k \leq l} a_{kl} x_k x_l = a_{11} x_1^2 + a_{12} x_1 x_2 + \dots + a_{1r} x_1 x_r + a_{22} x_2^2 + a_{23} x_2 x_3$$

is called a *quadratic form*—or here, for short, a *form*.

It is convenient to set

$$a_{lk} = a_{kl} \quad \text{for } k < l.$$

Definition 2.176. If

$$F(x_1, \dots, x_r) = \sum_{k,l} a_{kl} x_k x_l, \quad G(y_1, \dots, y_r) = \sum_{m,n} b_{mn} y_m y_n$$

are forms, then we say that F is *equivalent* to G , written

$$F \sim G,$$

if there are r^2 integers c_{kl} of determinant

$$|c_{kl}| = \begin{vmatrix} c_{11} & \cdots & c_{1r} \\ \vdots & \ddots & \vdots \\ c_{r1} & \cdots & c_{rr} \end{vmatrix} = \pm 1 \quad (2.31)$$

for which the r equations

$$x_k = \sum_l c_{kl} y_l \quad (2.32)$$

formally transform $F(x_1, \dots, x_r)$ into $G(y_1, \dots, y_r)$.

That is, we should have identically

$$\sum_{m,n} b_{mn} y_m y_n = \sum_{k,l} a_{kl} \sum_m c_{km} y_m \sum_n c_{ln} y_n = \sum_{m,n} y_m y_n \sum_{k,l} a_{kl} c_{km} c_{ln};$$

since, by (111), $y_m y_n$ and $y_n y_m$ on the right have the same coefficients (because $\sum_{k,l} a_{kl} c_{km} c_{ln} = \sum_{l,k} a_{lk} c_{ln} c_{km} = \sum_{k,l} a_{kl} c_{kn} c_{lm}$), this is equivalent to the conditions

$$b_{mn} = \sum_{k,l} a_{kl} c_{km} c_{ln},$$

that is,

$$b_{mn} = \sum_{k,l} c_{km} a_{kl} c_{ln}. \quad (2.33)$$

We say, for brevity: F goes into G under the transformation (c_{kl}) .

THEOREM 2.177. *We always have $F \sim F$.*

THEOREM 2.178 (Symmetry). *If $F \sim G$, then $G \sim F$.*

Proof: By (112), the equations in (113) may be solved for the variables y , the solutions having integral coefficients:

$$y_l = \sum_m d_{lm} x_m$$

and we have

$$|d_{lm}| = \pm 1$$

since

$$\delta_{km} = \sum_l c_{kl} d_{lm} = \sum_l d_{lm} c_{kl} = \delta_{km}.$$

THEOREM 2.179 (Transitivity). *If $F \sim G$ and $G \sim H$, then $F \sim H$.*

Proof: Suppose F goes into G under (c_{kl}) , and G into H under (f_{jl}) , so that

$$|c_{kl}| = |f_{jl}| = \pm 1.$$

From

$$x_k = \sum_l c_{kl} y_l, \quad y_l = \sum_m f_{lm} z_m$$

it follows that

$$x_k = \sum_l c_{kl} \sum_m f_{lm} z_m = \sum_m \left(\sum_l c_{kl} f_{lm} \right) z_m.$$

This represents an integral transformation carrying F into H ; it has determinant

$$\left| \sum_l c_{kl} f_{lm} \right| = |c_{kl}| \cdot |f_{lm}| = (\pm 1)(\pm 1) = \pm 1.$$

Theorems 173 to 175 show that the totality of forms falls into classes of mutually equivalent forms.

THEOREM 2.180. *If $F \sim G$, then F and G represent the same numbers.*

Proof: From $G(y_1, \dots, y_r) = c$, it follows that $F(x_1, \dots, x_r) = c$ for the eo ipso integral x_i given by (113).

Definition 2.181. F is called *positive definite*—or here, for short, *definite*—if $F > 0$ for all integral values of $x_1, \dots, x_r$ that do not all vanish simultaneously.

If $F \sim G$, then it of course follows that if F is definite, so is G .

Now I shall consider, in somewhat more detail, both binary forms ($r = 2$) and ternary forms ($r = 3$). Their discriminants $|a_{kl}|$ will always be denoted by d . For greater convenience, I shall write the binary forms as $ax^2 + 2bxy + cy^2$, so that

$$d = \begin{vmatrix} a & b \\ b & c \end{vmatrix} = ac - b^2.$$

Such a form is abbreviated as $\{a, b, c\}$.

THEOREM 2.182. $F = \{a, b, c\}$ is definite if and only if $d > 0$ and $a > 0$ (so that we also have $c > 0$).

Proof: 1) If $d > 0$ and $a > 0$, then for $x \neq 0$ or $y \neq 0$ we have

$$aF = (ax + by)^2 + dy^2 > 0.$$

Elementary Number Theory
(Continued)

Edmund Landau

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PART THREE

CHAPTER IV

§ 1

THEOREM 1 (179). *Every class of definite binary forms contains at least one form for which*

$$2|b| \leq a \leq c.$$

Proof. Let us fix upon a form $\{a_0, b_0, c_0\}$ belonging to the class. Let a be the smallest positive number representable by it (and therefore by any form of the class). Then, for suitable r and t , we have

$$a = a_0 r^2 + 2b_0 r t + c_0 t^2. \quad (115)$$

Here we have $(r, t) = 1$, for otherwise, if we set $(r, t) = v$, since v/a , it would then follow that

$$\frac{a}{v^2} = a_0 \left(\frac{r}{v}\right)^2 + 2b_0 \frac{r}{v} \frac{t}{v} + c_0 \left(\frac{t}{v}\right)^2$$

is representable by the form.

Therefore, by Theorem 66, we may choose s and u in such a way that

$$\begin{vmatrix} r & s \\ t & u \end{vmatrix} = ru - st = 1;$$

by Theorem 68, the most general such pair, s and u are of the form

$$s = s_0 + hr, \quad u = u_0 + ht,$$

where h is arbitrary.

By (115), the transformation $\begin{pmatrix} r & s \\ t & u \end{pmatrix}$ takes $\{a_0, b_0, c_0\}$ into $\{a, b, c\}$, where the first coefficient is actually our a (by (115), or else by direct computation, or else more elegantly: $G(1, 0) = F(r \cdot 1 + s \cdot 0, t \cdot 1 + u \cdot 0) = F(r, t)$, where G is the new form and F is the old one); furthermore (by (115), or else by direct computation), we have

$$\begin{aligned} b &= s(a_0 r + b_0 t) + u(b_0 r + c_0 t) \\ &= s_0(a_0 r + b_0 t) + u_0(b_0 r + c_0 t) + h(r(a_0 r + b_0 t) + t(b_0 r + c_0 t)); \end{aligned}$$

consequently, since h has coefficient $a_0 r^2 + 2b_0 r t + c_0 t^2 = a$, b takes on all values in a certain residue class mod a ; b can therefore be chosen in such a way that

$$|b| \leq \frac{a}{2}.$$

Since c can also be represented by the form $\{a, b, c\}$ (with $x = 0$ and $y = 1$), we have

$$a \leq c,$$

so that everything is proved. □

THEOREM 2 (180). *In every class of definite binary forms there is at least one form for which*

$$2|b| \leq a \leq \sqrt{\frac{|d|}{3}}.$$

Proof. From the inequalities of Theorem 179 it follows that

$$a \leq c, \quad ac \leq \frac{d+b^2}{4} \leq \frac{d+a^2}{4},$$

$$a^2 \leq ac \leq \frac{d+a^2}{4},$$

$$2|b| \leq a \leq \sqrt{\frac{|d|}{3}}.$$

□

THEOREM 3 (181). *Every definite binary form having discriminant 1 is equivalent to the form $x_1^2 + x_2^2$.*

Proof. By Theorem 180, every such form is equivalent to a form for which

$$2|b| \leq a \leq \sqrt{\frac{1}{3}},$$

that is,

$$b = 0, \quad a = 1, \quad c = 1.$$

□

Now for ternary forms!

THEOREM 4 (182). *$F = \sum_{i,j} a_{ij}x_i x_j$ is definite if and only if all of the following hold:*

$$a_{11} > 0, \quad b = \begin{vmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{vmatrix} > 0, \quad d > 0.$$

If F is definite, then we have, further, that

$$a_{11}F = (a_{11}x_1 + a_{12}x_2 + a_{13}x_3)^2 + K(x_2, x_3), \quad (116)$$

where $K(x_2, x_3)$ is the definite binary form $\{a_{11}a_{22} - a_{12}^2, a_{11}a_{23} - a_{12}a_{13}, a_{11}a_{33} - a_{13}^2\}$ with discriminant $a_{11} \cdot d$.

Proof of Theorem 182. (quite inelegant, in order to make it simple for the reader): (116) holds, in any case, for

$$K(x_2, x_3) = (a_{11}a_{22} - a_{12}^2)x_2^2 + 2(a_{11}a_{23} - a_{12}a_{13})x_2x_3 + (a_{11}a_{33} - a_{13}^2)x_3^2.$$

For

$$\begin{aligned} a_{11}F &= a_{11}^2x_1^2 + 2a_{11}a_{12}x_1x_2 + 2a_{11}a_{13}x_1x_3 + a_{11}a_{22}x_2^2 + 2a_{11}a_{23}x_2x_3 + a_{11}a_{33}x_3^2 \\ &= (a_{11}x_1 + a_{12}x_2 + a_{13}x_3)^2 + (a_{11}a_{22} - a_{12}^2)x_2^2 + 2(a_{11}a_{23} - a_{12}a_{13})x_2x_3 \\ &\quad + (a_{11}a_{33} - a_{13}^2)x_3^2 \\ (116) &= (a_{11}x_1 + a_{12}x_2 + a_{13}x_3)^2 + K(x_2, x_3). \end{aligned}$$

$K(x_2, x_3)$ has discriminant

$$\begin{aligned} C &= (a_{11}a_{22} - a_{12}^2)(a_{11}a_{33} - a_{13}^2) - (a_{11}a_{23} - a_{12}a_{13})^2 \\ &= a_{11}(a_{11}a_{22}a_{33} - a_{11}a_{23}^2 + 2a_{12}a_{23}a_{13} - a_{12}^2a_{33} - a_{13}^2a_{22}) = a_{11}d. \end{aligned}$$

1) If $a_{11} \leq 0$, then $F(1, 0, 0) = a_{11} \leq 0$, so that F is not definite.

2) If $a_{11} > 0$, then clearly $F(x_1, x_2, x_3)$ is definite if and only if $K(x_2, x_3)$ is definite. For if K is not definite, then $K(\bar{x}_2, \bar{x}_3) \leq 0$ for some suitable $\bar{x}_2$ and $\bar{x}_3$, not both 0; and also, therefore, with the additional condition that $\bar{x}_2$ and $\bar{x}_3$ be divisible by a_{11} (a proportionality factor); then x_1 can be determined as an integer from

$$a_{11}x_1 + a_{12}\bar{x}_2 + a_{13}\bar{x}_3 = 0,$$

and for the resulting triple $\bar{x}_1, \bar{x}_2, \bar{x}_3$, we have

$$a_{11}F(\bar{x}_1, \bar{x}_2, \bar{x}_3) = 0 + K(\bar{x}_2, \bar{x}_3) \leq 0,$$

so F is not definite.

On the other hand, if K is definite, then from $F \leq 0$ it follows, successively, that

$$K(x_2, x_3) \leq a_{11}F(x_1, x_2, x_3) \leq 0, \quad x_2 = x_3 = 0, \quad a_{11}x_1^2 \leq 0, \quad x_1 = 0.$$

By Theorem 178, applied to $K(x_2, x_3)$, it follows that in the case where $a_{11} > 0$, F is definite if and only if both

$$b = a_{11}a_{22} - a_{12}^2 > 0, \quad C = a_{11}d > 0,$$

that is, if and only if both

$$b > 0 \quad \text{and} \quad d > 0.$$

□

THEOREM 5 (183). *If $(c_{11}, c_{21}, c_{31}) = 1$, then the six remaining numbers c_{ij} can be chosen in such a way that*

$$\begin{vmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{vmatrix} = 1.$$

Proof. Let us set $(c_{11}, c_{21}) = g$, so that $(g, c_{31}) = 1$. Then let us choose c_{12} and c_{22} , by Theorem 66, in such a way that

$$c_{11}c_{22} - c_{12}c_{21} = g;$$

furthermore, let us choose u and v , by Theorem 66, in such a way that

$$gu - c_{31}v = 1.$$

Then we have

$$\begin{aligned} \begin{vmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{vmatrix} &= c_{33} \cdot g + (c_{11}c_{22} - c_{12}c_{21})u \\ &\quad + (c_{21}c_{32} - c_{31}c_{22})v + (c_{31}c_{12} - c_{11}c_{32})u \\ &= c_{33}g + gu = 1. \end{aligned}$$

□

THEOREM 6 (184). *Every class of definite ternary forms contains at least one form for which*

$$a_{11} \leq \frac{4}{3} \sqrt[3]{d}, \quad 2|a_{12}| \leq a_{11}, \quad 2|a_{13}| \leq a_{11}.$$

Proof. Let F be a fixed form belonging to the class. Let a_{11} be the smallest positive number that can be represented by F , and consequently by any form belonging to the class. Then, for suitable c_{13} , c_{23} , c_{31} , we have

$$a_{11} = F(c_{13}, c_{23}, c_{31}).$$

Here we have

$$(c_{13}, c_{23}, c_{31}) = 1,$$

since otherwise $\frac{a_{11}}{(c_{13}, c_{23}, c_{31})^2}$ would be representable.

Let $G = \sum b_{ij}x_i x_j$ be the form into which F is carried by the transformation (c_{ij}) , of determinant 1, constructed in accordance with Theorem 183; then we have

$$b_{11} = G(1, 0, 0) = F(c_{13}, c_{23}, c_{31}) = a_{11}.$$

$$d_{ik} = \begin{pmatrix} 1 & r & s \\ 0 & t & u \\ 0 & v & w \end{pmatrix},$$

where $tw - uv = 1$; then, for arbitrary r and s , we have

$$|d_{ik}| = 1.$$

Now we shall take care of t , u , v , and w . Let G go into H under (d_{ik}) . Then in H , the coefficient of y_1^2 equals $G(d_{11}, d_{21}, d_{31}) = G(1, 0, 0) = a_{11}$; let us set $H = \sum a_{ij}y_iy_j$; then, by (??), we have

$$a_{12} = \sum_{m,n} d_{m1}b_{mn}d_{n2} = \sum_n b_{1n}d_{n2} = tb_{11} + ub_{12} + vb_{13}, \quad (117)$$

$$a_{13} = \sum_{m,n} d_{m1}b_{mn}d_{n3} = \sum_n b_{1n}d_{n3} = sb_{11} + ub_{12} + wb_{13}; \quad (118)$$

the transformation

$$\begin{pmatrix} t & u \\ v & w \end{pmatrix} \quad (119)$$

therefore yields

$$\begin{aligned} a_{11}H(y_1, y_2, y_3) &= (a_{11}y_1 + a_{12}y_2 + a_{13}y_3)^2 + L(y_2, y_3), \\ a_{11}G(y_1, y_2, y_3) &= (b_{11}y_1 + b_{12}y_2 + b_{13}y_3)^2 + K(y_2, y_3), \end{aligned} \quad (120)$$

where K and L are definite in their respective variables. Since (119) carries the form $G(y_1, y_2, y_3)$ into $H(y_1, y_2, y_3)$, and since (120) also holds, it follows that $K(y_2, y_3)$ is taken into $L(y_2, y_3)$ by the transformation (119).

By Theorem 182, L has the discriminant $a_{11}d$, and its first coefficient is $a_{11}b_{22} - b_{12}^2$. Therefore, by Theorem 180, we may dispose of (119) by arranging matters so that

$$2|a_{12}| \leq a_{11}, \quad 2|a_{13}| \leq a_{11}.$$

Then, by (117) and (118), r and s can be chosen so that

$$|a_{12}| \leq \frac{a_{11}}{2}, \quad |a_{13}| \leq \frac{a_{11}}{2}.$$

We therefore obtain, since $a_{22} = H(0, 1, 0)$ is representable by H , and is therefore $\geq a_{11}$, that

$$a_{11} \leq a_{22} = a_{11}(b_{22}a_{11} - b_{12}^2) + a_{12}^2 \leq \frac{a_{11}^2}{4} + \frac{a_{11}^2}{4},$$

$$a_{22} \leq \frac{a_{11}^2}{4}, \quad a_{22} \leq \frac{3}{4}a_{11}^2,$$

$$a_{22} \leq \frac{3}{8}\sqrt[3]{d}, \quad a_{22} \leq \frac{3}{8}\sqrt[3]{d}, \quad a_{22} \leq \frac{3}{4}\sqrt[3]{d},$$

so that everything is proved. □

THEOREM 7 (185). *Every definite ternary form with discriminant 1 is equivalent to the form $x_1^2 + x_2^2 + x_3^2$.*

Consequently, every number representable by such a form can be written as a sum of three squares.

Proof. By Theorem 184, the given form is equivalent to a form in which

$$a_{11} \leq \frac{4}{3}, \quad 2|a_{12}| \leq a_{11}, \quad 2|a_{13}| \leq a_{11};$$

from this it follows that

$$a_{11} = 1, \quad a_{12} = 0, \quad a_{13} = 0.$$

The class therefore contains a form

$$G = a_{11}x_1^2 + 2a_{12}x_1x_2 + 2a_{13}x_1x_3 + K(x_2, x_3) = x_1^2 + K(x_2, x_3),$$

where $K(x_2, x_3) = a_{22}x_2^2 + 2a_{23}x_2x_3 + a_{33}x_3^2$ is definite and has discriminant 1. By Theorem 181, $K(x_2, x_3)$ goes into $x_2^2 + x_3^2$ by a suitable transformation $\begin{pmatrix} t & u \\ v & w \end{pmatrix}$ of determinant 1; consequently $\begin{pmatrix} 1 & 0 & 0 \\ 0 & t & u \end{pmatrix}$ takes G into $x_1^2 + x_2^2 + x_3^2$. $\square$

§ 2

THEOREM 8 (186). *If $n = x_1^2 + x_2^2 + x_3^2$, $n > 0$, then n is not of the form $4^a(8b + 7)$, $a \geq 0$, $b \geq 0$.*

Proof. 1) $n = 8b + 7$ cannot be written as the sum of three squares, since every perfect square is $\equiv 0, 1$, or $4 \pmod{8}$.

2) If the indecomposability of $4^a(8b + 7)$ has already been proved for some $a \geq 0$, then it follows for $a + 1$. For from

$$4^{a+1}(8b + 7) = x_1^2 + x_2^2 + x_3^2,$$

it would follow, because

$$x_1^2 + x_2^2 + x_3^2 \equiv 0 \pmod{4},$$

that x_1, x_2 , and x_3 must be even. Consequently, we would have

$$4^a(8b + 7) = \left(\frac{x_1}{2}\right)^2 + \left(\frac{x_2}{2}\right)^2 + \left(\frac{x_3}{2}\right)^2.$$

$\square$

§ 3

THEOREM 9 (187). *If $n > 0$ is not of the form $4^a(8b + 7)$, $a \geq 0$, $b \geq 0$, then n can be written as a sum of three squares.*

Proof. Without loss of generality, we may assume n to be odd (but $\not\equiv 7 \pmod{8}$) or else to be equal to twice an odd number. For from the decomposability of $\frac{n}{4}$, the same would follow for $4 \cdot \frac{n}{4}$, and if we divide out 4 as many times as we can from the given number n , we obtain a number in one of the two forms indicated.

Consequently, let

$$n \equiv 1, 2, 3, 5, \text{ or } 6 \pmod{8}.$$

By Theorem 185, it suffices to give, for n , a definite ternary form of discriminant 1 which represents n . By Theorem 182, we therefore have to specify nine numbers a_{11} , a_{12} , a_{13} , a_{22} , a_{23} , a_{33} , x_1 , x_2 , x_3 which satisfy the four conditions:

$$n = a_{11}x_1^2 + 2a_{12}x_1x_2 + 2a_{13}x_1x_3 + a_{22}x_2^2 + 2a_{23}x_2x_3 + a_{33}x_3^2,$$

$$a_{11} > 0,$$

$$a_{11}a_{22} - a_{12}^2 > 0,$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{vmatrix} = 1.$$

It will even do to take

$$a_{12} = 1, \quad a_{13} = 0, \quad a_{22} = n, \quad a_{23} = 0, \quad a_{33} = 1.$$

The three remaining unknowns thus have to satisfy the three conditions (the first of the previous conditions is already taken care of):

$$n = a_{11}x_1^2 + 2x_1x_2 + nx_2^2 + x_3^2,$$

(In fact, we have

$$\begin{vmatrix} a_{11} & 1 & 0 \\ 1 & n & 0 \\ 0 & 0 & 1 \end{vmatrix} = (a_{11}n - 1) \cdot 1 - 0 = a_{11}n - 1.$$

Here, taking $n > 1$ (which I may do, since $n = 1$ is trivial), $a_{11} > 0$ is a consequence of the other two conditions; for it follows from them that

$$a_{11} > 0.$$

□

PART FOUR

CLASS NUMBER OF BINARY QUADRATIC FORMS

INTRODUCTION

In this part I shall analyze the binary forms of the form $ax^2 + bxy + cy^2$. (In the foregoing, because of the analogy with the case of r , rather than 2, variables, it was more convenient to take the middle coefficient to be even.) Here, the number $d = b^2 - 4ac$ is what will be called the discriminant. (This is customary with binary forms; one should not be misled by the fact that for b even, the number $\frac{d}{4}$ was called the discriminant in the preceding part.) Even though our first investigation of equivalence can be carried over to the present part, I should nevertheless like to begin here anew and develop the various details afresh in order that Part Four be readable independently of Part Three.

In Chapter I, I shall begin with somewhat trivial considerations, which will show that only the case in which d is not a perfect square is of interest; in the other case, the form splits into the product of two linear forms. Moreover, we of course have $d = b^2 - 4ac \equiv 0$ or $1 \pmod{4}$. Henceforth we shall always take d to be an arbitrary non-square number $\equiv 0$ or $1 \pmod{4}$; then there will certainly be a form belonging to it, namely $\{1, 0, -\frac{d}{4}\}$ if d is even and $\{1, 1, \frac{1-d}{4}\}$ if d is odd.

In Chapters II and III it will turn out that the number of classes into which the forms of discriminant d fall is finite; the determination of this number will immediately reduce to the problem of determining the number of so-called primitive classes, that is, the classes containing a form for which $(a, b, c) = 1$.

If we did not know better, the problem of determining the class number would then already seem to be solved. But we can accomplish still more (unfortunately, or fortunately, depending on one's point of view; I, personally, am happy about it, for the theory in this fourth part, for which we have Gauss and Dirichlet to thank, is one of the most beautiful in all of mathematics); namely, we can find certain simple expressions for $h(d)$. This will come much later on.

Here I would like to tell, but only in anticipation of what will be done later on, how the class number is most easily shown to be finite in the first place, how, in the case $d < 0$, the class number is first “determined,” and how the final formula for it reads in the case $d = -p$, where $p > 3$.

The proof of finiteness is obtained by singling out a certain finite number of forms—let us, for the moment, call them reduced—and showing that each class contains at least one of these forms (and then, the pigeon-hole principle!). A (primitive or imprimitive) form is called reduced if

$$|b| \leq |a| \leq |c|.$$

It is easily shown that for any d there is only a finite number of such forms. Consequently the class number, and therefore in particular our $h(d)$, is finite.

In particular, we shall show that in the case $d < 0$ every class of forms having $a > 0$ contains one and only one form for which

$$-a < b \leq a < c \quad \text{or} \quad 0 < b = a = c. \quad (122)$$

This “determines” the number of classes, and in particular it determines $h(d)$, if of these forms we count only the primitive ones.

Now the beautiful formula reads:

$$h(d) = \frac{w}{2\pi} \sqrt{|d|} \sum_{n=1}^{\infty} \left(\frac{d}{n}\right) \frac{1}{n}. \quad (123)$$

First: From (123) it follows, since $h > 0$, that for every prime $p \equiv 3 \pmod{4}$ there are more quadratic residues mod p between 0 and $\frac{p}{2}$ than between $\frac{p}{2}$ and p ; for there are indeed as many non-residues between 0 and $\frac{p}{2}$ as there are residues between $\frac{p}{2}$ and p , since $\left(\frac{p-r}{p}\right) = -\left(\frac{r}{p}\right)$ for $p \nmid r$. (For $p \equiv 1 \pmod{4}$, on the other hand, there are equally many quadratic residues in both halves of the interval $0 < r < p$, since $\left(\frac{p-r}{p}\right) = \left(\frac{r}{p}\right)$ for $p \nmid r$.)

But no one has yet been able, using elementary methods, to prove this fact which, with the help of Theorem 79, can be expressed as follows:

For $p \equiv 3 \pmod{4}$, more of the numbers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ have remainders mod p less than $\frac{p}{2}$ than have remainders mod p greater than $\frac{p}{2}$.

Second: For $p \equiv 3 \pmod{4}$ and $p > 3$, it follows from what has been said (since obviously there are only primitive forms here) that the right-hand side of (123) equals the number of solutions of $b^2 - 4ac = -p$ satisfying the additional condition (122).

But no one has yet succeeded in proving this fact by use of elementary methods; and yet it is a statement that has nothing to do with either class numbers or quadratic forms.

Naturally, the second statement contains the first.

In Chapters IV and V, our longer route will next reduce the determination of $h(d)$ to the problem of finding the limit of the series

$$\sum_{n=1}^{\infty} \left(\frac{d}{n}\right) \frac{1}{n};$$

this is where the theorems of Part One, Chapter VI on the Kronecker symbol, and of Part One, Chapter VII, on Pell's equation, will be made use of. In Chapter VI, the sum of the series

$$\sum_{n=1}^{\infty} \left(\frac{d}{n}\right) \frac{1}{n}$$

will finally be found; and for the special case in which d is a fundamental discriminant, the well-known expression in closed form involving d will appear.

Chapter 1

FACTORABLE AND UNFACTORABLE FORMS

DEFINITION 10 (32). If a , b , and c are constants, then

$$F = F(x, y) = ax^2 + bxy + cy^2$$

is called a binary quadratic form, or here, for brevity, a *form*. It is written: $F = \{a, b, c\}$. The discriminant of the form is the number $d = b^2 - 4ac$.

We always have

$$d \equiv 0 \text{ or } 1 \pmod{4},$$

and there is a form for every such d , and even one with $a > 0$, namely $\{1, 0, -\frac{d}{4}\}$ and $\{1, 1, \frac{1-d}{4}\}$, respectively.

We always have

$$\begin{aligned} 4aF &= 4a^2x^2 + 4abxy + 4acy^2 = (2ax + by)^2 + (4ac - b^2)y^2 \\ &= (2ax + by)^2 - dy^2. \end{aligned} \tag{124}$$

THEOREM 11 (188). *If there exists a factorization*

$$F = (\varrho x + \sigma y)(\tau x + \nu y) \tag{125}$$

where ϱ , σ , τ , and ν are rational, then there exists a factorization

$$F = (rx + sy)(tx + uy) \tag{126}$$

(where r , s , t , and u are integers).

(This statement sounds quite harmless; at the bottom of it, however, lies a deep theorem of Gauss, which appears in Part Nine of Vorlesungen über Zahlentheorie.)

Proof. There follows from (125) an equation

$$mF = (rx + sy)(tx + uy) \tag{1.1}$$

in which $m > 0$. Consequently there is an equation of the type (1.1) in which $m > 0$ and $(m, r, s) = 1$, since otherwise we would merely have to replace m , r , s by the numbers $\frac{m}{(m,r,s)}$ and $\frac{r}{(m,r,s)}$, $\frac{s}{(m,r,s)}$. For the same reason, we may also assume that $(m, t, u) = 1$.

From (1.1), with $m > 0$, $(m, r, s) = 1$, and $(m, t, u) = 1$, however, it follows that $m = 1$, in other words (126). For otherwise there would be a prime $p|m$. From

$$ma = rt, \quad mb = ru + st, \quad mc = su$$

it would then follow that

$$p|rt, \quad p|ru + st, \quad p|su;$$

from $p|rt$ it would follow that $p|r$ or $p|t$; without loss of generality (symmetry!) let $p|r$; then we would have $p|st$; from $(m, r, s) = 1$ it would follow that $p \nmid s$, so that $p|t$; from $(m, t, u) = 1$ it would therefore follow that $p \nmid u$. But $p|t$, $p \nmid u$, and $p|su$ yield a contradiction. $\square$

THEOREM 12 (189). *There exists a factorization of the form (126) if and only if d is a perfect square.*

Proof. 1) From (126) it follows that

$$a = rt, \quad b = ru + st, \quad c = su,$$

$$d = b^2 - 4ac = (ru + st)^2 - 4rstu = (ru - st)^2.$$

2) Let $d = k^2$. Then it follows from (124) that

$$4aF = (2ax + by)^2 - k^2y^2 = (2ax + (b + k)y)(2ax + (b - k)y).$$

21) If $a \neq 0$, then it follows from this that there exists a factorization of the form (125), and therefore, by Theorem 188, one of the form (126).

22) If $a = 0$, then we have at once

$$F = y(bx + cy).$$

$\square$

Chapter 2

CLASSES OF FORMS

From now on, throughout the rest of this book, let

$$d \text{ be non-square, and } d \equiv 0 \text{ or } 1 \pmod{4}.$$

Consequently, for every form F having discriminant d , we certainly have $a \neq 0$ and $c \neq 0$.

THEOREM 13 (190). *If $d > 0$, then for suitable x and y , F represents both positive and negative numbers; if $d < 0$ and $a > 0$, then F represents no negative numbers, and represents 0 only for $x = y = 0$; if $d < 0$ and $a < 0$, then F represents no positive numbers, and represents 0 only for $x = y = 0$.*

Hence forms for which $d > 0$ are called indefinite; and those for which $d < 0$ are called definite or—more precisely—positive definite and negative definite, as the case may be.

Geometrically: If $d > 0$ and $k \neq 0$, then $F = k$ is a hyperbola (if x and y are construed as any real (not necessarily integral) coordinates); if $d > 0$, then $F = 0$ is a pair of straight lines, namely the pair of asymptotes belonging to all hyperbolas $F = k$ for $k \neq 0$; if $d < 0$, then $F = k$ is an ellipse when $ka > 0$, a so-called imaginary pair of lines when $k = 0$, and a so-called imaginary ellipse when $ka < 0$.

Proof. 1) Let $d > 0$. Then we have

$$F(1, 0) = a, \quad F(b, -2a) = ab^2 - b \cdot 2ba + c \cdot 4a^2 = a(4ac - b^2) = -da;$$

of these two numbers, certainly one is > 0 and the other is < 0 .

2) Let $d < 0$; from (124) it follows that, except when $x = y = 0$, we always have

$$aF > 0,$$

that is, F has the same sign as a ; indeed it follows from $aF \leq 0$ that

$$2ax + by = 0, \quad y = 0,$$

and finally that $x = 0$. □

Since the negative-definite forms, when multiplied by -1 , are transformed into positive-definite ones with the same discriminant, and vice versa, we shall only consider positive-definite forms in the case $d < 0$.

DEFINITION 14 (33). $F = \{a, b, c\}$ is said to be *equivalent* to $G = \{a_1, b_1, c_1\}$, written

$$F \sim G,$$

if there exist four integers, r, s, t , and u , for which $ru - st = 1$, such that the equations

$$x = rX + sY, \quad y = tX + uY \tag{128}$$

formally transform $F(x, y)$ into $G(X, Y)$. We also say: F goes into G under the transformation (128).

THEOREM 15 (191). *If F has discriminant d , then G also has discriminant d ; moreover, if $d < 0$ and $a > 0$, then we also have $a_1 > 0$. That is, if F is one of our allowable forms, then so is G .*

Proof.

$$\begin{aligned} a(rX + sY)^2 + b(rX + sY)(tX + uY) + c(tX + uY)^2 \\ = a_1X^2 + b_1XY + c_1Y^2, \end{aligned}$$

$$a_1 = ar^2 + bri + ct^2, \tag{2.1}$$

$$b_1 = 2ars + b(ru + st) + 2ctu, \tag{2.2}$$

$$c_1 = as^2 + bsu + cu^2. \tag{129-131}$$

From this we have

$$\begin{aligned} b_1^2 - 4a_1c_1 &= (2ars + b(ru + st) + 2ctu)^2 \\ &\quad - 4(ar^2 + bri + ct^2)(as^2 + bsu + cu^2) \\ &= b^2(ru + st)^2 + 4abrs(ru + st) + 4bctu(ru + st) + 4a^2r^2s^2 \\ &\quad + 4c^2t^2u^2 + 8acrsiu - 4a^2r^2s^2 - 4abrus^2 - 4acr^2u^2 \\ &\quad - 4abrt^2s - 4b^2rtus - 4act^2u^2 \\ &= b^2(ru - st)^2 - 4ac(ru - st)^2 = b^2 - 4ac = d. \end{aligned}$$

The fact that a_1 and a have the same sign for $d < 0$ follows from (2.1), since by (124) we have

$$a_1 = a \left(\left(r + \frac{bt}{2a} \right)^2 - \frac{d}{4a^2} t^2 \right),$$

so that a_1 and a have the same sign (indeed $a_1 \neq 0$, since $d < 0$ and $t = 0$ would imply $a_1 = ar^2 = a$). □

THEOREM 16 (192). $F \sim F$.

Proof. Take $r = u = 1, s = t = 0$. □

THEOREM 17 (193). *If $F \sim G$, then $G \sim F$.*

Proof. If $x = rX + sY$, $y = tX + uY$, where $ru - st = 1$, takes $F(x, y)$ into $G(X, Y)$, then the transformation $X = ux - sy$, $Y = -tx + ry$ (which has determinant $ru - st = 1$) takes $G(X, Y)$ into $F(x, y)$; indeed

$$r(ux - sy) + s(-tx + ry) = (ru - st)x = x,$$

$$t(ux - sy) + u(-tx + ry) = (ru - st)y = y.$$

□

THEOREM 18 (194). *If $F \sim G$ and $G \sim H$, then $F \sim H$.*

Proof. If $x = rX + sY$, $y = tX + uY$ takes F into G , and $X = r_1Z + s_1W$, $Y = t_1Z + u_1W$ takes G into H , then

$$x = (rr_1 + st_1)Z + (rs_1 + su_1)W, \quad y = (tr_1 + ut_1)Z + (ts_1 + uu_1)W$$

takes F into H ; and this transformation has determinant

$$\begin{aligned} & (rr_1 + st_1)(ts_1 + uu_1) - (rs_1 + su_1)(tr_1 + ut_1) \\ &= rr_1uu_1 + st_1ts_1 + rr_1ts_1 + st_1uu_1 - rs_1tr_1 - su_1ut_1 - rs_1ut_1 - su_1tr_1 \\ &= (ru - st)(r_1u_1 - s_1t_1) = 1. \end{aligned}$$

□

Chapter 3

THE FINITENESS OF THE CLASS NUMBER

Even though the finiteness of the class number will once again come out of what we do later on, we should like nonetheless to prove it beforehand as quickly as possible; in particular, for the case $d < 0$, we establish a deeper result mentioned in the Introduction (the “determinability” of the class number right off the bat).

THEOREM 19 (195). *Every class contains a form for which*

$$|b| \leq |a| \leq |c|. \quad (132)$$

Proof. Let a form $\{a_0, b_0, c_0\}$ belonging to the class be fixed. Let a be the number smallest in absolute value, other than 0, which is representable by $\{a_0, b_0, c_0\}$, or else one of the two such numbers having the smallest absolute value > 0 . Then we have

$$a = a_0 r^2 + b_0 r t + c_0 t^2$$

for suitable r and t . We certainly have $(r, t) = 1$, for otherwise $\frac{a}{(r, t)^2}$ would already be representable and smaller in absolute value than a . We can therefore find numbers s and u such that

$$ru - st = 1.$$

By (2.1), (128) takes $\{a_0, b_0, c_0\}$ into $\{a, b', c'\}$. The transformation

$$\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$$

of determinant 1 (in which h is, for the moment, arbitrary) takes $\{a, b', c'\}$ into $\{a, b, c\}$ by (2.1) and (2.2), where

$$b = 2ah + b'.$$

For suitable h it follows from this that

$$|b| \leq |a|.$$

□

THEOREM 20 (196). *Every class contains a form for which*

$$-a \leq b \leq a \leq c.$$

Proof. By Theorem 195, each class contains a form for which $|b| \leq |a|$. Since $c \neq 0$ and since it can be represented by $\{a, b, c\}$ (by setting $x = 0$ and $y = 1$), it follows, because of the minimality of $|a|$, that $|a| \leq |c|$. $\square$

THEOREM 21 (197). *The class number is finite.*

Proof. 1) For $d > 0$ it follows from (132) that

$$|ac| = b^2 - d = b^2 - 4ac > 4ac.$$

Hence we have

$$ac < 0, \quad 4a^2 \leq 4|ac| = -4ac = d - b^2 < d, \quad |a| \leq \sqrt{\frac{d}{4}}, \quad |b| \leq |a| \leq \sqrt{\frac{d}{4}};$$

a and b thus have only a finite set of possible values, and therefore so does c .

2) For $d < 0$ it follows from (132), since $a > 0$ and $c > 0$, that

$$|b| \leq a \leq c,$$

$$4a^2 \leq 4ac = -d + b^2 < |d| + a^2, \quad |b| \leq a \leq \sqrt{|d|};$$

so that c , too, has only a finite number of possible values. $\square$

THEOREM 22 (198). *For $d < 0$, the number of (positive-definite) classes of forms equals the number of solutions of*

$$b^2 - 4ac = d, \quad -a < b \leq a < c \quad \text{or} \quad 0 < b = a = c. \quad (133)$$

Proof. We have merely to show that each class contains exactly one such form.

1) By Theorem 196, each class contains at least one form for which

$$-a \leq b \leq a \leq c.$$

In order to remove the extra forms, those for which $b = -a$ and $a < c$, or $-b < 0$ and $a = c$, it suffices to show that

$$\{a, -a, c\} \sim \{a, a, c\}$$

and

$$\{a, -b, a\} \sim \{a, b, a\}.$$

The former follows from the fact that $\begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$ takes $\{a, -a, c\}$ into the form $a(X+Y)^2 - a(X+Y)Y + cY^2 = aX^2 + aXY + cY^2$. (Moreover, we even could have chosen the number b to lie in the interval $-|a| < b \leq |a|$ in the proof of Theorem 196.)

The latter follows from the fact that the transformation $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ of determinant 1 takes the form $\{a, -b, a\}$ into

$$aY^2 + bYX + aX^2 = aX^2 + bXY + aY^2.$$

2) The proof of the fact that every class contains at most one form of type (133) is not so simple. If $\{a, b, c\}$ and $\{a', b', c'\}$ are two such forms, we have to show that $a = a'$, $b = b'$, and $c = c'$.

Without loss of generality, let $a \leq a'$. Let the transformation $\begin{pmatrix} r & s \\ t & u \end{pmatrix}$ for which $ru - st = 1$ take $\{a, b, c\}$ into $\{a', b', c'\}$. Then, by (2.1) and (2.2), we have

$$a' = ar^2 + bri + ct^2, \quad (134)$$

$$b' = 2ars + b(ru + st) + 2ctu. \quad (135)$$

By (134), we have

$$a' \geq ar^2 - a|rt| + at^2, \quad (136)$$

so that (since $r^2 + t^2 \geq 2|rt|$)

$$a' = a \geq a|rt|,$$

and therefore

$$1 \geq |rt|. \quad (137)$$

Now we have $a = a'$; for otherwise we would have $rt = 0$, and it would follow (since r and t do not both vanish) that

$$a > a' = ar^2 + ct^2 \geq ar^2 + ta^2 = a.$$

Now let $c > a$ or $c' > a'$, and therefore (since $a = a'$), by symmetry, $c > a$. Then we have $t = 0$; for otherwise (since $ct^2 > at^2$) the sign $>$ would hold in (136), and therefore also in (137); and we would have $rt = 0$, $r = 0$, and

$$a = c \geq c.$$

In the case $c > a$ we therefore have $t = 0$ and $ru = 1$, so that, by (135),

$$b' \equiv b \pmod{2a};$$

since

$$-a < b \leq a$$

and

$$-a = -a' < b' < a' = a,$$

it follows that $b = b'$, and, finally, that

$$c = c'.$$

In the case $c = a$ and $c' = a'$, however, we have

$$c = c, \quad a = a', \quad b = \pm b',$$

and, since

$$b \geq 0, \quad b' \geq 0,$$

we therefore also have $b = b'$. □

DEFINITION 23 (34). $F = \{a, b, c\}$ is called *primitive* if $(a, b, c) = 1$; otherwise it is called *imprimitive*.

THEOREM 24 (199). *If $F \sim G$, then if F is primitive so is G , and if F is imprimitive so is G .*

Proof. From (2.1), (2.2), and (129–131), it follows that $(a, b, c)|(a_1, b_1, c_1)$; by symmetry we also have $(a_1, b_1, c_1)|(a, b, c)$, so that $(a, b, c) = (a_1, b_1, c_1)$. $\square$

The totality of classes of forms thus fall into classes of primitive forms and classes of imprimitive forms—for brevity, primitive and imprimitive classes.

THEOREM 25 (200). *If F is imprimitive, so that $(a, b, c) = g > 1$, then $g^2|d$, and $\{\frac{a}{g}, \frac{b}{g}, \frac{c}{g}\}$ is a primitive form of discriminant $\frac{d}{g^2}$. And conversely.*

$\frac{-d}{g^2}$ is necessarily > 0 if $a > 0$; and conversely.

Proof. $\left(\frac{b}{g}\right)^2 - 4\frac{a}{g} \cdot \frac{c}{g} = \frac{d}{g^2}$ is an integer, since $g^2|b^2$ and $g^2|4ac$. Moreover, $\left(\frac{a}{g}, \frac{b}{g}, \frac{c}{g}\right) = 1$. Conversely, if $\{a_1, b_1, c_1\}$ is primitive and has discriminant d_1 , then $\{ga_1, gb_1, gc_1\}$ is imprimitive and has discriminant g^2d_1 . $\square$

We thus obtain all of the classes having discriminant d from all of the primitive classes having discriminants of the form $\frac{d}{g^2}$ where $g > 0$ and $g^2|d$, by multiplying each class by g . Thus, if we had not yet proved the finiteness of the class number, it would nevertheless follow from the fact that, in the sequel, the finiteness of the number $f(d)$ of primitive classes will be proved once again. The number of (primitive and imprimitive) classes for any d is exactly

$$\sum_{g>0, g^2|d} f\left(\frac{d}{g^2}\right).$$

In any case it suffices, from now on, to consider only primitive forms. Consequently, we shall henceforth take

$$d \text{ non-square and } \equiv 0 \text{ or } 1 \pmod{4}, \quad b^2 - 4ac = d, \quad (a, b, c) = 1;$$

in case $d < 0$, we shall, in addition, take $a > 0$.

Chapter 4

PRIMARY REPRESENTATIONS BY FORMS

DEFINITION 26 (35). Let $k \neq 0$. Then

$$F(x, y) = k$$

is said to be a *proper representation* of k by the form F if $(x, y) = 1$; an *improper representation* if $(x, y) > 1$.

THEOREM 27 (201). Let $k > 0$, and let $F(x, y) = k$ be a proper representation. Then r , s , and l may be chosen in exactly one way so that

$$\frac{x}{y} = \frac{r}{s} \pmod{4k}, \quad 0 \leq l < 2k, \quad (138)$$

$$l \equiv d \pmod{4k}, \quad 0 \leq l < 2k, \quad (139)$$

and so that F goes into $\{k, l, m\}$ by the transformation $\begin{pmatrix} r & s \\ t & u \end{pmatrix}$ where m is the number which, in accordance with (139), is determined by

$$l^2 - 4km = d.$$

(We already know, by virtue of (2.1) and Theorem 191, that the first coefficient of the new form is independent of the choice of r and s satisfying (138), and that the second coefficient, namely l , satisfies the congruence in (139). The theorem therefore is that for exactly one choice of r and s satisfying (138) will the second coefficient belong to the interval $0 \leq l < 2k$. The point is that with each proper representation of k there is associated a well-determined number l belonging to the interval $0 \leq l < 2k$.)

Proof. By Theorem 68, all the solutions of (138) are of the form

$$r = r_0 + hk, \quad s = s_0 + hy,$$

where r_0, s_0 is any solution and h is arbitrary.

Let $F = \{a, b, c\}$; if $\{k, l, m\}$ represents the new form, then it follows from (2.2) that

$$\begin{aligned} l &= 2arx + b(xs + yr) + 2cys \\ &= 2ar_0x + b(xs_0 + yr_0) + 2cy_0 + h(2ax^2 + bxy + bxy + 2cy^2) = l_0 + 2hk; \end{aligned}$$

hence for exactly one h we have $0 \leq l < 2k$; from $l^2 - 4km = d$ it follows, finally, that

$$l^2 \equiv d \pmod{4k}.$$

□

DEFINITION 28 (36). For a given form F , two proper primary representations of k are said to be *associated* if the corresponding values of l are equal. The totality of proper primary representations associated with a given l satisfying (139) are said to form a *complete system* of proper primary representations.

THEOREM 29 (202). *Each complete system of proper primary representations of k by a form F contains exactly w representations, where $w = 2$ for $d < -4$, $w = 4$ for $d = -4$, and $w = 6$ for $d = -3$.*

Proof. By Theorem 201, each proper primary representation of k corresponds to exactly one form $\{k, l, m\}$ into which F goes by a transformation of determinant 1. By Theorem 192, there are exactly w transformations of determinant 1 which take $\{k, l, m\}$ into itself (automorphs); and by Theorem 191, each of these yields a proper primary representation of k . Conversely, every proper primary representation belonging to the same l arises in this way. □

THEOREM 30 (203). *The proper primary representations of k which belong to l and are furnished by a form F are in one-to-one correspondence with the transformations of determinant 1 which take F into $\{k, l, m\}$.*

Proof. If $x = rX + sY$, $y = tX + uY$ takes F into $\{k, l, m\}$, and if $\begin{pmatrix} r & s \\ t & u \end{pmatrix}$ and $\begin{pmatrix} r' & s' \\ t' & u' \end{pmatrix}$ yield the same proper primary representation of k , then by substituting $X = 1$, $Y = 0$, we have $x = r$, $y = t$ and $x = r'$, $y = t'$; hence $r = r'$, $t = t'$, and from $ru - st = 1 = r'u' - s't'$ and $r = r'$, $t = t'$ it follows that $u = u'$, $s = s'$. $\square$

DEFINITION 31 (37). A *representative system* of the (primitive) classes of forms (where, if $d < 0$, we take $a > 0$) shall mean a set of representatives, one from each class, having $a > 0$.

Such a representative, having $a > 0$, certainly exists in the case $d > 0$, for every form represents a positive number, and therefore represents some particular positive number properly, and is therefore equivalent to a form having this number as its first coefficient.

THEOREM 32 (204). *Let $k > 0$, and let $(k, d) = 1$. Then the number $\psi(k)$ of primary representations of k by all the forms belonging to a representative system is finite, and its value is given by*

$$\psi(k) = w \sum_{f|k} \left(\frac{d}{f} \right)$$

(Kronecker symbol!).

Proof. We first consider the proper primary representations. By the Remark in Theorem 97, the conditions

$$l^2 \equiv d \pmod{4k}, \quad 0 \leq l < 2k$$

have exactly $\chi(k)$ solutions, where f runs through all the square-free positive divisors of k . (Here the condition $(k, d) = 1$ was made use of.) For every such f , the form $\{k, l, m\}$, where m is determined from $l^2 - 4km = d$, is equivalent to exactly one form of the representative system. By means of this form, we obtain by Theorem 203 exactly w proper primary representations belonging to l . The number of proper primary representations of k by forms belonging to the representative system is therefore

$$w \sum_{f|k} \left(\frac{d}{f} \right).$$

By the second of the two remarks that precede Theorem 203, it follows that the number of primary representations of k by forms belonging to the representative system is

$$\psi(k) = w \sum_{\substack{g>0 \\ g^2|k}} \sum_{f|\frac{k}{g^2}} \left(\frac{d}{f} \right) = w \sum_{f|k} \left(\frac{d}{f} \right);$$

for every $n > 0$ can be uniquely written in the form $n = fg^2$, where f is square-free and $g > 0$; and then $n|k$ implies $g^2|k$, $f|\frac{k}{g^2}$, and conversely. $\square$

Chapter 5

REPRESENTATIONS OF $h(d)$ IN TERMS OF $K(d)$

We should now like to investigate

$$H(t, F) = \sum_{\substack{1 \leq k \leq t \\ (k, d)=1}} \psi(k, F),$$

where $\psi(k, F)$ is the number of primary representations of k by a fixed form F of the representative system; we ask whether—analogously to Theorem 205—

$$\lim_{t \rightarrow \infty} \frac{H(t, F)}{t} \quad (157)$$

exists. We shall find that:

- 1) It does.
- 2) The limit is independent of F ; thus it depends only on d .
- 3) It is > 0 .
- 4) It can be written explicitly.

If, for the moment, we denote it by $M(d)$, then the following is a consequence of what has just been stated.

If $F_1, F_2, \dots, F_H$ are representatives of a finite number of distinct classes (we behave as though we did not already know the class number to be finite), then we clearly have

$$\sum_{n=1}^H H(t, F_n) \leq H(t),$$

so that, by Theorem 205,

$$\sum_{n=1}^H \frac{H(t, F_n)}{t} \leq \frac{H(t)}{t}, \quad H \cdot M(d) = \sum_{n=1}^H \lim_{t \rightarrow \infty} \frac{H(t, F_n)}{t} \leq \lim_{t \rightarrow \infty} \frac{H(t)}{t} = \frac{w}{|d|} K(d).$$

Consequently, H is bounded, so that the class number is finite. Call it $h = h(d)$. Now, if F_n runs over the entire representative system, we have

$$\sum_{n=1}^h H(t, F_n) = H(t),$$

and therefore

$$h \cdot M(d) = \frac{w}{|d|} K(d). \quad (158)$$

This is precisely the goal of this chapter. The class number will once more be “determined”; to be sure, the heralded proof of our assertion regarding the limit in (157) will be somewhat tedious.

But (158) is far from being our final formula; for we shall manage to express the series

$$K(d) = \sum_{n=1}^{\infty} \left(\frac{d}{n} \right) \frac{1}{n}$$

in closed form. This, however, will take three full chapters more!

THEOREM 33 (206). *If x and y each run through a complete set of residues mod $|d|$, then exactly $|d|\varphi(|d|)$ of the d^2 numbers $F(x, y)$ which result are relatively prime to d .*

Remark. That the number is independent of the choice of the sets of residues is clear to begin with; for from $x \equiv x'$ and $y \equiv y'$, it follows that $F(x, y) \equiv F(x', y')$.

Proof. It suffices to show, for $p^f | d$, $f > 0$, that if x and y each run through a complete set of residues mod p^f , then $p \nmid F(x, y)$ exactly $p^{2f} \varphi(p^f)$ times. For if the canonical decomposition of $|d|$ is $|d| = \prod p_i^{f_i}$, then, since $(F, d) = 1$ is equivalent to $p_i \nmid F$ for all $p_i | d$, by Theorem 71 there are exactly $\prod p_i^{2f_i} \varphi(p_i^{f_i}) = |d|^2 \varphi(|d|)$ pairs of classes $x \equiv x_0 \pmod{|d|}$, $y \equiv y_0 \pmod{|d|}$.

Since $(a, b, c) = 1$ and $b^2 - 4ac = d$, we cannot have both $p|a$ and $p|c$ if $p|d$. Without loss of generality, let $p \nmid a$. (For, $a > 0$ is not used in the following proof.)

1) Let $p > 2$. Then, since $(p, 4a) = 1$, it depends precisely on $4aF = (2ax + by)^2 - dy^2$ not being divisible by p , that is, since $p|d$, on

$$2ax + by \not\equiv 0 \pmod{p}.$$

For each of our p^f values of y , since $p \nmid 2a$, a unique value of $x \pmod{p^f}$ satisfies $2ax + by \equiv 0 \pmod{p^f}$, so that p^{f-1} values of $x \pmod{p^f}$ satisfy $2ax + by \equiv 0 \pmod{p}$. Hence $p^f - p^{f-1} = \varphi(p^f)$ values of $x \pmod{p^f}$ satisfy $2ax + by \not\equiv 0 \pmod{p}$.

2) Let $p = 2$. Since $d \equiv 0$ or $1 \pmod{4}$, we have $f \geq 2$ for $d \equiv 0 \pmod{4}$, and d is odd for $f = 1$. If d is odd, then b is odd, and $F(x, y)$ is odd for all x, y . If $d \equiv 0 \pmod{4}$, then b is even, and $F(x, y) \equiv ax^2 + cy^2 \pmod{2}$; since $(a, b, c) = 1$ and b is even, at least one of a, c is odd, so that for half of the pairs $x, y \pmod{2^f}$ we have $2 \nmid F(x, y)$. $\square$

THEOREM 34 (207). *Let $m > 0$. Suppose given an ellipse or a sector of an hyperbola (the curvilinear triangle bounded by an arc of the hyperbola and two rays drawn from its endpoints to the center of the hyperbola); let I be its area. Suppose the figure stretched by an amount $\sqrt{t}$, $t > 0$ (that is, suppose we consider the set of points $\sqrt{t}\xi$, $\sqrt{t}\eta$ instead of the original set of points ξ , η). Let $U(t)$ denote the number of points with integral coordinates (so-called lattice points) within the extended figure (with each boundary point either counted or not, as we please) which satisfy the additional conditions:*

$$x \equiv x_0 \pmod{m}, \quad y \equiv y_0 \pmod{m}.$$

Then we have

$$\lim_{t \rightarrow \infty} \frac{U(t)}{t} = \frac{I}{m^2}.$$

Proof. In the plane of the original figure, let us lay out two mutually perpendicular systems of parallel lines, spaced $\frac{1}{\sqrt{t}}$ units apart, around the point x_0, y_0 . Let $W(t)$ be the number of squares in this net which are contained in the ellipse or in the sector of the hyperbola, as the case may be. Then

$$W(t) \cdot \frac{1}{t} \leq \frac{U(t)}{t} \leq (W(t) + \text{boundary squares}) \cdot \frac{1}{t},$$

and since the number of boundary squares grows only as $\sqrt{t}$, the result follows. $\square$

THEOREM 35 (208). *The following formulas hold for the function $H(t, F)$, defined above:*

$$\lim_{t \rightarrow \infty} \frac{H(t, F)}{t} = \begin{cases} \frac{2\pi}{w\sqrt{|d|}} & \text{for } d < 0, \\ \frac{\log \varepsilon}{w\sqrt{d}} & \text{for } d > 0, \end{cases}$$

where ε is defined as in Theorem 111.

Proof. By Theorem 206, it is sufficient to prove that the number of solutions $U(t) = U(t, F, x_0, y_0)$ of

$$0 \leq F(x, y) \leq t, \quad x \equiv x_0 \pmod{|d|}, \quad y \equiv y_0 \pmod{|d|},$$

which, in case $d > 0$, also satisfy conditions (146), has the property that

$$\lim_{t \rightarrow \infty} \frac{U(t)}{t} = \frac{2\pi}{|d|\sqrt{|d|}} \text{ for } d < 0, \quad \frac{2 \log \varepsilon}{|d|\sqrt{d}} \text{ for } d > 0.$$

($F(x, y) = 0$ is allowed, since this equation can only be satisfied by $x = y = 0$, because of the irrationality of $\sqrt{d}$.)

1) Let $d < 0$. By Theorem 207, we need only note that the ellipse

$$ax^2 + bxy + cy^2 \leq 1$$

(from which, after stretching, we obtain $F(x, y) \leq t$) has area $\frac{2\pi}{\sqrt{|d|}}$. The reader certainly knows this.

2) Let $d > 0$. By Theorem 207, it suffices to prove, using the abbreviations

$$\Lambda = 2a\xi + (b + \sqrt{d})\eta, \quad \bar{\Lambda} = 2a\xi + (b - \sqrt{d})\eta,$$

that the hyperbola sector

$$a\xi^2 + b\xi\eta + c\eta^2 \leq 1, \quad \Lambda > 0, \quad 1 \leq \frac{\Lambda}{\bar{\Lambda}} \leq \varepsilon^2 \tag{159}$$

has area $\frac{2 \log \varepsilon}{\sqrt{d}}$. $\square$

THEOREM 36 (209).

$$h(d) = \begin{cases} \frac{w\sqrt{|d|}}{2\pi} K(d) & \text{for } d < 0, \\ \frac{w\sqrt{d}}{\log \varepsilon} K(d) & \text{for } d > 0. \end{cases}$$

Proof. It follows from Theorem 205 and Theorem 208 (cf. (158)) that

$$h(d) \cdot \frac{2\pi}{w\sqrt{|d|}} = \frac{w}{|d|} K(d) \quad \text{for } d < 0, \quad h(d) \cdot \frac{\log \varepsilon}{w\sqrt{d}} = \frac{w}{|d|} K(d) \quad \text{for } d > 0.$$

Now we may forget completely the meaning of h , as well as the quadratic forms, and we have merely (merely!) to find the sum of the series

$$K(d) = \sum_{n=1}^{\infty} \left(\frac{d}{n} \right) \frac{1}{n}.$$

□

Chapter 6

GAUSSIAN SUMS

I shall assume that the reader is familiar with the following theorem from the classical theory of Fourier series:

THEOREM 37 (210). *Let $f(\xi)$ be defined for $0 \leq \xi \leq 1$ (as a real or a complex function; the case in which $f(\xi)$ is complex follows immediately from the real case); moreover, let $f(\xi)$ be continuous and let $f'(\xi)$ also exist and be continuous. If we set*

$$\alpha_0 = \int_0^1 f(\eta) d\eta, \quad \alpha_h = 2 \int_0^1 f(\eta) \cos 2\pi h\eta d\eta, \quad \beta_h = 2 \int_0^1 f(\eta) \sin 2\pi h\eta d\eta,$$

then we have

$$\frac{\alpha_0}{2} + \sum_{h=1}^{\infty} (\alpha_h \cos 2\pi h\xi + \beta_h \sin 2\pi h\xi) = \begin{cases} f(\xi) & \text{for } 0 < \xi < 1, \\ \frac{f(0) + f(1)}{2} & \text{for } \xi = 0, 1. \end{cases}$$

In particular, therefore (since $\alpha_{-h} = \alpha_h$ and $\beta_{-h} = -\beta_h$), we have

$$\begin{aligned} \frac{f(0) + f(1)}{2} &= \lim_{N \rightarrow \infty} \sum_{h=-N}^N \int_0^1 f(\eta) e^{-2\pi i h \eta} d\eta; \\ \frac{f(0) + f(1)}{2} &= \lim_{N \rightarrow \infty} \sum_{h=-N}^N \int_0^1 f(\eta) e^{-2\pi i h \eta} d\eta. \end{aligned} \tag{160}$$

THEOREM 38 (211). *Let $n > 0$. Then*

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \begin{cases} (1+i)\sqrt{n} & \text{for } n \equiv 0 \pmod{4}, \\ \sqrt{n} & \text{for } n \equiv 1 \pmod{4}, \\ 0 & \text{for } n \equiv 2 \pmod{4}, \\ i\sqrt{n} & \text{for } n \equiv 3 \pmod{4}. \end{cases}$$

Remark. There are proofs that are more elementary (cf. the Appendix to this chapter); the reader cannot, however, become accustomed early enough to the use of analysis; moreover, the following proof (using (160)) is the shortest of the many proofs of this theorem, which was one of Gauss's

greatest discoveries. All that is required, incidentally, for the determination of the class number, is the specialization

$$\sum_{r=0}^{p-1} e^{\frac{2\pi i r^2}{p}} = \begin{cases} \sqrt{p} & \text{for } p \equiv 1 \pmod{4}, \\ i\sqrt{p} & \text{for } p \equiv 3 \pmod{4}, \end{cases}$$

which will lead directly to Theorem 212. Nevertheless, I should not like to prove the formula

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \begin{cases} \sqrt{n} & \text{for } n \equiv 1 \pmod{4}, \\ i\sqrt{n} & \text{for } n \equiv 3 \pmod{4}, \end{cases}$$

for prime numbers only, since it follows for all positive odd numbers with just as little trouble, or rather (today!) with just as much ease.

Proof. Since

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \sum_{s=0}^{n-1} e^{\frac{2\pi i (s+n)^2}{n}},$$

it follows that

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \sum_{s=0}^{n-1} \sum_{k=-\infty}^{\infty} \int_0^1 e^{\frac{2\pi i (s+n\eta)^2}{n}} e^{-2\pi i k\eta} d\eta.$$

By (160), with

$$f(\eta) = e^{\frac{2\pi i (s+n\eta)^2}{n}} = f_s(\eta),$$

we have

$$\begin{aligned} \sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} &= \lim_{N \rightarrow \infty} \sum_{h=-N}^N \sum_{s=0}^{n-1} \int_0^1 e^{\frac{2\pi i (s+n\eta)^2}{n}} e^{-2\pi i h\eta} d\eta \\ &= \lim_{N \rightarrow \infty} \sum_{h=-N}^N \int_0^n e^{\frac{2\pi i \xi^2}{n}} e^{-\frac{2\pi i h\xi}{n}} d\xi \end{aligned}$$

(setting $\xi = s + n\eta$).

$$= \lim_{N \rightarrow \infty} \sum_{h=-N}^N \int_0^n e^{\frac{2\pi i \xi^2}{n}} e^{-\frac{2\pi i h\xi}{n}} d\xi. \quad (161)$$

Now it is well known that

$$\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1$$

converges; it is not necessary for us to know the value, since it will come out of what follows. Hence, if we set $\xi = \sqrt{n}x$, we have, on the one hand,

$$\int_{-\infty}^{\infty} e^{\frac{2\pi i \xi^2}{n}} e^{-\frac{2\pi i h\xi}{n}} d\xi = \sqrt{n} \int_{-\infty}^{\infty} e^{2\pi i x^2} e^{-2\pi i h x} dx = \sqrt{n} \cdot e^{-\frac{\pi i h^2}{2}};$$

and on the other hand,

$$\sum_{k=-\infty}^{\infty} e^{\frac{2\pi i (s+kn)^2}{n}} = \sum_{k=-\infty}^{\infty} e^{\frac{2\pi i s^2}{n}} e^{4\pi i s k} e^{2\pi i k^2} = \sum_{k=-\infty}^{\infty} e^{\frac{2\pi i s^2}{n}};$$

since

$$\sum_{k=-\infty}^{\infty} e^{2\pi i(s+kn)\frac{h}{n}} = \sum_{k=-\infty}^{\infty} e^{\frac{2\pi i sh}{n}} e^{2\pi i kh} = \begin{cases} n & \text{if } n|h, \\ 0 & \text{otherwise.} \end{cases}$$

Hence, because of (161), we have

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \sqrt{n} \cdot \gamma \cdot (1 + i^{-n}),$$

where γ is an absolute constant. The absolute constant γ is determined by setting $n = 1$ in the above:

$$1 = \sqrt{1} \cdot \gamma \cdot (1 - i), \quad \gamma = \frac{1}{1 - i} = \frac{1 + i}{2}.$$

It therefore follows that

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \sqrt{n} \cdot \frac{1 + i}{2} \cdot (1 + i^{-n}),$$

and in this formula we have

$$(1 + i)(1 + i^{-n}) = \begin{cases} 2(1 + i) & \text{for } n \equiv 0 \pmod{4}, \\ 2 & \text{for } n \equiv 1 \pmod{4}, \\ 0 & \text{for } n \equiv 2 \pmod{4}, \\ 2i & \text{for } n \equiv 3 \pmod{4}. \end{cases}$$

□

THEOREM 39 (212).

$$\sum_{r=1}^{p-1} \left(\frac{r}{p}\right) e^{\frac{2\pi ir}{p}} = \begin{cases} \sqrt{p} & \text{for } p \equiv 1 \pmod{4}, \\ i\sqrt{p} & \text{for } p \equiv 3 \pmod{4}. \end{cases}$$

Remark. Except for sign, this is quite easily proven (only after years of effort did Gauss succeed in determining the sign).

Namely, if we set

$$\varrho = \sum_{r=1}^{p-1} \left(\frac{r}{p}\right) e^{\frac{2\pi ir}{p}},$$

for the sake of abbreviation, then $\left(\frac{r}{p}\right)$ and ϱ have period p for $p \nmid r$ we have

$$\varrho^2 = \sum_{r=1}^{p-1} \sum_{s=1}^{p-1} \left(\frac{r}{p}\right) \left(\frac{s}{p}\right) e^{\frac{2\pi i(r+s)}{p}},$$

the sum being taken over a reduced set of residues; hence, if s also runs through such a set, we have

$$\varrho^2 = \sum_{t=1}^{p-1} \sum_{r=1}^{p-1} \left(\frac{r \cdot tr}{p}\right) e^{\frac{2\pi ir(1+t)}{p}} = \sum_{t=1}^{p-1} \left(\frac{t}{p}\right) \sum_{r=1}^{p-1} e^{\frac{2\pi ir(1+t)}{p}},$$

since rt also runs through a reduced set of residues when t does; hence we have

$$\varrho^2 = \sum_{t=1}^{p-1} \left(\frac{t}{p}\right) \sum_{r=1}^{p-1} e^{\frac{2\pi ir(1+t)}{p}} = \sum_{t=1}^{p-1} \left(\frac{t}{p}\right) (-1) = \left(\frac{-1}{p}\right) p,$$

since

$$\sum_{r=0}^{p-1} e^{\frac{2\pi ir(1+t)}{p}} = \begin{cases} p & \text{for } t = p-1, \\ 0 & \text{for } 1 \leq t \leq p-2; \end{cases}$$

it follows that

$$\begin{aligned} \varrho^2 &= \left(\frac{-1}{p}\right) p = \begin{cases} p & \text{for } p \equiv 1 \pmod{4}, \\ -p & \text{for } p \equiv 3 \pmod{4}, \end{cases} \\ \varrho &= \begin{cases} \pm\sqrt{p} & \text{for } p \equiv 1 \pmod{4}, \\ \pm i\sqrt{p} & \text{for } p \equiv 3 \pmod{4}. \end{cases} \end{aligned}$$

These remarks, incidentally, are not made use of in the following complete proof of Theorem 212.

Proof. If a runs over the quadratic residues and b over the quadratic non-residues mod p in the interval $0 < r < p$, then we have

$$\sum_a e^{\frac{2\pi ia}{p}} = \sum_{r=1}^{p-1} \frac{1 + \left(\frac{r}{p}\right)}{2} e^{\frac{2\pi ir}{p}}, \quad \sum_b e^{\frac{2\pi ib}{p}} = \sum_{r=1}^{p-1} \frac{1 - \left(\frac{r}{p}\right)}{2} e^{\frac{2\pi ir}{p}}.$$

On the other hand, we have

$$\sum_{r=0}^{p-1} e^{\frac{2\pi ir^2}{p}} = \sum_a e^{\frac{2\pi ia}{p}} + 2 \sum_b e^{\frac{2\pi ib}{p}},$$

so that

$$\varrho = \sum_{r=1}^{p-1} \left(\frac{r}{p} \right) e^{\frac{2\pi i r}{p}} = \sum_a e^{\frac{2\pi i a}{p}} - \sum_b e^{\frac{2\pi i b}{p}}.$$

By Theorem 79, the a coincide with the remainders of $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ upon division by p . The remainders of s^2 , for $1 \leq s \leq p-1$, therefore yield each of the a exactly twice. Consequently, by Theorem 211,

$$\sum_{r=0}^{p-1} e^{\frac{2\pi i r^2}{p}} = \sum_a e^{\frac{2\pi i a}{p}} + 2 \sum_b e^{\frac{2\pi i b}{p}} = \begin{cases} \sqrt{p} & \text{for } p \equiv 1 \pmod{4}, \\ i\sqrt{p} & \text{for } p \equiv 3 \pmod{4}. \end{cases}$$

□

APPENDIX

Introduction

Because of the importance of Theorem 211, I should now like to present three more proofs, at least for odd n , which are quite different from the above and from each other.

The first, due to Kronecker, makes use of complex function theory, in particular of Cauchy's Theorem; it also yields a proof, incidentally, for even n .

The second proof is due to I. Schur and makes use of matrix algebra.

The third proof is due to Mertens, and its calculations involve only (though extensively) finite trigonometric sums.

Incidentally, Theorem 211 is trivial for $n \equiv 2 \pmod{4}$; for then we have

$$e^{\frac{2\pi i(s+\frac{n}{2})^2}{n}} = e^{\frac{2\pi i s^2}{n}} e^{2\pi i s} e^{\frac{\pi i n}{2}} = -e^{\frac{2\pi i s^2}{n}},$$

so that the terms in the sum cancel each other out in pairs.

The case $n \equiv 0 \pmod{4}$ could be reduced to the case $4 \nmid n$ in an elementary manner; but I shall omit the lengthy proof of this statement since, in any case, it is only Theorem 212, that is, Theorem 211 for odd (and indeed prime) n which is later made use of.

§ 1 Kronecker's Proof

Let $n > 0$. For $0 < \varrho < \frac{1}{2}$, $\omega > 1$, let us consider the integral

$$I = \int \frac{e^{\frac{2\pi i \xi}{n}}}{e^{2\pi i \xi} - 1} d\xi,$$

which is taken in the positive direction around the rectangle with vertices $\pm\omega \pm i\omega$, in which—in order to avoid the points 0 (a pole of the integrand) and $\frac{n}{2}$ (a pole, in case n is even)—semicircular indentations have been made, the semicircles being of radius ϱ with centers at 0 and $\frac{n}{2}$, respectively. The integrand is regular along this path, and inside the path has the poles s , where $1 \leq s \leq n-1$. (For $n = 1$ or 2 there are no poles at all inside.)

I is therefore independent of ϱ and ω (that is, it depends only on n); since the residue of the integrand at s is clearly $\frac{1}{n}e^{\frac{2\pi i s^2}{n}}$, it follows from Cauchy's Theorem that

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = nI,$$

where the symbol $\sum'$ indicates that if n is even, then the term for which $s = \frac{n}{2}$ (which would, incidentally, be 0) is omitted from the sum.

The sum of the integrals over both of the straight-line segments of the boundary on the left $\xi = i\eta$, $\varrho \leq \eta \leq \omega$ and $\xi = -i\eta$, $\varrho \leq \eta \leq \omega$, is

$$-\frac{i}{n} \int_{\varrho}^{\omega} \frac{e^{-\frac{2\pi i \eta^2}{n}}}{e^{2\pi i \eta} - 1} d\eta + \frac{i}{n} \int_{\varrho}^{\omega} \frac{e^{\frac{2\pi i \eta^2}{n}}}{e^{2\pi i \eta} - 1} d\eta = \frac{2}{n} \int_{\varrho}^{\omega} \frac{\sin \frac{2\pi \eta^2}{n}}{e^{2\pi i \eta} - 1} d\eta;$$

as $\omega \rightarrow \infty$, this expression approaches

$$\frac{2}{n} \int_{\varrho}^{\infty} \frac{\sin \frac{2\pi \eta^2}{n}}{e^{2\pi i \eta} - 1} d\eta.$$

Since

$$e^{2\pi i(\xi + \frac{n}{2})} = e^{2\pi i \xi} \cdot e^{\pi i n} = e^{2\pi i \xi}, \quad e^{\frac{2\pi i(\xi + \frac{n}{2})^2}{n}} = e^{\frac{2\pi i \xi^2}{n}} e^{2\pi i \xi} e^{\frac{\pi i n}{2}} = i^n e^{\frac{2\pi i \xi^2}{n}},$$

and

$$e^{2\pi i(-\xi)} = e^{-2\pi i \xi}, \quad e^{\frac{2\pi i(-\xi)^2}{n}} = e^{\frac{2\pi i \xi^2}{n}},$$

the sum of the integrals over both of the straight boundary segments $\xi = \omega + i\eta$, $\omega - \varrho \leq \eta \leq \omega$ and $\xi = -\omega + i\eta$, $\omega - \varrho \leq \eta \leq \omega$ on the right is

$$\frac{1}{n} \int_{\varrho}^{\omega} \frac{e^{\frac{2\pi i(\omega + i\eta)^2}{n}}}{e^{2\pi i(\omega + i\eta)} - 1} d\eta + \frac{1}{n} \int_{\varrho}^{\omega} \frac{e^{\frac{2\pi i(-\omega + i\eta)^2}{n}}}{e^{2\pi i(-\omega + i\eta)} - 1} d\eta.$$

as $\omega \rightarrow \infty$, this expression approaches 0.

The integrals over the horizontal segments of the boundary approach 0 as $\omega \rightarrow \infty$; since for $\xi = t + i\eta$, $2\omega \leq t \leq 0$ we have

$$\left| \frac{e^{\frac{2\pi i \xi^2}{n}}}{e^{2\pi i \xi} - 1} \right| \leq \frac{e^{-\frac{4\pi t \eta}{n}}}{|1 - e^{-2\pi \eta}|} \leq \frac{1}{1 - e^{-2\pi}} \cdot e^{-\frac{4\pi \omega}{n}},$$

so that

$$\left| \int \frac{e^{\frac{2\pi i \xi^2}{n}}}{e^{2\pi i \xi} - 1} d\xi \right| \leq \frac{1}{1 - e^{-2\pi}} \cdot e^{-\frac{4\pi\omega}{n}} \cdot 2\omega \rightarrow 0;$$

and for $\xi = t - i\eta$, $0 \leq t \leq 2\omega$ we have

$$\left| \frac{e^{\frac{2\pi i \xi^2}{n}}}{e^{2\pi i \xi} - 1} \right| \leq \frac{e^{\frac{4\pi t \eta}{n}}}{|e^{2\pi i \eta} - 1|} \leq \frac{1}{e^{2\pi} - 1} \cdot e^{\frac{8\pi\omega}{n}}.$$

We therefore have

$$\sum_{s=0}^{n-1} ' e^{\frac{2\pi i s^2}{n}} = J_1 + J_2,$$

where J_1 and J_2 denote the integrals over the semicircles around 0 and $\frac{n}{2}$ respectively.

We now consider the limit as $\varrho \rightarrow 0$. We have

$$\lim_{\varrho \rightarrow 0} J_1 = \int_{(0)} \frac{e^{\frac{2\pi i \xi^2}{n}}}{e^{2\pi i \xi} - 1} d\xi;$$

furthermore, if $f(\xi)$ has a pole of order one with residue α at ξ_0 , then by integrating in the negative direction around a semicircle (0) with center at ξ_0 and radius ϱ we have

$$\lim_{\varrho \rightarrow 0} \int_{(0)} f(\xi) d\xi = -\pi i \alpha.$$

ELEMENTARY NUMBER THEORY

Conclusion (Pages 206–256)

Edmund Landau

Translated by Jacob E. Goodman
With Exercises by Paul T. Bateman and Eugene E. Kohlbecker

Appendix to Chapter VI

§1 Conclusion of the Analytic Proof

(Picking up from page 206 of the text.)

We have, for $0 < \varrho < \delta$,

$$\lim_{\varrho \rightarrow 0} \int \frac{f(\xi)}{\xi} d\xi = -\alpha_0 \pi i,$$

$$\epsilon = \varrho e^{i\psi}, \quad 0 < \psi < \pi,$$

for

$$O = \alpha_0 + \eta(\epsilon),$$

where $\eta(\xi)$ is regular at 0; and therefore, since

$$\lim_{\varrho \rightarrow 0} \int_{\mathfrak{p}_0} \frac{\eta(\xi)}{\xi} d\xi = 0$$

(length of path $\rightarrow 0$, and integrand uniformly bounded for small ϱ), we have

$$\lim_{\varrho \rightarrow 0} \int_{\mathfrak{p}_0} \frac{f(\xi)}{\xi} d\xi = \alpha_0 \lim_{\varrho \rightarrow 0} \int_{\pi}^0 \frac{d\epsilon}{\epsilon} = \alpha_0 \lim_{\varrho \rightarrow 0} (-\pi i) = -\alpha_0 \pi i.$$

Consequently (in case n is odd $\frac{1}{\xi}$ is regular and we have simply $\lim_{\varrho \rightarrow 0} I_0 = 0$),

$$\begin{aligned} \int_0^1 \frac{f(x)}{x} dx &= \frac{1}{2\pi i} \cdot \frac{2\pi i}{n} \sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} \\ &= \frac{1}{n} \sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}}, \end{aligned}$$

in case $2 \nmid n$,

$$\begin{aligned} \frac{2\pi i}{n} \cdot \frac{n}{2} &= \pi i, \\ \frac{1}{n} \sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} &= \frac{1}{2} (1 + (-1)^{\frac{n}{2}}), \end{aligned}$$

so that, in any case,

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = 2i(1 + (-i)^n) \int_0^\infty e^{-\frac{2\pi n t^2}{1}} dt = 2i(1 - (-i)^n) \int_0^\infty e^{-2\pi n t^2} dt.$$

The integral on the right can be evaluated by the specialization $n = 1$; we have

$$1 = 2i(1 - i) \int_0^\infty e^{-2\pi t^2} dt,$$

so that

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \frac{1 + (-i)^n}{1 - i} \sqrt{n}$$

these are the four values of our theorem, depending on the residue class mod 4.

§2 Schur's Proof

I shall not take for granted any knowledge of matrix calculus, and shall develop:

- 1) The concept: A matrix is a square array

$$\mathfrak{A} = \begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1n} \\ \vdots & \ddots & \vdots \\ \alpha_{n1} & \cdots & \alpha_{nn} \end{pmatrix} = (\alpha_{kl})$$

of n^2 elements. (The numbering may of course also run, say, from 0 to $n - 1$.)

- 2) The definition of multiplication:

$$(\alpha_{kl})(\beta_{kl}) = \left(\sum_m \alpha_{km} \beta_{ml} \right).$$

- 3) The definition

$$\mathfrak{A}^2 = \mathfrak{A}\mathfrak{A}.$$

- 4) The notation

$$e_{kl} = \begin{cases} 1 & \text{for } k = l, \\ 0 & \text{for } k \neq l. \end{cases}$$

- 5) The definition of the characteristic function of a matrix $\mathfrak{A} = (\alpha_{kl})$; it is the determinant

$$\Phi(\xi) = |\xi e_{kl} - \alpha_{kl}|.$$

$\Phi(\xi)$ is a polynomial of exactly n -th degree with leading coefficient 1, and therefore $= \prod_{r=1}^n (\xi - \xi_r)$; the ξ_r are called the characteristic roots of the matrix.

The coefficient of ξ^{n-1} is $-\sum_k \alpha_{kk}$, so that

$$\sum_{r=1}^n \xi_r = \sum_k \alpha_{kk}.$$

$\sum_k \alpha_{kk}$ is called the *trace* of the matrix.

- 6) The THEOREM: The characteristic roots of $\mathfrak{A}^2$ are the ξ_r^2 ; and therefore those of $(\mathfrak{A}^2)^*$ are the ξ_r^4 .

PROOF: The statement reads:

$$|\xi e_{kl} - \sum_m \alpha_{km} \alpha_{ml}| = \prod_{r=1}^n (\xi - \xi_r^2).$$

Indeed, let η be a number for which $\eta^2 = \xi$. Then the matrix product

$$\begin{aligned} (\eta e_{kl} + \alpha_{kl})(\eta e_{kl} - \alpha_{kl}) &= \left(\sum_m (\eta e_{km} + \alpha_{km})(\eta e_{ml} - \alpha_{ml}) \right) \\ &= \left(\sum_m (\eta^2 e_{km} e_{ml} + \eta \alpha_{km} e_{ml} - \eta e_{km} \alpha_{ml} - \alpha_{km} \alpha_{ml}) \right) \\ &= \left(\eta^2 e_{kl} - \sum_m \alpha_{km} \alpha_{ml} \right) = (\xi e_{kl} - \sum_m \alpha_{km} \alpha_{ml}), \end{aligned}$$

so that

$$|\eta e_{kl} + \alpha_{kl}| \cdot |\eta e_{kl} - \alpha_{kl}| = |\xi e_{kl} - \sum_m \alpha_{km} \alpha_{ml}|$$

and in this we have

$$\begin{aligned} |\eta e_{kl} + \alpha_{kl}| &= \Phi(-\eta) = \prod_{r=1}^n (-\eta - \xi_r), \\ |\eta e_{kl} - \alpha_{kl}| &= (-1)^n |-\eta e_{kl} + \alpha_{kl}| = (-1)^n \Phi(\eta) = (-1)^n \prod_{r=1}^n (\eta - \xi_r) \\ &= \prod_{r=1}^n (-\eta + \xi_r); \end{aligned}$$

so that

$$|\xi e_{kl} - \sum_m \alpha_{km} \alpha_{ml}| = \prod_{r=1}^n (\eta^2 - \xi_r^2) = \prod_{r=1}^n (\xi - \xi_r^2).$$

Now let $n > 0$ be odd, and let

$$S = \sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}}.$$

We first show that

$$|S| = \sqrt{n};$$

this will still be easy.

$$\begin{aligned} |S|^2 &= S\bar{S} = \sum_{s,t=0}^{n-1} e^{\frac{2\pi i(s^2-t^2)}{n}} \\ &= \sum_{s,t=0}^{n-1} e^{\frac{2\pi i(s+t)(s-t)}{n}}, \end{aligned}$$

the sum being taken over any two complete sets of residues mod n . Since, when t is fixed, $s+t$ also runs through such a set when s does, we have

$$\begin{aligned} |S|^2 &= \sum_{s,t=0}^{n-1} e^{\frac{2\pi i(s+t)(s-t)}{n}} = \sum_{s,t=0}^{n-1} e^{\frac{2\pi i \cdot s \cdot (s-2t)}{n}} \\ &= \sum_{s,t=0}^{n-1} e^{\frac{2\pi i s(s-2t)}{n}}. \end{aligned}$$

In this formula,

$$\sum_{t=0}^{n-1} e^{\frac{2\pi i s(s-2t)}{n}} = \begin{cases} n & \text{for } n \mid 2s, \text{ that is, } n \mid s, \\ 0 & \text{otherwise,} \end{cases}$$

so that

$$|S|^2 = n, \quad |S| = \sqrt{n}.$$

Now we set

$$\epsilon = e^{\frac{2\pi i}{n}},$$

and we consider the n -by- n matrix

$$\mathfrak{U} = (\epsilon^{kl}) \quad (k \text{ and } l = 0, 1, \dots, n-1).$$

If $\xi_1, \dots, \xi_n$ are its characteristic roots, then our

$$S = \sum_{k=0}^{n-1} \epsilon^{kk} = \sum_{r=1}^n \xi_r$$

is the trace of $\mathfrak{U}$.

We have

$$\mathfrak{U}^2 = \left(\sum_{m=0}^{n-1} \epsilon^{km} \epsilon^{ml} \right) = \left(\sum_{m=0}^{n-1} \epsilon^{(k+l)m} \right) = (\delta_{kl}),$$

where

$$\delta_{kl} = \sum_{m=0}^{n-1} \epsilon^{(k+l)m} = \begin{cases} n & \text{for } n \mid k+l, \\ 0 & \text{otherwise.} \end{cases}$$

It follows further from this that

$$(\mathfrak{U}^2)^2 = \left(\sum_{m=0}^{n-1} \delta_{k+m} \delta_{m+l} \right) = (n^2 \epsilon_{kl});$$

for if $k = l$ we have

$$\sum_{m=0}^{n-1} \delta_{k+m} \delta_{m+k} = \sum_{m=0}^{n-1} \delta_{k+m}^2 = n$$

(since n divides $k+m$ for exactly one m); for $k \neq l$ we have

$$\sum_{m=0}^{n-1} \delta_{k+m} \delta_{m+l} = 0,$$

since we never have $n \mid k+m$ and $n \mid m+l$ simultaneously.

$(\mathfrak{U}^2)^*$ has the ξ_r^4 as its roots, by 6). Moreover, $(\mathfrak{U}^2)^*$ has the characteristic function

$$|\xi e_{kl} - n^2 e_{kl}| = (\xi - n^2)^n.$$

Consequently, $(\mathfrak{U}^2)^*$ has its n roots all $= n^2$. We therefore have

$$\xi_r^4 = n^2, \quad \xi_r = \sqrt{n}, \quad \nu = 0, 1, 2, \text{ or } 3.$$

If $\xi_r/n^{1/2}$ ($r = 0, 1, 2, 3$) has multiplicity m_r , then

$$S = \sum_r \xi_r = \sqrt{n} \sum_{r=0}^3 i^r m_r = \sqrt{n} (m_0 - m_2 + i(m_1 - m_3)).$$

From

$$|S| = \sqrt{n}$$

it follows that

$$(m_0 - m_2)^2 + (m_1 - m_3)^2 = 1.$$

We therefore have

$$m_0 - m_2 = 0, \quad m_1 - m_3 = \pm 1, \quad \text{or} \quad m_0 - m_2 = \pm 1, \quad m_1 - m_3 = 0,$$

$$S = \nu\sqrt{n}, \quad \text{where } \nu = 1 \text{ and } \nu = 1 \text{ or } i.$$

Since $\mathfrak{U}^2$ has trace

$$\sum_{r=0}^{n-1} \delta_{rr} = n,$$

it follows that

$$n = \sum_{r=1}^n \xi_r^2 = n(m_0 - m_1 + m_2 - m_3);$$

we therefore have the four equations

$$\begin{aligned} m_0 + m_1 + m_2 + m_3 &= n, \\ m_0 - m_1 + m_2 - m_3 &= 1, \\ m_0 + m_1 - m_2 - m_3 &= \nu, \\ m_0 - m_1 - m_2 + m_3 &= \bar{\nu}, \end{aligned}$$

for m_0, m_1, m_2 , and m_3 . From this it follows that

$$\begin{aligned} 2(m_0 - m_2) &= \nu + \bar{\nu} = 2\Re(\nu), \\ 4m_0 &= n + 1 + \nu + \bar{\nu}. \end{aligned}$$

Since m_0 is an integer, we must have

$$n \equiv 1 \pmod{4} \quad \text{for } \nu = 1, n \equiv 3 \pmod{4} \quad \text{for } \nu = i,$$

so that in any case

$$\bar{S} = \left(\frac{-1}{n}\right)S.$$

It remains to be shown that we must have

$$\nu = 1.$$

This is achieved by computing

$$|\mathfrak{U}|$$

in two different ways. On the one hand, since

$$|\xi e_{kl} - \epsilon^{kl}| = \prod_{r=1}^n (\xi - \xi_r),$$

we have, setting $\xi = 0$,

$$|\mathfrak{U}| = \pm \prod_{r=1}^n \xi_r = \pm n^{\frac{n}{2}} \cdot i^{m_1 - m_3},$$

since

$$\begin{aligned} m_1 - m_3 &= 0 & \text{for } n \equiv 1 \pmod{4}, \\ m_1 - m_3 &= \nu & \text{for } n \equiv 3 \pmod{4}, \end{aligned}$$

and

$$\begin{aligned} i^\nu &= 1 & \text{for } n \equiv 1 \pmod{4}, \\ 2m_1 &= n + 1 & \text{for } n \equiv 3 \pmod{4}, \\ \nu &= \sqrt{-1}, \end{aligned}$$

it follows that

$$|\mathfrak{U}| = n^{\frac{n}{2}} \cdot i^{\frac{n-1}{2}} \cdot n^{-\frac{n}{4}} \cdot i^{\frac{n}{4}} = n^{\frac{n}{4}} \cdot i^{\frac{n-1}{4}}.$$

On the other hand, by a well-known formula in determinants, we have

$$\begin{aligned} |\mathfrak{U}| &= \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \epsilon & \epsilon^2 & \cdots & \epsilon^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \epsilon^{n-1} & \epsilon^{2(n-1)} & \cdots & \epsilon^{(n-1)^2} \end{vmatrix} = \prod_{0 \leq l < k \leq n-1} (\epsilon^k - \epsilon^l) \\ &= \prod_{0 \leq l < k \leq n-1} \epsilon^{\frac{k+l}{2}} (\epsilon^{\frac{k-l}{2}} - \epsilon^{-\frac{k-l}{2}}) = \prod_{0 \leq l < k \leq n-1} (2i) \sin \frac{\pi(k-l)}{n}, \end{aligned}$$

since, in this product, every sine is positive, it follows by comparison with the other formula that

$$\nu = 1.$$

§3 Mertens' Proof

For this proof we must first establish the fact that

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \begin{cases} \sqrt{n} & \text{for } n \equiv 1 \pmod{4}, \\ +i\sqrt{n} & \text{for } n \equiv 3 \pmod{4}. \end{cases}$$

This is more difficult than the proof given earlier in §2 of the mere fact that

$$|S| = \sqrt{n};$$

for $n = p$, to be sure, we know from the Remark preceding Theorem 212 and from the proof of that theorem that the determination of the value of S , up to sign, is quite simple. (As already noted, the case $n = p$ would have sufficed for the applications to be made later.)

We first show that if, for $n > 0$, we set

$$\varphi(m, n) = \sum_s e^{\frac{2\pi i m s^2}{n}},$$

over any complete set of residues mod n , then

$$\varphi(mN_1, N_1)\varphi(mN_1, N_2) = \varphi(m, N_1N_2) \quad \text{for } N_1 > 0, N_2 > 0, (m, N) = 1.$$

Indeed, we have

$$\begin{aligned}
\varphi(mN_1, N_2)\varphi(mN_2, N_1) &= \sum_{s_1, s_2} e^{2\pi i m \left(\frac{N_1 s_1^2}{N_2} + \frac{N_2 s_2^2}{N_1} \right)} \\
&= \sum_{s_1, s_2} e^{\frac{2\pi i m (N_1^2 s_1^2 + N_2^2 s_2^2)}{N_1 N_2}} \\
&= \sum_{s_1, s_2} e^{\frac{2\pi i m (N_1 s_1 + N_2 s_2)^2}{N_1 N_2}} = \varphi(m, N_1 N_2)
\end{aligned}$$

by Theorem 73.

For $m = 1$ we have

$$\varphi(1, 1) = 1.$$

If $n > 1$ and $n = \prod p_j^{l_j}$ is its canonical decomposition, then repeated application of the above functional relation yields

$$\varphi(1, n) = \prod_j \varphi(p_j^{l_j}, p_j^{-l_j} n).$$

For $p \geq 2$ and $p \nmid m$, we now have

$$\varphi(m, p) = p \sum_{t=0}^{p^{l-1}-1} e^{\frac{2\pi i m t^2}{p^{l-1}}} = p \varphi(m, p^{l-1});$$

for

$$\begin{aligned}
\varphi(m, p^l) &= \sum_{s=0}^{p^l-1} e^{\frac{2\pi i m s^2}{p^l}} = \sum_{t=0}^{p-1} \sum_{z=0}^{p^{l-1}-1} e^{\frac{2\pi i m (tp^{l-1}+z)^2}{p^l}} \\
&= \sum_{t=0}^{p-1} \sum_{z=0}^{p^{l-1}-1} e^{\frac{2\pi i m (t^2 p^{2l-2} + 2tzp^{l-1} + z^2)}{p^l}} \\
&= \sum_{z=0}^{p^{l-1}-1} e^{\frac{2\pi i m z^2}{p^l}} \sum_{t=0}^{p-1} e^{\frac{2\pi i \cdot 2mtz}{p}}.
\end{aligned}$$

In this formula, the inner sum is p , in case $p \mid 2mz$, that is, in case $p \mid z$; otherwise it is 0. Consequently

$$\varphi(m, p^l) = p \sum_{v=0}^{p^{l-1}-1} e^{\frac{2\pi i m v^2}{p^{l-2}}} = p \varphi(m, p^{l-2}).$$

From this it follows, for $p \nmid 2m$ and $l = 1$, that

$$\varphi(m, p^l) = \begin{cases} p^{\frac{l}{2}} & \text{for } l \text{ even,} \\ p^{\frac{l-1}{2}} \varphi(m, p) & \text{for } l \text{ odd.} \end{cases}$$

It follows, for odd $n > 1$, that

$$\varphi(1, n) = \sqrt{n} \prod_{p|n} \left(\frac{\varphi(1, p)}{\sqrt{p}} \right),$$

where the product is taken over those primes that go an odd number of times into n .

Now we can easily show that

$$\varphi(m, p) = \left(\frac{m}{p}\right) \varphi(1, p) \quad \text{for } p \nmid 2m.$$

For if

$$\epsilon = e^{\frac{2\pi i}{p}},$$

and if a runs over the quadratic residues in the interval $0 < a < p$, then we have

$$\varphi(m, p) = \sum_{a=0}^{p-1} e^{\frac{2\pi i m a^2}{p}} = 1 + 2 \sum_a \epsilon^{ma};$$

and from here on, everything continues as in the proof of Theorem 212, together with the Remark preceding it, except only that here ϵ represents any primitive p -th root of unity, while there it represented a particular one.

Thus

$$\varphi(1, p) = \left(\frac{-1}{p}\right) \varphi(1, p) = 0 \cdot \varphi(1, p), \quad \epsilon > 0,$$

so that

$$S^2 = \varphi(1, n) = \left(\frac{-1}{n}\right)^* \cdot n,$$

since we have already established directly, in §2, that

$$|S| = |\varphi(1, n)| = \sqrt{n}.$$

(It also follows directly, of course, since

$$\sqrt{p} \geq \sqrt{2},$$

that

$$= n$$

and

$$\sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{n}} = \pm i \sqrt{n}$$

without $|S| = \sqrt{n}$ first being shown.)

Consequently: Let $n > 0$ be odd. We already know that

$$S = \pm \sqrt{n} \quad \text{or} \quad \pm i \sqrt{n},$$

and we should like to show that the upper sign always holds.

Let us define R by

$$(1 + i)R = \sum_{s=0}^{n-1} e^{\frac{2\pi i s^2}{4n}}.$$

Then we have

$$\begin{aligned}
(1+i)^2 R^2 &= \sum_{r=0}^{n-1} e^{\frac{2\pi i r^2}{4n}} \cdot \sum_{t=-r}^{n-1-r} e^{\frac{2\pi i t^2}{4n}} \\
&= \sum_{r=0}^{n-1} e^{\frac{2\pi i r^2}{4n}} \cdot \sum_{t=-r}^{n-r-1} e^{\frac{2\pi i t^2}{4n}} = S + \sum_{t=-r}^{n-r-1} e^{\frac{2\pi i t^2}{4n}} \\
&= \sum_{r=0}^{n-1} \sum_{t=-r}^{n-r-1} e^{\frac{2\pi i (r^2+t^2)}{4n}} = \sum_{r=0}^{n-1} \sum_{t=-r}^{n-r-1} e^{\frac{2\pi i ((r+t)^2-2rt)}{4n}} \\
&= (1+i)S.
\end{aligned}$$

R is therefore real, and we have to show merely (merely! this is where the computation really begins in earnest) that

$$R > 0.$$

Let us set

$$\omega = \frac{\pi}{4n}$$

for the sake of brevity. Then we have

$$\sum_{s=0}^{8n-1} e^{\frac{2\pi i s^2}{8n}} = \sum_{s=0}^{4n-1} e^{\frac{2\pi i s^2}{8n}} + \sum_{s=4n}^{8n-1} e^{\frac{2\pi i s^2}{8n}}.$$

Here, on the one hand, we have

$$\begin{aligned}
\sum_{s=0}^{4n-1} e^{\frac{2\pi i s^2}{8n}} &= \sum_{t=0}^{4n-1} e^{\frac{2\pi i (4n+t)^2}{8n}} = \sum_{t=0}^{4n-1} e^{\frac{2\pi i (16n^2+8nt+t^2)}{8n}} \\
&= \sum_{t=0}^{4n-1} (-1)^t e^{\frac{2\pi i t^2}{8n}},
\end{aligned}$$

and on the other hand,

$$\begin{aligned}
\sum_{s=4n}^{8n-1} e^{\frac{2\pi i s^2}{8n}} &= \sum_{u=1}^{4n} e^{\frac{2\pi i (4n+u)^2}{8n}} = \sum_{u=1}^{4n} e^{\frac{2\pi i (16n^2-8nu+u^2)}{8n}} \\
&= \sum_{u=1}^{4n} e^{\frac{2\pi i u^2}{8n}} = \sum_{u=0}^{4n-1} (-1)^u e^{\frac{2\pi i u^2}{8n}},
\end{aligned}$$

so that, on the one hand, we have

On the one hand,

$$(1+(-1)^s) e^{\frac{2\pi i s^2}{8n}} = 2 \sum_{s=0}^{4n-1} e^{\frac{2\pi i s^2}{8n}} < 2 \sum_{s=0}^{4n-1} e^{\frac{2\pi i s^2}{8n}} = (1+i)R,$$

and on the other hand,

$$\sum_{s=0}^{8n-1} e^{\frac{2\pi i s^2}{8n}} \cdot \sum_{s=0}^{4n-1} e^{\frac{2\pi i s^2}{8n}};$$

and consequently

$$(1+i)R = \sum_{s=0}^{8n-1} e^{\frac{2\pi i s^2}{8n}} - 2 \sum_{s=0}^{4n-1} e^{\frac{2\pi i s^2}{8n}}.$$

Since R is real, it follows that

$$R = \sum_{s=1}^{4n-1} \cos(8s^2\omega) = \sum_{s=1}^{4n-1} \sin(8s^2\omega).$$

Now

$$\begin{aligned} \sin(\alpha + \beta) \sin(\alpha - \beta) &= (\sin \alpha \cos \beta + \cos \alpha \sin \beta)(\sin \alpha \cos \beta - \cos \alpha \sin \beta) \\ &= \sin^2 \alpha \cos^2 \beta - \cos^2 \alpha \sin^2 \beta \\ &= \sin^2 \alpha (1 - \sin^2 \beta) - (1 - \sin^2 \alpha) \sin^2 \beta \\ &= \sin^2 \alpha - \sin^2 \beta. \end{aligned}$$

Setting

$$\alpha = (2s+1)^2\omega, \quad \beta = (2s-1)^2\omega,$$

we have, since

$$\alpha + \beta = 8s^2 + 2\omega, \quad \alpha - \beta = 8s\omega,$$

that

$$\begin{aligned} \sin^2((2s+1)^2\omega) - \sin^2((2s-1)^2\omega) &= \sin(8s^2 + 2)\omega \cdot \sin 8s\omega \\ &= \sin 8s^2\omega \cos 2\omega + \cos 8s^2\omega \sin 2\omega; \end{aligned}$$

for $1 \leq s \leq 4n-1$ (we have $\sin 8s\omega > 0$ since $0 < 8s\omega \leq 32n\omega = \pi$); it follows, by summation over $s = 1, 2, \dots, 4n-1$, that

$$R(\cos 2\omega + \sin 2\omega) = \sin^2 \omega - \frac{\sin^2((2s+1)^2\omega)}{\sin 8s\omega},$$

so that, by so-called summation by parts,

$$\begin{aligned} R(\cos 2\omega + \sin 2\omega) &= \sin^2 \omega - \frac{\sin^2 \omega}{\sin 8\omega} \\ &\quad + \sum_{s=1}^{4n-2} \sin^2((2s+1)^2\omega) \left(\frac{1}{\sin 8s\omega} - \frac{1}{\sin 8(s+1)\omega} \right) \\ &\quad + \frac{\sin^2((8n-1)^2\omega)}{\sin(8(4n-1)\omega)}. \end{aligned}$$

The right-hand side is positive; for, first of all, we have

$$\sin^2 \omega - \frac{\sin 2\omega}{\sin 8\omega} = \frac{\sin^2 \omega \sin 8\omega - \sin \omega \sin \omega}{\sin 8\omega} > 0;$$

and second, in the sum, every term

$$\frac{1}{\sin 8s\omega} - \frac{1}{\sin 8(s+1)\omega} > 0,$$

since the sine function is increasing in the first quadrant; and third,

$$\sin^2((8n-1)^2\omega) > 0.$$

Moreover, $\cos 2\omega + \sin 2\omega$ is > 0 ; and we therefore have

$$R > 0.$$

Chapter 6

REDUCTION TO FUNDAMENTAL DISCRIMINANTS

Definition 7.1. Let d continue to be non-square and $\equiv 0$ or $1 \pmod{4}$. d is called a *fundamental discriminant* if it is not divisible by the square of any odd prime and is either odd or $\equiv 8$ or $12 \pmod{16}$.

Theorem 7.213. Every $d \equiv 0$ or $1 \pmod{4}$ which is non-square can be written, and written uniquely, in the form fm^2 , where $m > 0$ and f is a fundamental discriminant.

Proof. 1) Let d be odd. If $d = fm^2$ at all, in the sense of the theorem, then m^2 must be the greatest square that divides d . Then f is actually a fundamental discriminant; for, first, $f \equiv d \equiv 1 \pmod{4}$, second, f is non-square, and third, f is square-free.

2) Let d be even.

2₁) I shall first show that d may be written in the form fm^2 . In any case, we have $d = qr^2$, where $r > 0$ and q is square-free and non-square.

If $q \equiv 1 \pmod{4}$, then q is a fundamental discriminant.

If $q \equiv 2$ or $3 \pmod{4}$, then r is even (since $4 \mid d$), so that $d = 4q \left(\frac{r}{2}\right)^2$, and $4q$ is a fundamental discriminant; for $4q$ is first of all $\equiv 0 \pmod{4}$, second, non-square, third, not divisible by the square of any odd prime, and fourth, $\equiv 8$ or $12 \pmod{16}$.

2₂) I shall now prove uniqueness. From the fact that $d = fm^2$, $m > 0$, and f is a fundamental discriminant, it follows that m^2 is divisible by the greatest odd square contained in d .

If f is odd, then f is square-free, so that m is the above r , and f is the above q .

If f is even, so that $f \equiv 8$ or $12 \pmod{16}$, then $\frac{f}{4} \nmid 1$, so that $2m$ is the above r , and f is the above $4q$. $\square$

Theorem 7.214. Let $d = fm^2$ be the decomposition of our d in accordance with Theorem 213. Then we have

$$K(d) = \prod_{p \mid m} \left(1 - \left(\frac{f}{p}\right) \frac{1}{p}\right) K(f).$$

Remark. Since the product on the right contains only a finite number of factors, it follows that the determination of $K(d)$ is reduced to the determination of $K(f)$, so that in the next chapter we may restrict our attention to fundamental discriminants.

Proof. If $\sum_{n=1}^{\infty} b_n$ converges and if $a_1 + \cdots + a_r$ is a finite sum, then we certainly have

$$\begin{aligned} (a_1 + \cdots + a_r)(b_1 + b_2 + \cdots \text{ ad inf.}) \\ = (a_1 b_1 + a_1 b_2 + \cdots) + (0 + a_2 b_1 + 0 + a_2 b_2 + \cdots) \\ + (0 + 0 + a_3 b_1 + 0 + 0 + a_3 b_2 + \cdots) + \cdots = \sum_{s=1}^r \sum_{n=1}^{\infty} a_s b_n. \end{aligned}$$

The right-hand side of our assertion, namely

$$\prod_{p|m} \left(1 - \left(\frac{f}{p} \right) \frac{1}{p} \right) \sum_{n=1}^{\infty} \left(\frac{f}{n} \right) \frac{1}{n},$$

is therefore

$$\begin{aligned} &= \sum_{n=1}^{\infty} \sum_{\substack{s|m \\ (s,n)=1}} \mu(s) \left(\frac{f}{s} \right) \left(\frac{f}{n} \right) \frac{1}{sn} \\ &= \sum_{n=1}^{\infty} \sum_{\substack{s|m \\ (s,n)=1}} \mu(s) \left(\frac{f}{sn} \right) \frac{1}{sn} \\ &= \sum_{\substack{n=1 \\ (n,m)=1}}^{\infty} \left(\frac{f}{n} \right) \frac{1}{n} = \sum_{\substack{n=1 \\ (n,m)=1}}^{\infty} \left(\frac{fm^2}{n} \right) \frac{1}{n} \\ &= \sum_{\substack{n=1 \\ (n,m)=1}}^{\infty} \left(\frac{d}{n} \right) \frac{1}{n} = \sum_{n=1}^{\infty} \left(\frac{d}{n} \right) \frac{1}{n} = K(d). \end{aligned}$$

□

Chapter 8

THE DETERMINATION OF $K(d)$ FOR FUNDAMENTAL DISCRIMINANTS

Let d always represent a fundamental discriminant throughout Chapters VIII and IX.

If $\xi > 0$, then by $\sqrt{\xi}$ I always mean the positive number whose square is ξ . In this chapter, by way of exception, if $\xi < 0$, the symbol $\sqrt{\xi}$ will be given a definite value, namely $i\sqrt{|\xi|}$.

Theorem 8.215. *If $n > 0$ and if r runs through a complete set of positive residues mod $|d|$, then*

$$\sum_r \left(\frac{d}{r}\right) e^{\frac{2\pi i n r}{|d|}} = \left(\frac{d}{n}\right) \sqrt{d}.$$

Remark. The fact that this sum is independent of the choice of residue classes follows immediately from the fact that $\left(\frac{d}{r}\right)$ (by Theorem 99, 4)) and $e^{\frac{2\pi i n r}{|d|}}$ each have period $|d|$ (with respect to the positive variable r).

Proof. The numbers $-4, 8, -8$, and $(-1)^{\frac{p-1}{2}}p$, where $p > 2$, are all fundamental discriminants; let them for the moment be called *prime discriminants*.

It clearly suffices to prove all of the following four statements:

- 1) d may be written in the form $\prod q$, where the q are prime discriminants and every pair of factors is relatively prime. (That is, at most one of the numbers $-4, 8, -8$ appears, and each $(-1)^{\frac{p-1}{2}}p$ appears at most once.)
- 2) Every non-empty $\prod q$, where the q are prime discriminants which are pairwise relatively prime, is a fundamental discriminant.
- 3) If the theorem is true for two fundamental discriminants d_1 and d_2 for which $(d_1, d_2) = 1$, then it holds for $d_1 d_2$ (which, by 1) and 2), is certainly a fundamental discriminant).
- 4) The theorem is true for prime discriminants.

PROOF OF 1): If d is odd, then $d \equiv 1 \pmod{4}$ and is square-free, so that

$$d = \prod_{p|d} (-1)^{\frac{p-1}{2}} p,$$

where every $p > 2$.

If $d \equiv 8 \pmod{16}$, then $\frac{d}{8}$ is odd and square-free, so that for a suitable choice of sign we have

$$d = \pm 8 \prod_{\substack{p|d \\ p>2}} (-1)^{\frac{p-1}{2}} p.$$

If $d \equiv 12 \pmod{16}$, then $\frac{d}{-4} \equiv 1 \pmod{4}$ and is square-free, so that

$$d = (-4) \prod_{\substack{p|d \\ p>2}} (-1)^{\frac{p-1}{2}} p.$$

PROOF OF 2): $\prod q$ is not divisible by the square of any odd prime, is non-square, and is $\equiv 1 \pmod{4}$ or $\equiv 8$ or $12 \pmod{16}$.

PROOF OF 3): From

$$\sum_{r_1} \left(\frac{d_1}{r_1} \right) e^{\frac{2\pi i n r_1}{|d_1|}} \cdot \sum_{r_2} \left(\frac{d_2}{r_2} \right) e^{\frac{2\pi i n r_2}{|d_2|}} = \left(\frac{d_1}{n} \right) \sqrt{d_1} \cdot \left(\frac{d_2}{n} \right) \sqrt{d_2}$$

it follows that

$$\sum_{r_1, r_2} \left(\frac{d_1}{r_1} \right) \left(\frac{d_2}{r_2} \right) e^{2\pi i n \left(\frac{r_1}{|d_1|} + \frac{r_2}{|d_2|} \right)} = \left(\frac{d_1}{n} \right) \left(\frac{d_2}{n} \right) \sqrt{d_1} \sqrt{d_2}.$$

For $p \nmid d_1 d_2$ and $p > 2$, it follows from Theorem 81 that

$$\left(\frac{d_1}{p} \right) \left(\frac{d_2}{p} \right) = \left(\frac{d_1 d_2}{p} \right);$$

for $p \nmid d_1 d_2$ and $p = 2$, it follows from Definition 20 that

$$\left(\frac{d_1}{2} \right) \left(\frac{d_2}{2} \right) = \left(\frac{d_1}{2} \right) \left(\frac{d_2}{2} \right) = \left(\frac{d_1 d_2}{2} \right);$$

for $p \mid d_1 d_2$ it follows from Definition 20 that

$$\left(\frac{d_1}{p} \right) \left(\frac{d_2}{p} \right) = 0 = \left(\frac{d_1 d_2}{p} \right).$$

For every p we therefore have

$$\left(\frac{d_1}{p} \right) \left(\frac{d_2}{p} \right) = \left(\frac{d_1 d_2}{p} \right).$$

Consequently, by Definition 20, we have

$$\left(\frac{d_1}{n} \right) \left(\frac{d_2}{n} \right) = \left(\frac{d_1 d_2}{n} \right),$$

so that by (163) and Theorem 99, 3), we have

$$\sum_{r_1, r_2} \left(\frac{d_1 d_2}{r_1 |d_2| + r_2 |d_1|} \right) e^{\frac{2\pi i n (r_1 |d_2| + r_2 |d_1|)}{|d_1 d_2|}} = \left(\frac{d_1 d_2}{n} \right) \sqrt{d_1} \sqrt{d_2}. \quad (164)$$

By Theorem 73, $r_1|d_2| + r_2|d_1|$ runs through a complete positive system of residues $r \pmod{|d_1d_2|}$; by Theorem 99, 4) we have

$$\left(\frac{d_1d_2}{r_1|d_2| + r_2|d_1|}\right) = \left(\frac{d_1d_2}{r}\right), \quad e^{\frac{2\pi in(r_1|d_2| + r_2|d_1|)}{|d_1d_2|}} = e^{\frac{2\pi inr}{|d_1d_2|}}.$$

(164) therefore yields

$$\sum_r \left(\frac{d_1d_2}{r}\right) e^{\frac{2\pi inr}{|d_1d_2|}} = \left(\frac{d_1d_2}{n}\right) \sqrt{d_1} \sqrt{d_2}.$$

It therefore suffices to prove only that

$$\sqrt{d_1} \sqrt{d_2} = \sqrt{d_1d_2}.$$

Since

$$\sqrt{d_1} \sqrt{d_2} = \begin{cases} -\sqrt{d_1d_2} & \text{for } d_1 < 0, d_2 < 0, \\ \sqrt{d_1d_2} & \text{otherwise,} \end{cases}$$

I therefore have to show that

$$\left(\frac{d_1}{d_2}\right) \left(\frac{d_2}{d_1}\right) = \begin{cases} -1 & \text{for } d_1 < 0, d_2 < 0, \\ 1 & \text{otherwise.} \end{cases}$$

Since $(d_1, d_2) = 1$, let d_1 be odd, without loss of generality. Then, by Theorem 98, 1), we have

$$\left(\frac{d_1}{d_2}\right) \left(\frac{d_2}{d_1}\right) = \left(\frac{d_1}{|d_2|}\right) \left(\frac{|d_2|}{d_1}\right) (-1)^{\frac{d_1-1}{2} \cdot \frac{|d_2|-1}{2}}.$$

For $d_2 > 0$, this reads

$$\left(\frac{d_1}{d_2}\right) = (-1)^{\frac{d_1-1}{2} \cdot \frac{d_2-1}{2}} = \begin{cases} 1 & \text{in case } d_1 > 0, \\ (-1)^{\frac{d_1-1}{2}} & \text{in case } d_1 < 0, \end{cases}$$

for $d_2 < 0$, it reads

$$\left(\frac{d_1}{|d_2|}\right) (-1)^{\frac{d_1-1}{2}} = \begin{cases} (-1)^{\frac{d_1-1}{2}} & \text{for } d_1 > 0, \\ -1 & \text{for } d_1 < 0. \end{cases}$$

PROOF OF 4): Let d be a prime discriminant. We shall investigate the cases $(n, d) > 1$ and $(n, d) = 1$; in the latter case it suffices to carry out the proof for $n = 1$, that is, to show that

$$\sum_r \left(\frac{d}{r}\right) e^{\frac{2\pi ir}{|d|}} = \sqrt{d};$$

for, if we set $e^{\frac{2\pi i}{|d|}} = \omega$, then for $(n, d) = 1$ we have

$$\sum_r \left(\frac{d}{r}\right) e^{\frac{2\pi inr}{|d|}} = \sum_r \left(\frac{d}{rn}\right) \left(\frac{d}{n}\right) \omega^{nr} = \left(\frac{d}{n}\right) \sum_r \left(\frac{d}{r}\right) \omega^r.$$

In case $(n, d) > 1$, the statement of the theorem reads

$$\sum_r \left(\frac{d}{r}\right) \omega^{nr} = 0.$$

4₁) For $d = -4$, we have

$$\begin{aligned} \sum_r \left(\frac{-4}{r}\right) e^{\frac{2\pi i r}{4}} &= i^1 + i^2 + i^3 + i^4 = i - 1 - i + 1 = 0 \quad \text{in case } n = 2m, \\ &= i - (-1) - (-i) + 1 = 2i = \sqrt{-4} \quad \text{in case } n = 1. \end{aligned}$$

4₂) For $d = 8$, it follows from Theorem 93 (since $\omega^4 = 1$, $\omega = i$) that

$$\sum_r \left(\frac{8}{r}\right) e^{\frac{2\pi i r}{8}} = \sum_r \left(\frac{8}{r}\right) \omega^r = \omega + \omega^3 - \omega^5 - \omega^7 = \omega(1 - \omega^4) - \omega^5(1 - \omega^4) = \dots$$

This reads

$$e^{\frac{2\pi i}{8}} (1 + (-1)^n - 2(-1)^{\frac{n}{2}}) = 0 \quad \text{in case } n = 2m,$$

and

$$\begin{aligned} \frac{1+i}{\sqrt{2}} \cdot \sqrt{2} &= 1+i = \sqrt{2} \cdot e^{\frac{\pi i}{4}} \quad \text{in case } n = 1, d = 8, \\ \frac{1-i}{\sqrt{2}} \cdot \sqrt{2} &= 1-i = \sqrt{-8} \quad \text{in case } n = 1, d = -8. \end{aligned}$$

4₃) For $d = (-1)^{\frac{p-1}{2}} p$, we have

in case $(n, d) > 1$, since $p \mid n$ and because $\left(\frac{d}{r}\right)$ is a character of the second kind mod $|d|$:

$$\sum_r \left(\frac{d}{r}\right) e^{\frac{2\pi i n r}{|d|}} = 0;$$

in case $n = 1$, by Theorem 98, 1) and Theorem 212:

$$\sum_r \left(\frac{d}{r}\right) e^{\frac{2\pi i r}{p}} = \sum_r \left(\frac{d}{r}\right) \omega^r = \sum_r \left(\frac{r}{p}\right) \omega^r = \sum_r \left(\frac{r}{p}\right) e^{\frac{2\pi i r}{p}} = \sqrt{d}.$$

(The entire machinery of Chapter VI was necessary only for subcase 4₃) of the fourth statement.) $\square$

Theorem 8.216. For $0 < \varphi < 2\pi$, we have

$$\sum_{n=1}^{\infty} \frac{\sin n\varphi}{n} = \frac{\pi - \varphi}{2}, \quad (165)$$

$$\sum_{n=1}^{\infty} \frac{\cos n\varphi}{n} = -\log \left(2 \sin \frac{\varphi}{2} \right). \quad (166)$$

Proof. This I assume known from analysis. Otherwise, the inexperienced reader may derive (165) from the Fourier expansion of the very well-behaved right-hand side; (166) preferably not in this way, because of the singularities $\varphi = 0$ and $\varphi = 2\pi$, but rather from the fact that for $|\xi| < 1$, with the exception of $\xi = 1$ (ξ complex), we have, by suitable normalization of the imaginary part of the logarithm (this normalization being arbitrary, since it drops out immediately),

$$\sum_{n=1}^{\infty} \frac{\xi^n}{n} = -\log(1 - \xi),$$

from which, if we set $\xi = e^{i\varphi}$, our result follows by comparison of the real parts. $\square$

Theorem 8.217. *If d is a fundamental discriminant, then*

$$K(d) = \begin{cases} -\frac{1}{\sqrt{d}} \sum_{r=1}^{|d|-1} \left(\frac{d}{r}\right) \log \sin \frac{\pi r}{|d|} & \text{for } d > 0, \\ \frac{\pi}{|d|\sqrt{|d|}} \sum_{r=1}^{|d|-1} r \left(\frac{d}{r}\right) & \text{for } d < 0. \end{cases}$$

Proof. By Theorem 215, we have

$$\begin{aligned} \sum_r \left(\frac{d}{r}\right) \sum_{n=1}^{\infty} \frac{1}{n} e^{\frac{2\pi i n r}{|d|}} &= \sum_{n=1}^{\infty} \frac{1}{n} \left(\frac{d}{n}\right) \sqrt{d} \\ &= \sqrt{d} \sum_{n=1}^{\infty} \left(\frac{d}{n}\right) \frac{1}{n} = \sqrt{d} \cdot K(d). \end{aligned} \quad (167)$$

(since $z = \sum_{n=1}^{\infty} \frac{1}{n} e^{\frac{2\pi i n r}{|d|}}$ on the right converges, by Theorem 216, since $\left|\frac{2\pi r}{|d|}\right| < 2\pi$.)

1) Let $d > 0$. By (166), it follows from (167), since the left-hand side is real, that

$$K(d) = -\frac{1}{\sqrt{d}} \sum_{r=1}^{|d|-1} \left(\frac{d}{r}\right) \log \sin \frac{\pi r}{|d|}.$$

2) Let $d < 0$. By (165), it follows from (167), since the left-hand side is pure imaginary, that

$$K(d) = \frac{\pi}{|d|\sqrt{|d|}} \sum_{r=1}^{|d|-1} r \left(\frac{d}{r}\right). \quad \square$$

Chapter 9

FINAL FORMULAS FOR THE CLASS NUMBER

Theorem 9.218. *Let d be a fundamental discriminant. Then we have*

$$h(d) = \begin{cases} -\frac{\sqrt{d}}{2 \log \varepsilon} \sum_{\substack{0 < r < |d| \\ (\frac{d}{r})=1}} \log \sin \frac{\pi r}{|d|} & \text{for } d > 0, \\ \frac{w}{2\pi} \sqrt{|d|} \left(\sum_s - \sum_t \right) & \text{for } d < 0, \end{cases}$$

where s runs through the numbers in the interval $0 < r < |d|$ for which $(\frac{d}{r}) = 1$ and t through those numbers in the same interval for which $(\frac{d}{r}) = -1$.

Remark. 1) For $d = (-1)^{\frac{p-1}{2}} p$, where $p > 2$, it follows from Theorem 98, 1) that this means that s runs through the quadratic residues and t through the quadratic non-residues mod p between 0 and p .

2) In case $d > 0$, it follows from Theorem 101 that the right-hand side of the statement also equals

$$\frac{\sqrt{d}}{2 \log \varepsilon} \sum_{\substack{0 < r < |d|/2 \\ (\frac{d}{r})=1}} \log \frac{\sin \frac{\pi r}{|d|}}{\sin \frac{\pi(|d|-r)}{|d|}}.$$

Proof. 1) Let $d > 0$. By Theorems 209 and 217, we have

$$\begin{aligned} h(d) &= \frac{\sqrt{d}}{\log \varepsilon} K(d) = \frac{\sqrt{d}}{\log \varepsilon} \cdot \left(-\frac{1}{\sqrt{d}} \right) \sum_{r=1}^{|d|-1} \left(\frac{d}{r} \right) \log \sin \frac{\pi r}{|d|} \\ &= -\frac{1}{\log \varepsilon} \left(\sum_{(\frac{d}{r})=1} - \sum_{(\frac{d}{r})=-1} \right) \log \sin \frac{\pi r}{|d|}. \end{aligned}$$

2) Let $d < 0$. By Theorems 209 and 217, we have

$$\begin{aligned} h(d) &= \frac{w\sqrt{|d|}}{2\pi} K(d) = \frac{w\sqrt{|d|}}{2\pi} \cdot \frac{\pi}{|d|\sqrt{|d|}} \sum_{r=1}^{|d|-1} r \left(\frac{d}{r} \right) \\ &= \frac{w}{2|d|} \left(\sum_s s - \sum_t t \right) = \frac{w}{2\pi} \sqrt{|d|} \left(\sum_s - \sum_t \right). \end{aligned} \quad \square$$

For $d < 0$ I shall develop—for general purposes, first of all, and secondly to carry out what was promised in the Introduction—still another, and even simpler, formula for $h(d)$, and—since the elementary transformation of (169) could be carried out, to be sure, but would be quite laborious—I shall do this by once again bringing in the series $K(d)$.

Theorem 9.219. *Let $d < 0$ be a fundamental discriminant. Then*

$$h(d) = \frac{w}{2 \left(2 - \left(\frac{d}{2} \right) \right)} \sum_{r=1}^{|d|-1} \left(\frac{d}{r} \right).$$

Remark. For $d = -p$, $p \equiv 3 \pmod{4}$ and $p > 3$, this is precisely what was asserted in (123), since

$$\frac{w}{2 \left(2 - \left(\frac{-p}{2} \right) \right)} = \frac{2}{2 - \left(\frac{-p}{2} \right)} = \begin{cases} 1 & \text{for } p \equiv 7 \pmod{8}, \\ \frac{1}{3} & \text{for } p \equiv 3 \pmod{8}. \end{cases}$$

Proof. By (165) we have, for $2n < p < 4n$,

$$\sum_{n=1}^{\infty} \frac{\sin np}{n} = \sum_{n=1}^{\infty} \frac{\sin n(p - 2\pi)}{n} = \frac{\pi - (p - 2\pi)}{2} = \frac{3\pi - p}{2}.$$

The series on the left converges for $p = 2\pi$ as well; by Theorem 215 (with $2n$ in place of n), we therefore have

$$\sum_r \left(\frac{d}{r} \right) \sum_{n=1}^{\infty} \frac{1}{n} e^{\frac{2\pi i \cdot 2n \cdot r}{|d|}} = \sqrt{d} \sum_{n=1}^{\infty} \left(\frac{d}{2n} \right) \frac{1}{n} = 2\sqrt{d} \sum_{n=1}^{\infty} \left(\frac{d}{n} \right) \frac{1}{2n}.$$

(I have, to be sure, replaced $\left(\frac{d}{2} \right)$ by the incorrect value $-\left(\frac{d}{2} \right)$ instead of 0, for $2 \mid d$; this is harmless, however, since for d even we certainly have $\left(\frac{d}{r} \right) = 0$ since $4 \mid d$). It therefore follows, by (168), that

$$\begin{aligned} \frac{2\pi}{|d|} \sum_{r=1}^{|d|-1} r \left(\frac{d}{r} \right) &= 2\sqrt{d} \left(2 - \left(\frac{d}{2} \right) \right) K(d), \\ \frac{\pi}{|d|\sqrt{|d|}} \sum_{r=1}^{|d|-1} r \left(\frac{d}{r} \right) &= \left(2 - \left(\frac{d}{2} \right) \right) K(d), \end{aligned}$$

so that, by Theorem 209, we have

$$h(d) = \frac{w\sqrt{|d|}}{2\pi} \cdot \frac{\pi}{|d|\sqrt{|d|}} \sum_{r=1}^{|d|-1} r \left(\frac{d}{r} \right) \cdot \frac{1}{\left(2 - \left(\frac{d}{2} \right) \right)} = \frac{w}{2 \left(2 - \left(\frac{d}{2} \right) \right)} \sum_{r=1}^{|d|-1} \left(\frac{d}{r} \right). \quad \square$$

APPENDIX

EXERCISES

Exercises for Part One

Exercises for Chapter I

1. Suppose $b > 0$ and $b \nmid a$. Put $r_0 = a$ and $r_1 = b$ and determine $r_2, r_3, \dots, r_n$ by the relations:

$$\begin{aligned}r_0 &= q_2 r_1 + r_2, \\r_1 &= q_3 r_2 + r_3, \\&\vdots \\r_{n-2} &= q_n r_{n-1} + r_n, \\r_{n-1} &= q_{n+1} r_n.\end{aligned}$$

$$(r_1 > r_2 > r_3 > \dots > r_{n-1} > r_n > 0).$$

Show that r_n , the last non-zero remainder, is the greatest common divisor of a and b . This process for finding the greatest common divisor is known as the Euclidean Algorithm.

2. For arbitrary a and b , not both zero, consider all numbers of the form $ax + by$, where x and y take on all integer values. Let d be the smallest positive number in this collection. Show that $d = (a, b)$.
3. Prove that $(ma, mb) = m(a, b)$, where $m > 0$ and a and b are not both zero.
4. Deduce Theorem 15 from either Theorem 14 or Ex. 3.
5. Show that the greatest common divisor of $a + b$ and $a - b$ is either 1 or 2 if $(a, b) = 1$.
6. Suppose $F_n = 2^{2^n} + 1$, for $n = 0, 1, 2, \dots$. Show that if $k > 0$, then $F_k \mid (F_{k+n} - 2)$. Infer that any two of the numbers $F_0, F_1, F_2, \dots$ are relatively prime.
7. Show that if $ad - bc = 1$, then $(a + b, c + d) = 1$.
8. Prove that if n is odd, then $n^2(n^2 - 1)$ is divisible by 24.

9. Show that in the so-called Fibonacci series, $1, 2, 3, 5, 8, \dots$, in which each term is the sum of the two preceding terms, two consecutive terms are always relatively prime.
10. Show that the sum $1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$ is not an integer if $n > 1$.
(*Hint:* Let 2^l be the highest power of 2 not exceeding n . Then there is no other number between 1 and n inclusive which is divisible by 2^l .)

Exercises for Chapter II

1. Prove that if $a > 3$ and $n \geq 2$, then $a^n - 1$ is composite.
2. Prove that $\tau(a)$ is odd if and only if a is a perfect square.
3. Suppose $n > 0$. Show that $\tau(2^n - 1) = \tau(n)$ and $\tau(2^n + 1) > \tau^*(n)$, where $\tau^*(n)$ is the number of odd positive divisors of n .
4. Use Ex. 6 of Chap. I to give another proof of Theorem 18.
5. Suppose k to be a given integer greater than 2.
 - a) If $q_1 - 1, q_2 - 1, \dots, q_r - 1$ are divisible by k , show that $q_1 q_2 \cdots q_r - 1$ is divisible by k .
 - b) If $n > 0$, show that there is a prime p such that $k \nmid (p - 1)$ and $p \mid (nk - 1)$.
 - c) By the method of proof of Theorem 18, show that there are infinitely many primes p such that $k \nmid (p - 1)$.
6. If $f(n)$ is a non-constant polynomial in n with integral coefficients, then $f(n)$ is composite for infinitely many values of n .
(*Hint:* Let a be such that $d = |f(a)| > 1$. Then $d \mid f(dx + a)$ for every x .)
7. Prove that if ξ and η are real, then $[\xi + \eta] = [\xi] + [\eta]$ or $[\xi] + [\eta] + 1$.
8. Use Theorem 27 and Ex. 7 to prove that if $m > 0$ and $n > 0$, then $\binom{m+n}{m}$ is divisible by $m!n!$. Infer that the product of any n consecutive integers is always divisible by $n!$.
9. If $1 \leq r \leq p^n$, $p^k \parallel r!$, and $p^{k+1} \nmid r!$, show that $p^n \mid \{r!(p^n - r)!\}^{-1}$ is divisible by p^{n-k} but not by p^{n-k+1} .

Exercises for Chapter III

1. Suppose $r \geq 2$, $a_1 > 0, \dots, a_r > 0$, and let w be the least positive common multiple of $a_1, \dots, a_r$. Show that if we write $a_j = \frac{w}{b_j}$, then $(b_1, \dots, b_r) = 1$.
2. Suppose $r \geq 2$ and $a_1, \dots, a_r$ are not all zero. Let $d_2 = a_1$ and $d_n = (a_n, d_{n-1})$ for $2 \leq n \leq r$. Show that $d_r = (a_1, \dots, a_r)$.
3. If $r \geq 2$ and $a_1 > 0, \dots, a_r > 0$, let us denote (temporarily) the least positive common multiple of $a_1, \dots, a_r$ by $\{a_1, \dots, a_r\}$. Show that if $a > 0$, $b > 0$, and $c > 0$, then

- a) $(a, \{b, c\}) = \{(a, b), (a, c)\};$
- b) $\{a, (b, c)\} = (\{a, b\}, \{a, c\});$
- c) $(\{a, b\}, \{a, c\}, \{b, c\}) = \{(a, b), (a, c), (b, c)\}.$

Exercises for Chapter IV

1. If $a > 0$ and $b > 1$, show that $\frac{S(ab)}{ab} = \frac{S(a)}{a} \cdot \frac{S(b)}{b}.$
2. Show that an odd integer divisible by no more than two primes cannot be a perfect number.
(*Hint:* Show that $S(a) < 2a$ for such an odd number a .)
3. Prove that the sum of the reciprocals of the positive divisors of a perfect number is equal to 2.
4. If n is an odd perfect number, show that $n = p^\alpha m^2$, where $p \equiv \alpha \equiv 1 \pmod{4}$ and $p \nmid m$. If m is given, show that there is at most one prime-power p^α such that $p \nmid m$ and $p^\alpha m^2$ is perfect.
5. It is implicit in the proof of Theorem 31 that if $a > 0$, $b > 0$, and $(a, b) = 1$, then $S(ab) = S(a)S(b)$. Prove more generally that if $a > 0$ and $b > 0$, then

$$S(a)S(b) = \sum_{d|(a,b)} dS\left(\frac{ab}{d^2}\right).$$

(*Hint:* First treat the case where a and b are powers of the same prime.)

6. The number-theoretic function $\lambda(a)$ (Liouville function) is defined thus:
 $\lambda(a) = 1$ if $a = 1$ or if a is the product of an even number of primes (not necessarily distinct), while $\lambda(a) = -1$ if a is the product of an odd number of primes (not necessarily distinct). Prove the following:
 - a) If $a > 0$, $b > 0$, and $(a, b) = 1$, then $\lambda(ab) = \lambda(a)\lambda(b)$.
 - b) $\sum_{d|a} \lambda(d) = \begin{cases} 1 & \text{if } a = k^2 \text{ for some } k > 0, \\ 0 & \text{if } a \neq k^2 \text{ for all } k. \end{cases}$
 - c) For $x \geq 1$ we have $\left| \sum_{n=1}^x \frac{\lambda(n)}{n} \right| \leq 1$.
 - d) For $x \geq 1$ we have $|\sum_{n=1}^x \lambda(n)| \leq x$.
7. If $a > 0$, show that $\sum_{d|a} \mu(d)S\left(\frac{a}{d}\right) = a$.
8. Show that $\Lambda(a) = \sum_{d|a} \mu(d) \log\left(\frac{a}{d}\right) = -\sum_{d|a} \mu(d) \log d$.
9. Let $G(a)$ be any number-theoretic function. Denote by $F(a)$ the number-theoretic function

$$F(a) = \sum_{d|a} \mu(d)G\left(\frac{a}{d}\right).$$

Show that $G(a) = \sum_{d|a} F(d)$.

10. Let $f(a)$ be a number-theoretic function which never takes the value zero. Denote by $g(a)$ the number-theoretic function

$$g(a) = \prod_{d|a} f(d).$$

Show that we have

$$f(a) = \prod_{d|a} g\left(\frac{a}{d}\right)^{\mu(d)}.$$

11. Show that $\varphi(5186) = \varphi(5187) = \varphi(5188) = 2592$.
12. If $n > 0$, show that $\varphi(n) \mid n!$ if and only if n is of one of the forms $1, 2^a, 2^a 3^b$, where $a > 0, b > 0$.
13. Prove that if $a > 0$ and $b > 0$, then $\varphi(ab) = \varphi(a)\varphi(b)\frac{c}{\varphi(c)}$, where c is the product of those primes that divide both a and b .
14. If $n > 0$, then $\tau(1) + \tau(2) + \cdots + \tau(n) = \left[\frac{n}{1}\right] + \left[\frac{n}{2}\right] + \cdots + \left[\frac{n}{n}\right]$.
(*Hint*: Count in two ways the number of solutions of $xy \leq n, x > 0, y > 0$.)
15. If $m > 0$ and $k = \lfloor \sqrt{n} \rfloor$, show that

$$\tau(1) + \tau(2) + \cdots + \tau(n) = 2 \left(\left[\frac{n}{1}\right] + \left[\frac{n}{2}\right] + \cdots + \left[\frac{n}{k}\right] \right) - k^2.$$

(*Hint*: If $xy \leq n, x > 0$, and $y > 0$, then either $x \leq k$ or $y \leq k$, and possibly both.)

Exercises for Chapter V

1. If $a = c_0 + c_1g + \cdots + c_ng^n$, prove that

$$a \equiv c_0 + c_1 + \cdots + c_n \pmod{g-1}.$$

(In particular, any number is congruent modulo 9 to the sum of its decimal digits; cf. Theorem 8.)

2. If $a > 4$ and a is not a prime, show that $(a-1)! \equiv 0 \pmod{a}$.
3. If $k > 0$ and $n(k-1)$ is even, show that there exist integers x and y relatively prime to k such that $x + y \equiv n \pmod{k}$.

(*Hint*: First consider the case in which k is a power of a prime and then use Theorem 70.)

4. Given that $a > 0, b > 0, (a, b) = 1$, and $aa' + bb' = 1$. Show that if $n \geq 0$ the diophantine equation $ax + by = n$ has

$$\left[\frac{n}{ab}\right] \text{ or } \left[\frac{n}{ab}\right] + 1$$

solutions in non-negative integers x and y .

5. Given that $a > 0$, $b > 0$, $(a, b) = 1$. Show that if $n > ab - a - b$ there exist non-negative integers x and y such that $ax + by = n$, but that if $n = ab - a - b$, this is not the case.
6. Show that $(a + b)^p \equiv a^p + b^p \pmod{p}$.
7. Prove that if $m^2 + n^2 \equiv 0 \pmod{p}$, where $p > 2$, then $m^4 + n^4 \equiv 0 \pmod{p^2}$.
8. Given that $r > 1$, $s > 1$, and $rs > p$. Then if $p \nmid a$, we can find integers x and y such that

$$ax \equiv y \pmod{p}, \quad 1 \leq |x| < r, \quad 1 \leq |y| < s.$$

(*Hint*: Consider the numbers $au - v$, where $0 \leq u \leq r - 1$ and $0 \leq v \leq s - 1$. Since $rs > p$, two of these numbers must be congruent modulo p .)

9. Given that $p > 2$.
 - a) Prove that $p!$ and $(p - 1)! - 1$ are relatively prime.
 - b) Prove that if $n > 0$ and $n \equiv (p - 1)! - 1 \pmod{p!}$, then the $p - 2$ integers preceding n and the p integers following n are composite.
10. If $p > 2$, use Wilson's Theorem to show that
11. If $1 \leq j \leq p - 2$ and if s_j is the sum of the products of the numbers $1, 2, \dots, p - 1$ taken j at a time, show that $s_j \equiv 0 \pmod{p}$.
12. If $l > 2$ show that

$$a^{2^{l-2}} \equiv 1 \pmod{2^l}$$

for a odd.

(*Hint*: Use mathematical induction.)

13. If $p > 2$ and $l > 0$ or if $p = 2$ and $0 \leq l \leq 2$, define $\chi(p^l) = \varphi(p^l)$. If $p = 2$ and $l > 2$, define $\chi(p^l) = \frac{1}{2}\varphi(p^l)$. If $m > 1$ and if $m = p_1^{l_1} \cdots p_r^{l_r}$ is its canonical factorization, define $\chi(m)$ as the least common multiple of $\chi(p_1^{l_1}), \dots, \chi(p_r^{l_r})$. Show that if $(a, m) = 1$ then $a^{\chi(m)} \equiv 1 \pmod{m}$. For this reason $\chi(m)$ is sometimes called the universal exponent of m .
14. Given that $m > 1$ and m is odd. Consider the following four statements:
 - (i) m is prime,
 - (ii) $\varphi(m) \mid (m - 1)$,
 - (iii) $\chi(m) \mid (m - 1)$,
 - (iv) $2^{m-1} \equiv 1 \pmod{m}$.
 - a) Show that if one of these four statements holds, then those that follow it also hold.
 - b) Show that if $m = 341, 645, 1387$, or 1905 , then (iv) holds but (iii) does not. Show that if $m = 561, 1105$, or 1729 , then (iii) holds but (ii) does not. (There are no known examples in which (ii) holds but (i) does not.)
 - c) Show that if (iv) holds when $m = k$, then it also holds when $m = 2k - 1$.

Exercises for Chapter VI

1. Show that the congruence $ax^2 + bx + c \equiv 0 \pmod{p}$, where p is odd and $p \nmid a$, has a solution if and only if $b^2 - 4ac$ is a quadratic residue modulo p .
2. If $\left(\frac{2}{p}\right) = -1$, show that $\sum_{d|n} \left(\frac{d}{p}\right) = 0$.
3. Show that $\sum_{n=1}^{p-1} \left(\frac{n}{p}\right) = 0$.
4. Show that the product of the quadratic residues of an odd prime p is $\equiv (-1)^{\frac{p+1}{2}} \pmod{p}$.
(*Hint:* Proceed as in the first proof of Wilson's Theorem.)
5. If $p \equiv 3 \pmod{4}$ and r is the number of quadratic non-residues among the numbers $1, 2, \dots, \frac{p-1}{2}$, then $\left(\frac{r}{p}\right) \cdot r! \equiv (-1)^r \pmod{p}$.
6. Show that the conclusion of Theorem 86 can be written

$$\left(\frac{p}{q}\right) = \left(\frac{(-1)^{\frac{q-1}{2}} q}{p}\right).$$

7. If $p > 2$, show that the number of solutions of the congruence $x^4 \equiv 16 \pmod{p}$ is 8 if $p \equiv 1 \pmod{8}$, 4 if $p \equiv 5 \pmod{8}$, and 2 if $p \equiv 3, 7 \pmod{8}$.
(*Hint:* Use the identity $x^8 - 16 = (x^2 - 2)(x^2 + 2)((x - 1)^2 + 1)((x + 1)^2 + 1)$.)
8. Show that for every prime p the congruence $x^4 - 11x^2 + 1 \equiv 0 \pmod{p}$ has a solution.
9. Show that the number of solutions of $ax^2 + bx + c \equiv 0 \pmod{p}$, where $p > 2$ and $p \nmid a$, is $1 + \left(\frac{b^2 - 4ac}{p}\right)$.
10. Show that if $p > 3$ then the sum of the quadratic residues of p between 0 and p is divisible by p if and only if $p \equiv 1 \pmod{4}$.
11. If $p > 2$, show that $\sum_{n=1}^{p-1} n \left(\frac{n}{p}\right) = 0$ if $p \equiv 1 \pmod{4}$.
12. If $p > 3$, show that $\sum_{n=1}^{p-1} n^2 \left(\frac{n}{p}\right) = p \sum_{n=1}^{p-1} n \left(\frac{n}{p}\right)$.
13. If $p > 2$, show that the sum of the quadratic non-residues of p between 0 and p is greater than, equal to, or less than the sum of the quadratic residues according as $p \equiv 1, 5$, or $3 \pmod{8}$.
14. If $p \equiv 3 \pmod{4}$ and $p > 3$, show that

$$\sum_{r=1}^{\frac{p-1}{2}} \left(\frac{r}{p}\right) = 0 \text{ if } p \equiv 3 \pmod{8}, \quad \text{and} \quad \sum_{r=1}^{\frac{p-1}{2}} \left(\frac{r}{p}\right) = -1 \text{ if } p \equiv 7 \pmod{8}.$$

15. Show that the Kronecker symbol may be expressed in terms of the Jacobi symbol as follows: If $d \equiv 0$ or $1 \pmod{4}$ and is not a square and $m > 0$, then

$$\left(\frac{d}{m}\right) = \begin{cases} 0 & \text{if } (m, d) > 1, \\ \left(\frac{m}{|d|}\right) & \text{if } (m, d) = 1, d \equiv 1 \pmod{4}, \\ \left(\frac{2}{m}\right)^b \left(\frac{m}{|d|/4}\right) & \text{if } (m, d) = 1, d \equiv 0 \pmod{4}, \end{cases}$$

where $b = 0$ or 1 according as $d/4 \equiv 1$ or $2 \pmod{4}$.

16. a) Show that the fifth assertion of Theorem 99 may be modified to read: $\left(\frac{d}{p}\right) = -1$ for suitable p .
- b) If n is not a square, show that there are an infinite number of odd p such that $\left(\frac{n}{p}\right) = -1$.
(*Hint:* Apply a) with $d = h^2n$, where h is the product of the first r primes, r being an arbitrary positive integer.)
- c) If the congruence $x^2 \equiv n \pmod{p}$ is solvable for all sufficiently large p , show that n is a square.

Exercises for Chapter VII

- Let a , b , and c be given and either $b^2 - ac < 0$ or $b^2 - ac = a$ positive square. Show that for any given k the equation $ax^2 + 2bxy + cy^2 = k$ has only a finite number of solutions.
- Let a , b , and c be given and either $b^2 - ac = 0$ or $b^2 - ac$ positive and not a square. Show that there exists a non-zero k such that the equation $ax^2 + 2bxy + cy^2 = k$ has an infinite number of solutions.
- If k is given, show that there are an infinite number of positive values of d for which the equation $x^2 - dy^2 = k$ is solvable.
- If d is divisible by 4 or by any prime congruent to 3 modulo 4, show that the equation $x^2 - dy^2 = -1$ has no solutions.
- Let d be positive and not a square and suppose that the equation $x^2 - dy^2 = -1$ has solutions. If x_0, y_0 is that solution for which y_0 has the smallest positive value and $x_0 > 0$, show that the general solution of $x^2 - dy^2 = -1$ is given by the formula

$$\pm(x_0 + y_0\sqrt{d})^{2t+1} = x + y\sqrt{d},$$

while the general solution of $x^2 - dy^2 = 1$ is given by the formula

$$\pm(x_0 + y_0\sqrt{d})^{2t} = x + y\sqrt{d}.$$

- Show that the equation $x^2 - 34y^2 = -1$ has no solutions.
(*Hint:* The equation $x^2 - 34y^2 = 1$ has the solution $x = 35, y = 6$.)

7. If d is divisible by any prime congruent to 3 modulo 4, show that the equation $x^2 - dy^2 = -4$ has no solutions.
8. Suppose $d > 0$, $d \equiv 0, 1 \pmod{4}$, d not a square and suppose the equation $x^2 - dy^2 = -4$ has solutions. If x_0, y_0 is that solution for which y_0 has the smallest positive value and $x_0 > 0$, show that the general solution of $x^2 - dy^2 = -4$ is given by the formula

$$\pm \left(\frac{x_0 + y_0 \sqrt{d}}{2} \right)^{2t+1} = \frac{x + y \sqrt{d}}{2},$$

while the general solution of $x^2 - dy^2 = 4$ is given by the formula

$$\pm \left(\frac{x_0 + y_0 \sqrt{d}}{2} \right)^{2t} = \frac{x + y \sqrt{d}}{2}.$$

9. Suppose $d \equiv 0 \pmod{4}$. Show that any solution of the equation $x^2 - dy^2 = 4$ is of the form $x = 2u$, $y = v$, where $u^2 - \frac{d}{4}v^2 = 1$. Similarly for the equation $x^2 - dy^2 = -4$, if it has solutions.
10. Suppose $d \equiv 1 \pmod{8}$. Show that any solution of the equation $x^2 - dy^2 = -4$ is of the form $x = 2u$, $y = 2v$, where $u^2 - dv^2 = 1$. Similarly for the equation $x^2 - dy^2 = 4$, if it has solutions.
11. Suppose $d \equiv 5 \pmod{8}$. If x_0, y_0 is that solution of $x^2 - dy^2 = 4$ for which y_0 has the smallest value and $x_0 > 0$, and if u_0, v_0 is that solution of $x^2 - dy^2 = 1$ for which v_0 has the smallest positive value and $u_0 > 0$, show that

$$\frac{x_0 + y_0 \sqrt{d}}{2} = u_0 + v_0 \sqrt{d}$$

if x_0 and y_0 are even, but that

$$\frac{x_0 + y_0 \sqrt{d}}{2} = \left(\frac{u_0 + v_0 \sqrt{d}}{2} \right)^3$$

if x_0 and y_0 are odd. Similarly for the equation $x^2 - dy^2 = -4$, if it has solutions.

Exercises for Part Two

Exercises for Chapter I

1. Show that there is a positive constant a such that there is a prime number between n and an for every positive n .
2. If $\epsilon > 0$ show that $\sum p^{-1}(\log p)^{-1-\epsilon}$ converges, the sum being extended over all prime numbers in increasing order.

3. Show that if m runs through the composite numbers in increasing order, then $\sum \{m - \varphi(m)\}^{-2}$ converges.

(Hint: If m has a prime factor not exceeding $m^{\frac{1}{2}}$, then $m - \varphi(m) = m \prod_{p|m} (1 - \frac{1}{p}) \geq m^{\frac{1}{2}}$. If not, m is either the square of a prime number or the product of two distinct prime numbers.)

4. (a) Show that $\sum \left\{ \log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right\}$ converges, the sum being extended over all prime numbers in increasing order.
 (b) If $k > 3$ show that

$$\sum_{p \leq \xi} \frac{1}{p} = \log \log \xi + \sum_p \left\{ \log \left(1 - \frac{1}{p} \right) + \frac{1}{p} \right\} + o(1).$$

(Cf. the second proof of Theorem 114.)

5. For $\xi > 0$ define $\vartheta(\xi) = \sum_{p \leq \xi} \log p$.

Show that there are positive constants a' and a'' such that $a'\xi < \vartheta(\xi) < a''\xi$ for $\xi \geq 2$.

(Hint: Use the fact that $\{\pi(\xi) - \pi(\sqrt{\xi})\} \log \sqrt{\xi} \leq \vartheta(\xi) \leq \pi(\xi) \log \xi$.)

6. Show that

$$\left| \sum_{p \leq n} \frac{\log p}{p} - \log n \right|$$

is bounded for $n = 1, 2, \dots$

(Hint: Use Theorem 27 to show that

$$n \sum_{p \leq n} \frac{\log p}{p} - \sum_{p \leq n} \log p \leq \log n! \leq n \sum_{p \leq n} \frac{\log p}{p-1},$$

and then use Ex. 5 and the obvious inequalities $e^n > n^n (n!)^{-1} \geq 1$.)

7. In the notation of Ex. 5 show that

$$\lim_{\xi \rightarrow \infty} \vartheta(\xi) \log \xi \cdot \{\pi(\xi)\}^{-1} = 1.$$

(Hint: Show that

$$\vartheta(\xi) \log \xi \leq \pi(\xi) \leq \vartheta(\xi) + o(\xi) \log o(\xi)$$

if $\xi = w^\vartheta$, and then take $w = \log \xi$ for large ξ .)

8. If $\epsilon > 0$ show that the number of distinct prime factors of n is less than $(1 + \epsilon) \log n \log^{-1} \log n$ for all sufficiently large n .

(Hint: If n has r distinct prime factors, where $r > 2$, then $n = p_1 p_2 \cdots p_r$ and accordingly $\log n \geq \sum_{i=1}^r \log p_i \geq c \cdot r \log r$ for some $c > 0$.)

9. If $t > 0$ and $\xi \geq 2$ show that

$$\prod_{t < p \leq \xi} \left(1 - \frac{1}{p} \right)^{-1} < c \log \xi \cdot \prod_{p \leq t} \left(1 - \frac{1}{p} \right)^{-1}.$$

(Cf. the proof of the second part of Theorem 115.)

Exercises for Chapter II

1. Suppose given a set of N objects and certain properties $A_1, A_2, \dots, A_r$ pertaining to them. Let $N(A_i)$ of the objects have property A_i , $N(A_i, A_j)$ of the objects have both properties A_i and A_j , $N(A_i, A_j, A_k)$ of the objects have the three properties A_i, A_j , and A_k , $\dots$, $N(A_1, A_2, \dots, A_r)$ have all r properties $A_1, A_2, \dots, A_r$. Let Z be the number of the objects that have none of the properties $A_1, A_2, \dots, A_r$. Show that

$$Z = N + \sum_{n=1}^r (-1)^n N_n,$$

where

$$N_n = \sum N(A_{i_1}, A_{i_2}, \dots, A_{i_n}) \quad (1 \leq i_1 < i_2 < \dots < i_n \leq r).$$

2. Under the conditions of the preceding problem, show that if m is even and $0 < m \leq r$ then

$$N + \sum_{n=1}^{m-1} (-1)^n N_n \leq Z \leq N + \sum_{n=1}^m (-1)^n N_n.$$

3. Given that $\xi > 0$ and that $a_1, a_2, \dots, a_r$ are given integers no two of which are equal. Let $P(\xi)$ denote the number of $n \leq \xi$ such that $n + a_1, n + a_2, \dots, n + a_r$ are all prime numbers. By the method used in proving Theorem 119 show that for $r \geq 2$

$$P(\xi) < B\xi(\log \xi)^{-r}(\log \log \xi)^r,$$

where B is a positive number depending only on $r, a_1, a_2, \dots, a_r$.

(*Hint:* First show that the result is trivial if $a_1, a_2, \dots, a_r$ represent every residue class modulo some prime number. Also note that any prime dividing $a_i - a_j$ for some i and j ($i \neq j$) plays an exceptional role similar to that played by the prime 2 in the proof of Theorem 119.)

4. Show that Theorem 120 is still true if the general term of the series is $p^{-1}(\log p)^q$ instead of p^{-1} , where $q < 1$.

Exercises for Chapter III

1. If $m > 0$ and $\chi(m)$ is defined as in Ex. 13 of Part One, Chap. V, show that there exists a number g that belongs to the exponent $\chi(m)$ modulo m . Infer that the numbers that occur as exponents modulo m are precisely the divisors of $\chi(m)$.
2. Let $m > 1$. Show that the following five assertions are equivalent:
 - (a) m is 2, 4, a power of an odd prime number, or twice a power of an odd prime,
 - (b) $\chi(m) = \varphi(m)$,
 - (c) there exists a number g such that every integer relatively prime to m is congruent modulo m to a power of g ,
 - (d) if a belongs to the exponent 2 modulo m , then $a \equiv -1 \pmod{m}$,
 - (e) the product of the elements in a reduced residue system modulo m is congruent to -1 modulo m .

3. If $k_1 > 0$, $\chi_1(a)$ is a character modulo k_1 , $k_2 > 0$, and $\chi_2(a)$ is a character modulo k_2 , then $\chi_1(a)\chi_2(a)$ is a character modulo the least common multiple of k_1 and k_2 .
4. If $k_1 > 0$, $k_2 > 0$, $(k_1, k_2) = 1$, and $\chi(a)$ is a character modulo k_1k_2 , then $\chi(a)$ is expressible uniquely in the form $\chi_1(a)\chi_2(a)$, where $\chi_1(a)$ is a character modulo k_1 and $\chi_2(a)$ is a character modulo k_2 .
(Hint: Let $u_1 \equiv 1 \pmod{k_1}$, $u_1 \equiv 0 \pmod{k_2}$, $u_2 \equiv 0 \pmod{k_1}$, $u_2 \equiv 1 \pmod{k_2}$. Then $\chi(a) = \chi(u_1a + u_2)\chi(u_1 + u_2a)$.)
5. Let $\chi(a)$ be a character modulo k and $\chi_0(a)$ the principal character modulo k . Let k_1 and k_2 be divisors of k , $\chi_1(a)$ a character modulo k_1 , and $\chi_2(a)$ a character modulo k_2 such that

$$\chi(a) = \chi_0(a)\chi_1(a) = \chi_0(a)\chi_2(a).$$

Show that there exists a character $\chi_3(a)$ modulo (k_1, k_2) such that

$$\chi(a) = \chi_0(a)\chi_3(a).$$

6. Let $k > 0$, where k has r distinct odd prime factors. Show that the number of real characters (characters of the first and second kinds) is 2^r if $4 \nmid k$, 2^{r+1} if $4 \mid k$ but $8 \nmid k$, and 2^{r+2} if $8 \mid k$.
7. Let $\chi(a)$ be a non-principal character modulo k (character of the second or third kind) and

$$S(m) = \sum_{a=1}^m \chi(a).$$

If d is a number such that $d \geq |S(m)|$ for every $m \geq 1$, show that

$$|L(1, \chi)| < 1 + d \left(1 + \frac{1}{2} + \cdots + \frac{1}{k} \right).$$

Infer that $|L(1, \chi)| < \log k$.

(Hint: $L(1, \chi) = \sum_{m=1}^{\infty} S(m) \left(\frac{1}{m} - \frac{1}{m+1} \right)$.)

8. Show that the series (71) and (73) actually converge uniformly for $s \geq \epsilon$, provided $\epsilon > 0$.
9. If $\chi(a)$ is a character put $g = g(s, w) = \sum_a \chi(a)\Lambda(a)(a^s \log a)^{-w}$ for $s > 1$. Show that $e^g = L(s, \chi)$ for $s > 1$.
(Hint: Show that the derivative of $e^{-g}L(s, \chi)$ is zero.)
10. Show that if $0 < r < 1$ and v and ψ are real then

$$(1-r)^3 |1 - re^{i\psi}|^4 |1 - re^{2i\psi}| \leq 1.$$

(Hint: Use the relation $\log |1 - re^{i\psi}| = -\sum_{n=1}^{\infty} \frac{r^n}{n} \cos n\psi$.)

Also show that if $\sqrt{2}|A - 1| < 1$, the above inequality can be used in place of Theorem 149 for the purpose of proving Theorem 151. (Theorem 149 is the special case $A = \frac{1}{5}$. The case $A = 1$ is also sometimes used.)

11. If $R > 0$ use Ex. 9 to show that

$$\prod_{\chi} L(s, \chi) \geq 1$$

for real $s > 1$, where the product is extended over all the characters modulo k . Show that this inequality can be used to give another proof of Theorem 151.

(*Hint:* The conjugate of a character of the third kind is itself a character of the third kind.)

12. a) Let $k > 0$. If $\chi_0(a)$ is the principal character modulo k , show that (in the notation of Ex. 9) $g(s, \chi_0) - \log(s - 1)$ has a finite limit as $s \rightarrow 1$ from the right. If $\chi(a)$ is a non-principal character modulo k , show that $g(s, \chi)$ has a finite limit as $s \rightarrow 1$ from the right.
- b) Let $(l, k) = 1$, $l > 0$. Show that for $s > 1$ (with the summations as in Theorem 154)

$$\frac{1}{h} \sum_{\chi} \frac{\chi(l)}{g(s, \chi)} = \sum_{\substack{a \equiv l \\ (\text{mod } k)}}^{\infty} \frac{\Lambda(a)}{a^s \log a}.$$

- c) Prove Theorem 155 by using b) and the functions $g(s, \chi)$ instead of (82) and the functions $-\frac{L'(s, \chi)}{L(s, \chi)}$. In fact show that

$$\sum_{\substack{p \equiv l \\ (\text{mod } k) \\ p \leq \xi}} \frac{1}{p} - \frac{1}{h} \log \log \xi - \frac{1}{h} \log \frac{1}{s - 1}$$

has a finite limit as $s \rightarrow 1$ from the right.

- d) Show that $\sum_{p \equiv 1 \pmod{4}} \frac{1}{p}$ diverges.

13. Using Theorem 155 and Exercise 9 of Part I, Chap. V, show that, given any positive number r , however large, there exist an infinite number of primes q such that the difference between q and any other prime is greater than r in absolute value.

Exercises for Part Three

Exercises for Chapter I

1. If $b > 0$, $b' > 0$, and $ba' - ab' = \pm 1$, show that $\frac{a}{b}$ and $\frac{a'}{b'}$ are neighbors in the Farey series of order n for any n such that $b + b' > n = \max(b, b')$.
(Cf. the proof of Theorem 156.)
2. If $\frac{a}{b}$ and $\frac{a'}{b'}$ are neighbors in the Farey series of some order, then $\frac{a+a'}{b+b'}$ is the only fraction between them in the Farey series of order $b + b'$.
3. Suppose that for each fraction $\frac{a}{b}$ with $(a, b) = 1$ and $b > 0$ we construct the circle with radius $(2b^2)^{-1}$ and center at $\frac{a}{b} + i(2b^2)^{-1}$ in the complex plane. Show that no two of these circles intersect and that two such circles are tangent if and only if the corresponding fractions are neighbors in the Farey series of some order.
4. If ξ is a given irrational number, show that there are an infinite number of fractions $\frac{a}{b}$ with $(a, b) = 1$, $b > 0$, and $|\frac{a}{b} - \xi| < (2b^2)^{-1}$.

Exercises for Chapter II

1. Show directly (that is, without using any of the results of this chapter) that if $n = x^2 + y^2$, $x > 0$, $y > 0$, $(x, y) = 1$ has more than one solution, then n is composite.
2. Show directly that if some $p \equiv 3 \pmod{4}$ divides n with exactly odd multiplicity, then n is not expressible as a sum of two squares.
3. By using the identity $2(x^2 + y^2) = (x + y)^2 + (x - y)^2$ but using none of the results of this chapter, show that $U(2n) = U(n)$.

4. Prove that every $p \equiv 1 \pmod{4}$ is expressible as a sum of two squares by use of Exercise 8 of Part One, Chapter V instead of the methods of this chapter.

(*Hint:* In the notation of the problem mentioned, take $r = s = [\sqrt{p}] + 1$ and choose a so that $a^2 + 1 \equiv 0 \pmod{p}$.)

5. By using only the result of the preceding problem and the identity

$$(x_1^2 + y_1^2)(x_2^2 + y_2^2) = (x_1x_2 + y_1y_2)^2 + (x_1y_2 - x_2y_1)^2,$$

show that a positive n is expressible as a sum of two squares if no $p \equiv 3 \pmod{4}$ divides n with exactly odd multiplicity.

6. If $n > 0$, if no $p \equiv 3 \pmod{4}$ divides n with exactly odd multiplicity, and if m is defined as in Theorem 164, show that the number of solutions of $n = x^2 + y^2$, $x \geq y \geq 0$ is equal to the number of solutions of $m = xy$, $x \geq y > 0$.

(*Hint:* Show that both quantities are equal to $\left[\frac{\tau(m)+1}{2}\right]$. Cf. Exercise 2 of Part One, Chap. II.)

7. For $n > 0$ put $b_n = 1$ if $U(n) > 0$ and $b_n = 0$ if $U(n) = 0$. Show that $\sum_{n=1}^{\infty} b_n n^{-s}$ converges if $s > 1$ and diverges if $s \leq 1$.

8. If $x > 0$ show that

$$\pi x - A\sqrt{x} - B < Q(x) < \pi x + A\sqrt{x} + B.$$

Exercises for Chapter III

1. Show that in the proof of Theorem 168 it is possible to dispense with the preliminary proof that m is odd. (If this is done, the inequality $|v| < \frac{m}{2}$ must be replaced by $|v| \leq \frac{m}{2}$ and later the possibility that $n = m$ must be excluded by a separate argument.)
2. Show that if $n_1 > 0$, $n_2 > 0$, and $(n_1, n_2) = 1$, then

$$\frac{Q(n_1 n_2)}{8} = \frac{Q(n_1)}{8} \cdot \frac{Q(n_2)}{8}.$$

3. Let $R(n)$ be the number of solutions of $x_1^2 + x_2^2 + x_3^2 + x_4^2 = n$, $(x_1, x_2, x_3, x_4) = 1$. If $n > 0$ show that

$$Q(n) = \sum_{d|n} R\left(\frac{n}{d}\right), \quad R(n) = \sum_{d|n} \mu(d) Q\left(\frac{n}{d}\right).$$

(Cf. Theorems 38 and 162.)

4. Use the results of the two preceding exercises to show that if $n_1 > 0$, $n_2 > 0$, and $(n_1, n_2) = 1$, then

$$\frac{R(n_1 n_2)}{8} = \frac{R(n_1)}{8} \cdot \frac{R(n_2)}{8}.$$

5. Let u be a positive odd number and w the largest square-free divisor of u . Using the results of the preceding two exercises, show that

$$\begin{aligned} R(u) &= 8u\{S(w)w^{-3}\}, \\ R(2u) &= 24u\{S(w)w^{-1}\}, \\ R(4u) &= 16u\{S(w)w^{-1}\}, \\ R(2^l u) &= 0 \quad \text{if } l > 2. \end{aligned}$$

6. Show that there are infinitely many positive n for which

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = n, \quad x_1 > x_2 > x_3 > x_4 > 0$$

has no solutions.

7. Show that if n is sufficiently large then

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = n, \quad x_1 > x_2 > x_3 > x_4 > 245^2$$

has solutions.

Exercises for Chapter IV

1. Use Theorem 181 to give another proof that if $n > 0$ and no $p \equiv 3 \pmod{4}$ divides n with exactly odd multiplicity, then n is expressible as a sum of two squares.

(*Hint:* Without loss of generality it may be assumed that n is square-free. Then there exist integers b and c such that $b^2 \equiv -1 + cn$. Hence the definite binary form $nx_1^2 + 2bx_1x_2 + cx_2^2$ is equivalent to $x_1^2 + x_2^2$.)

2. If each of two positive integers is expressible as a sum of two squares, their product is also. Show that the analogous assertion for three squares is false.

3. Show that if $p \equiv 5 \pmod{12}$ and $p > 17$, then p is expressible as a sum of three distinct positive squares.

(*Hint:* Use the identity $9(a^2 + b^2) = (2a - b)^2 + (2a + 2b)^2 + (2b - a)^2$.)

4. Using only the results of this chapter show that for $n > 0$ there exist solutions of $x_1^2 + x_2^2 + x_3^2 + x_4^2 = n$, $(x_1, x_2, x_3, x_4) = 1$ if and only if $8 \nmid n$.

5. Show that for given m and n the system

$$\begin{aligned}x_1^2 + x_2^2 + x_3^2 + x_4^2 &= n, \\x_1 + x_2 + x_3 + x_4 &= m\end{aligned}$$

has a solution if and only if either

- (i) $4n - m^2 = 0$ and $4 \mid m$ or
- (ii) $4n - m^2 > 0$, $m \equiv n \pmod{2}$, and $4n - m^2$ is not of the form $4^a(8b+7)$, $a \geq 0$, $b \geq 0$.

6. If $x > 0$ let $N(x)$ be the number of positive integers not exceeding x and not expressible as a sum of three squares. Show that

$$N(x) = 3 \sum_{l=0}^{r-1} 4^{-l} + \theta \cdot 4^{-r},$$

where $r = \left\lceil \frac{\log x - \log 7}{\log 4} \right\rceil$, and infer that

$$A \frac{x}{\sqrt{\log x}} < N(x) < B \frac{x}{\sqrt{\log x}}.$$

7. Use the arguments of this chapter to show that if $n > 4$ and $n \equiv 1, 2, 3, 5$, or $6 \pmod{8}$, there exist solutions of

$$x_1^2 + x_2^2 + x_3^2 = n, \quad (x_1, x_2, x_3) = 1.$$

Also show that if $n \equiv 0, 4$, or $7 \pmod{8}$ no such solutions exist.

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Index of Conventions

$a, b, c, d, \dots$	denote integers (throughout, except as noted), 11
$p, p', \dots, p_r, p_0, \dots$	denote primes (pp. 18–250), 18
n, n_1, n_2, d, d_1, d_2	is > 0 (pp. 37–52), 37
d	is not a perfect square and is $\equiv 0$ or $1 \pmod{4}$ (pp. 70–75, 172–250), 70, 172
$a, a', a'', \dots$	positive constants > 0 (pp. 88–103), 88
i	$= \sqrt{-1}$ (pp. 109–250), 109
ξ, η, x, y, z	not necessarily integral (pp. 115–125), 115
$a, b, c, d, a_1, a_2, a_3, a_4$	are > 0 (pp. 135–140), 135
u, v, m, n	are odd integers > 0 (pp. 146–150), 146
d	is a fundamental discriminant (pp. 221–229), 221

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